CARIBBEAN AIRLINES

ATR 72-212A '600 VERSION'



MINIMUM EQUIPMENT LIST

Copy No.:……..

Doc. No.: BW-ATR-MEL-0039

CARIBBEAN AIRLINES

ATR 72-212A '600 VERSION'

MINIMUM EQUIPMENT LIST

Doc. No.: BW-ATR-MEL-0039

CIVIL AVIATION AUTHORITY

Mailing Address: P.O. Box 2163, National Mail Centre, Golden Grove Road, Piarco
Republic of Trinidad & Tobago. Tel: (868) 66TTCAA (88222). Fax: (868) 669-5397
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OUR REF: CAA/968/26/FOI

19 October, 2026.

Captain Daniel Galy
Chief Pilot (Ag)
Caribbean Airlines Limited,
Iere House
Golden Grove Road
PIARCO

Dear Captain Galy,

APPROVAL: ATR 72-212A '600 VERSION' MINIMUM EQUIPMENT LIST (MEL)
- DOC. NO: BW-ATR-MEL-0039, REVISION No: 19 dated 12 June, 2026

Reference No: ***OY: 160/2026*** dated 09 October, 2026.

In accordance with the Trinidad and Tobago Civil Aviation Regulations ***No.3:34 (4)***, the Trinidad and Tobago Civil Aviation Authority (***TTCAA***) hereby approves **Revision No: 19,** dated 12 June, 2026 of the Caribbean Airlines ATR 72-212A '600 Version' Minimum Equipment List (MEL) **-** Doc No: **BW-ATR-MEL-0039.**

The ATR 72 '600 Version' Minimum Equipment List, Doc No: BW-ATR-MEL-0039 shall be current and appropriate for the prescribed functions, and the TTCAA must be advised of any amendments to the document.

Please find attached the Original List of Effective Pages dated and signed by an authorized TTCAA Inspector.

Yours respectfully,
TRINIDAD AND TOBAGO CIVIL AVIATION AUTHORITY



Captain Hollis A. Gyan
/f/ Director General Civil Aviation

xc. Mr. Cary Price – Director-General, TTCAA.
Mr. Gerard Mark – Executive Manager, Safety Regulations, TTCAA.
Mr. Ronald Bidaisee – Manager, Flight Standards (Ag), TTCAA.
Captain Raj Narwani – Flight Operations Inspector, TTCAA.
Captain Rene Seow – Director of Operations, CAL.
Captain Simon Laughlin – Manager Flight Standard and Training (Ag), CAL.
Captain David Seignoret – ATR Fleet Manager, CAL.
Mr. Dion Nagee – Executive Manager Quality and Air Safety, CAL.
Mr. Dinesh Rampersad – Manager Engineering (Ag), CAL.
Mr. Derry Gosine – Quality Manager, Maintenance (Ag), CAL.
Mr. Shyam Kamal – Quality Manager, Operations (Ag), CAL.

Board Members: Mr. Kimchand Rampaul *(Chairman)*, Mr. Yves Jacques Nicholson (*Deputy Chairman*), Mr. Trevor Benjamin, Mr. Brandon Baksh, Mr. Doolar Ramlal, Ms. Sarah Awai-King, Mr. Javan Edwards, Mr. Cary Price *(Director-General)*

Caribbean Airlines

FLIGHT OPERATIONS PUBLICATIONS FILING INSTRUCTIONS

DOCUMENT TITLE : ATR 72-600 MINIMUM EQUIPMENT LIST

DOCUMENT NO. : BW-ATR-MEL-0039

REVISION NO. : 19 DATE: 12 June, 2026

HIGHLIGHTS OF CHANGE

MEL Revision 19 incorporates changes in accordance with ATR:

- Schedule Revision – DIFF- 12 JUN 2026 EASA Approved
  - BW6_MMEL_75_ALL_L_SI_diff_Rev19.0

The following changes are the Highlights of Changes for Revision 19.

PAGE	EXPLANATION OF CHANGE
0.5 Record of Revisions Approval	Updated.
0.7 List of Effective Pages	Updated.
0.8 Distribution List / Access	Updated.
0.9 PREAMBLE: GENERAL	
1.02 Principle	Editorial correction.
2.1 MMEL Revision	A new section included, to highlight descriptions relating to MMEL Revision(s).
2.2 Presentation of the MEL	Section number revised.
2.3 How to Read the MEL Operations & Maintenance Procedures Parts	Content improvement providing clearer guidance on multiple dispatch conditions.
2.4 How to Read the MEL Maintenance Procedures Part	Section number revised.
2.5 Definitions	Section number revised and content improved.
2.6 CAT 2 Automatic Approach	Section number revised.
2.7 Glossary of Standard Nomenclature	Section number revised. Glossary updated to harmonize with other operational documents.
0.10 Historical Data	Updated.
0.11 List of Modifications (LOM)	Updated.

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Insert updated pages dated 12 June 2026, following the sequence on the List of Effective Pages. On the List of Effective Pages new pages are identified by *'n'*, '*' for revised pages and *'d'* for pages to be deleted.

Note: When updating your manual, please audit your book using the current List of Effective Pages to ensure that your manual is current.

Caribbean Airlines

FLIGHT OPERATIONS PUBLICATIONS FILING INSTRUCTIONS

DOCUMENT TITLE : ATR 72-600 MINIMUM EQUIPMENT LIST

DOCUMENT NO. : BW-ATR-MEL-0039

REVISION NO. : 19 DATE: 12 June, 2026

PAGE	EXPLANATION OF CHANGE
0.15 ALERT MESSAGES LIST (AML)	
20 AML CAUTION MESSAGES	
ENGINE	
ENG BOOST	Link correction.
ENG EEC	Link correction.
ENG START NO NH	Link correction.
FLIGHT CONTROL	
FLAPS JAM	Editorial correction.
AIR	
AIR BLEED 1	Links added.
AIR BLEED 2	Link added.
AIR BLEED 1 OVHT	Link added.
AIR BLEED 2 OVHT	Link added.
ANTI-ICE	
ICING	Removed from EWD message list.
AVIONICS	
DU DISCREPANCY	Editorial correction.
XPDR 1 + 2	Editorial correction.

System/Sequence Number	Remarks
ATA 21 – AIR CONDITIONING	
-31-01	Editorial: Harmonization of ‘altitude’ measure unit within operational documentation.
-31-04-01	Editorial: Harmonization of ‘altitude’ measure unit within operational documentation.
-31-04-02	Editorial: Harmonization of ‘altitude’ measure unit within operational documentation.
-31-04-03	Editorial: Harmonization of ‘altitude’ measure unit within operational documentation.
-50-01	ETOPS/NON-ETOPS configuration management for integration of MOD 4711.
-51-03	Editorial: Proviso updated for clarity.

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FLIGHT OPERATIONS PUBLICATIONS FILING INSTRUCTIONS

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DOCUMENT NO. : BW-ATR-MEL-0039

REVISION NO. : 19 DATE: 12 June, 2026

System/Sequence Number	Remarks
ATA 22 – AUTOMATIC FLIGHT CONTROL SYSTEM	
TOC. -1	Updated.
-18-01	Configuration management for integration of MOD 4711 – Addition of (O) procedures.
-18-03	Editorial: Operational/Maintenance procedure link corrected.
-50-11	Editorial correction.
ATA 23 – COMMUNICATIONS	
TOC. -1, -2	Updated.
-11-01	Removed deletion symbol.
-12-01-01	Editorial: “ACAS” changed to “TCAS”.
-44-01-05	Editorial correction.
-44-01-07B	Editorial correction.
-52-01	ETOPS/NON-ETOPS configuration management for integration of MOD 4711.
ATA 24 – ELECTRICAL POWER	
TOC. -1, -2	Updated.
-22-02	Editorial correction of Item Title.
-30-01	ETOPS/NON-ETOPS configuration management for integration of MOD 4711.
-30-02	ETOPS/NON-ETOPS configuration management for integration of MOD 4711.
ATA 25 – EQUIPMENT	
TOC. -2	Updated.
-11-01	Removed deletion symbol.
-11-02	(O) procedures added for clarity.
-11-03	(M) procedure updated to add an (M) procedure task.
-31-01-04	Editorial correction.
-65-10	New Item.
ATA 26 – FIRE PROTECTION	
TOC. -1, -2	Updated.
-17-01-01A	Addition of (O) and (M) procedures for sub-item “A”. Addition of placarding requirement.
-17-01-01B	(O) procedure updated for clarity.
-18-01	Repagination of Item.

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FLIGHT OPERATIONS PUBLICATIONS FILING INSTRUCTIONS

DOCUMENT TITLE : ATR 72-600 MINIMUM EQUIPMENT LIST

DOCUMENT NO. : BW-ATR-MEL-0039

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System/Sequence Number	Remarks
ATA 26 – FIRE PROTECTION *(Cont'd)*	
-19-01B	Addition of (O) and (M) procedures for sub-item "B". Addition of placarding requirement
-21-01	Repagination of Item.
-25-00	Repagination of Item.
-30-01	Repagination of Item.
ATA 27 – FLIGHT CONTROLS	
TOC. -1, -2	Updated.
-23-01B	(O) procedures updated.
-23-02	Repagination of Item.
-25-01	Repagination of Item.
-32-01	Repagination of Item..
-32-02	Repagination of Item.
-32-03	Repagination of Item.
-35-01	Repagination of Item.
-36-01	Repagination of Item.
-36-02	Repagination of Item.
-36-03	Repagination of Item.
-37-01	Repagination of Item.
-37-02	Repagination of Item.
-51-01	Repagination of Item.
-55-01	Repagination of Item.
-55-02	Repagination of Item.
-61-01	Repagination of Item.
-65-01	Repagination of Item.
-70-01-01	Repagination of Item.
-70-01-02	Repagination of Item.
ATA 28 – FUEL SYSTEM	
-21-02	ETOPS/NON-ETOPS configuration management for integration of MOD 4711.
-21-03	ETOPS/NON-ETOPS configuration management for integration of MOD 4711.

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FLIGHT OPERATIONS PUBLICATIONS FILING INSTRUCTIONS

DOCUMENT TITLE : ATR 72-600 MINIMUM EQUIPMENT LIST

DOCUMENT NO. : BW-ATR-MEL-0039

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System/Sequence Number	Remarks
ATA 28 – FUEL SYSTEM *(Cont'd)*	
-21-05	ETOPS/NON-ETOPS configuration management for integration of MOD 4711.
-23-01	ETOPS/NON-ETOPS configuration management for integration of MOD 4711.
-25-01-01	Editorial: (M) procedure updated to harmonize the procedure title with (M) procedures.
-42-01	ETOPS/NON-ETOPS configuration management for integration of MOD 4711.
-42-03	ETOPS/NON-ETOPS configuration management for integration of MOD 4711.
ATA 29 – HYDRAULIC SYSTEM	
-33-01	Editorial update for clarity.
-33-02	Editorial update for clarity.
ATA 30 – ICE AND RAIN PROTECTION	
-11-01	'ETOPS' operations harmonized as per aircraft model applicability.
-11-02	'ETOPS' operations harmonized as per aircraft model applicability.
-21-01	ETOPS/NON-ETOPS configuration management for integration of MOD 4711.
-21-02	'ETOPS' operations harmonized as per aircraft model applicability.
-30-01	'ETOPS' operations harmonized as per aircraft model applicability.
-31-04	(O) procedure update to improve clarity.
-31-05B	'ETOPS' operations harmonized as per aircraft model applicability.
-53-01	'ETOPS' operations harmonized as per aircraft model applicability.
-61-01	'ETOPS' operations harmonized as per aircraft model applicability.
-70-01	Editorial correction for Windshield.
-80-01	(M) procedure updated.

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FLIGHT OPERATIONS PUBLICATIONS FILING INSTRUCTIONS

DOCUMENT TITLE : ATR 72-600 MINIMUM EQUIPMENT LIST

DOCUMENT NO. : BW-ATR-MEL-0039

REVISION NO. : 19 DATE: 12 June, 2026

System/Sequence Number	Remarks
ATA 31 – INDICATING AND RECORDING SYSTEMS	
TOC. -1,	Updated.
-21-01	Updated Item title.
-31-03-01	Editorial: APM word harmonization.
-31-03-02	Item title updated for clarity.
-32-04	Editorial correction.
-60-02-04	Editorial correction.
-60-03-01	Editorial correction.
-60-03-02	Dispatch condition updated – PF changed to PM in proviso.
-60-03-07	Editorial correction of Item title.
-60-06	ETOPS/NON-ETOPS configuration management for integration of MOD 4711.
ATA 32 – LANDING GEAR	
-31-03	'ETOPS' operations harmonized as per aircraft model applicability.
-61-01	(O) procedure updated for clarity.
ATA 33 – LIGHTS	
TOC. -1, -2	Updated.
-16-01-01	Removed deletion symbol.
-16-01-02A	Dispatch condition updated for clarity.
-42-01A	Dispatch condition updated for clarity.
-46-01A	Dispatch condition updated for clarity.
-49-01A	Dispatch condition updated for clarity.
-49-02A	Dispatch condition updated for clarity.
-51-01A	Dispatch condition updated for clarity.

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FLIGHT OPERATIONS PUBLICATIONS FILING INSTRUCTIONS

DOCUMENT TITLE : ATR 72-600 MINIMUM EQUIPMENT LIST

DOCUMENT NO. : BW-ATR-MEL-0039

REVISION NO. : 19 DATE: 12 June, 2026

System/Sequence Number	Remarks
ATA 34 - NAVIGATION	
TOC. -1, -2, -3, -4	Updated.
-11-01	ETOPS/NON-ETOPS configuration management with integration of MOD 4711.
-20-01	ETOPS/NON-ETOPS configuration management with integration of MOD 4711. Addition of new (O) procedure – (M) procedure update to add AHRS NOT ALIGN alert scenarios.
-22-01-01	ETOPS/NON-ETOPS configuration management with integration of MOD 4711.
-22-01-02	Editorial correction.
-36-01	Dispatch updated to clarify the dispatch with only one ILS inoperative – New (O) procedure.
-42-01	Editorial update " ACAS" changed to "TCAS".
-48-01-01	CAL improved the proviso and (O) procedures to comply with the Operations Manual General – Part A, Doc. No.: BW-OM-0001.
-48-01-02	CAL improved the proviso and (O) procedures to comply with the Operations Manual General – Part A, Doc. No.: BW-OM-0001.
-48-01-03	CAL improved the proviso and (O) procedures to comply with the Operations Manual General – Part A, Doc. No.: BW-OM-0001.
-48-01-04	CAL improved the proviso and (O) procedures to comply with the Operations Manual General – Part A, Doc. No.: BW-OM-0001.
-48-01-05A	CAL improved the proviso and (O) procedures to comply with the Operations Manual General – Part A, Doc. No.: BW-OM-0001.
-48-01-05B	CAL improved the proviso and (O) procedures to comply with the Operations Manual General – Part A, Doc. No.: BW-OM-0001.

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FLIGHT OPERATIONS PUBLICATIONS FILING INSTRUCTIONS

DOCUMENT TITLE : ATR 72-600 MINIMUM EQUIPMENT LIST

DOCUMENT NO. : BW-ATR-MEL-0039

REVISION NO. : 19 DATE: 12 June, 2026

System/Sequence Number	Remarks
ATA 34 – NAVIGATION *(Cont'd)*	
-50-01	Repagination of item.
-50-02	Repagination of item.
-50-03	Repagination of item.
-52-01A	Removed Item not in MMEL.
-52-02A	Editorial update "ACAS" changed to "TCAS".
-52-02B	Editorial update "ACAS" changed to "TCAS".
-52-03	Repagination of item.
-52-04	Editorial correction.
-57-01-01	Editorial: title update. Editorial update (O) procedure "ACAS" changed to "TCAS".
-58-01	ETOPS/NON-ETOPS configuration management with integration of MOD 4711.
-61-01	Repagination of item.
-61-02A	Editorial correction.
-61-02C	Repagination of item.
ATA 35 -OXYGEN	
TOC. -1, -2	Updated.
-65-01	Addition of new conditions of dispatch.
ATA 36 – PNEUMATIC SYSTEM	
-11-02-01	ETOPS/NON-ETOPS configuration management with integration of MOD 4711.
-22-01	Editorial correction.
ATA 52 - DOORS	
TOC. -2	Updated.
-11-02	Editorial correction.
-50-01-04	Editorial title update.

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FLIGHT OPERATIONS PUBLICATIONS FILING INSTRUCTIONS

DOCUMENT TITLE : ATR 72-600 MINIMUM EQUIPMENT LIST

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System/Sequence Number	Remarks
ATA 61 - PROPELLERS	
-21-06	(O) procedure updated for clarity.
-22-02-01	(O) procedure updated to add step to check AFU.
-22-02-03	(O) procedure updated to add step to check AFU.
-42-01	ETOPS/NON-ETOPS configuration management with integration of MOD 4711.
ATA 73 – ENGINE FUEL AND CONTROL	
TOC. -1, -2	Updated.
-23-01	Editorial update to clarify the rotary selector position.
-23-02-01	Editorial sub-title update and typo correction.
ATA 74 - IGNITION	
-21-01	ETOPS/NON-ETOPS configuration management with integration of MOD 4711.
ATA 77 – ENGINE INDICATING	
TOC. -1, -2	Updated.
-00-02	ETOPS/NON-ETOPS configuration management with integration of MOD 4711.
-11-01-01	Removed Item not in MMEL.
-11-01-02	ETOPS/NON-ETOPS configuration management with integration of MOD 4711.
-11-02	Repagination of Item.
-13-01-01	Dispatch condition updated for clarity.
-13-01-02	ETOPS/NON-ETOPS configuration management with integration of MOD 4711.
-13-01-03	New item.

0.2 TABLE OF CONTENTS

0.3 RECORD OF REVISIONS

0.4 RECORD OF TEMPORARY REVISIONS

0.5 RECORD OF REVISIONS APPROVAL

0.6 RECORD OF TEMPORARY REVISIONS APPROVAL

0.7 LIST OF EFFECTIVE PAGES

0.8 DISTRIBUTION & ACCESS

0.9 **PREAMBLE**

1. GOALS AND PRINCIPLE

1.01 Goals

1.02 Principle

2. GENERAL

2.1 Presentation of the MEL

2.2 How to read the MEL Operational Procedures Part (MOP)

2.3 How to read the MEL Maintenance Procedures Part (MMP)

2.4 Definitions

2.5 CAT 2 Automatic Approach

2.6 Glossary of Standard Nomenclature

0.10 MANUAL ORGANIZATION AND ADMINISTRATION

0.11 LIST OF MODIFICATIONS

0.12 CROSS REFERENCE TABLE

0.13 NON-ESSENTIAL DEFERRAL PROGRAM

0.14 COMMERCIAL DESIGNATION

0.15 **ALERT MESSAGE LIST**

1. INTRODUCTION

1.01 General

1.02 Reading the Table

WARNING MESSAGE 10

Engine

Elec

Flight Control

Landing Gear

Fire Protection

Air

Miscellaneous

CAUTION MESSAGE 20

Engine

Fuel

Elec

Hydraulic

Flight Control

Landing Gear

Doors

Air

Anti-Ice

Auto Pilot

MPC

Avionics

Miscellaneous

MFC

0.16 ATA CHATPER INDEX

SECTION 1 - MINIMUM EQUIPMENT LIST (MEL) & OPERATIONAL & MAINTENANCE PROCEDURES

- ATA SECTIONS

REVISION NO	ISSUE DATE	INSERTED BY	DATE INSERTED
9	09 JUN. 2022		
10	29 SEP. 2022		
10a (CAL)	06 FEB. 2023		
10b (CAL)	11 APR. 2023		
10c (CAL)	01 MAY 2023		
10d (CAL)	02 MAY 2023		
10e (CAL)	03 JUL. 2023		
10f (CAL)	03 NOV. 2023		
10g (CAL)	06 MAY 2024		
10h (CAL)	15 AUG. 2024		
12	27 JUN. 24		
12a (CAL)	07 APR. 25		
14	30 APR. 25		
17	20 OCT. 25		
19	12 JUN. 26		

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REV. NO	ISSUE DATE	DATE INSERTED	INSERTED BY	DATE REMOVED	REMOVED BY	INCORPORATED INTO MANUAL BY REV. NO.

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REV. NO	SIGNATURE	DATE
9	INCORPORATED	09 JUN. 22
10	Chief Pilot (Ag.) - Captain Gordon Forbes	29 SEP. 22
10a (CAL)	Chief Pilot (Ag.) - Captain Gordon Forbes	06 FEB. 23
10b (CAL)	Chief Pilot (Ag.) - Captain Gordon Forbes	11 APR. 23
10c (CAL)	Chief Pilot (Ag.) - Captain Gordon Forbes	01 MAY 23
10d (CAL)	Chief Pilot (Ag.) - Captain Gordon Forbes	02 MAY 23
10e (CAL)	Chief Pilot (Ag.) - Captain Gordon Forbes	03 JUL. 23
10f (CAL)	Chief Pilot (Ag.) - Captain Christopher Kelly	06 NOV. 23
10g (CAL)	Chief Pilot (Ag.) - Captain Christopher Kelly	14 MAY 24
10h (CAL)	Chief Pilot (Ag.) - Captain Christopher Kelly	11 SEP. 24
12	Chief Pilot - Captain Renè Seow	04 Apr. 25
12a (CAL)	Chief Pilot - Captain Renè Seow	07 Apr. 25
14	Chief Pilot - Captain Renè Seow	05 Jun. 25
17	Chief Pilot - Captain Renè Seow	26 Nov. 25
19	Chief Pilot - 	05 Oct. 26

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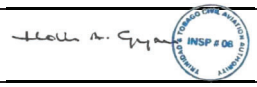
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Approved by TTCAA	
Signature:	Date 19 October, 2026

DESCRIPTION	PAGE		REVISION NO.	DATE
FRONT MATTER - SECTION 0				
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	TOC.0.2-2		9	09 Jun. 22
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	LEP.0.7-18	*	19	12 Jun. 26
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Approved by TTCAA			
Signature:		Date	19 October, 2026

DESCRIPTION	PAGE		REVISION NO.	DATE
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	PRE.0.9-8	*	19	12 Jun. 26
	PRE.0.9-9	*	19	12 Jun. 26
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	PRE.0.9-20	*	19	12 Jun. 26
	PRE.0.9-21	*	19	12 Jun. 26

Approved by TTCAA	
Signature: 	Date 19 October, 2026

DESCRIPTION	PAGE		REVISION NO.	DATE
FRONT MATTER - SECTION 0				
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	PRE.0.9-24	*	19	12 Jun. 26
	PRE.0.9-25	*	19	12 Jun. 26
	PRE.0.9-26	*	19	12 Jun. 26
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	NEF.0.13-3		9	09 Jun. 22
	NEF.0.13-4		9	09 Jun. 22
	NEF.0.13-5		9	09 Jun. 22

Approved by TTCAA	
Signature:	Date 19 October, 2026

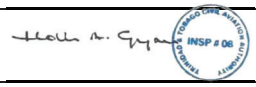
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	AML.20.0.15-13		12	27 Jun. 24
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	21-7		12	27 Jun. 24
	21-8	*	19	12 Jun. 26
	21-9		9	09 Jun. 22
	21-10		9	09 Jun. 22
	21-11	*	19	12 Jun. 26
	21-12	*	19	12 Jun. 26
	21-13	*	19	12 Jun. 26
	21-14		9	09 Jun. 22
	21-15		9	09 Jun. 22
	21-16	*	19	12 Jun. 26
	21-17		12	27 Jun. 24
	21-18		12	27 Jun. 24
	21-19	*	19	12 Jun. 26
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	21-25		9	09 Jun. 22
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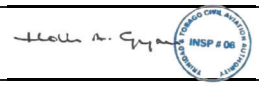
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	22-9		12	27 Jun. 24
	22-10		9	09 Jun. 22
	22-11		9	09 Jun. 22
	22-12		9	09 Jun. 22
	22-13		9	09 Jun. 22
	22-14		9	09 Jun. 22
	22-15		17	20 Oct. 25
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	22-18	*	19	12 Jun. 26
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	23-10	*	19	12 Jun. 26
	23-11		10b (CAL)	11 Apr. 23
	23-12	*	19	12 Jun. 26
	23-13		12	27 Jun. 24
	23-14		10b (CAL)	11 Apr. 23
	23-15		10b (CAL)	11 Apr. 23
	23-16		10b (CAL)	11 Apr. 23
	23-17		10b (CAL)	11 Apr. 23
	23-18		10b (CAL)	11 Apr. 23
	23-19		10b (CAL)	11 Apr. 23
	23-20	*	19	12 Jun. 26
	23-21		12	27 Jun. 24
	23-22		12	27 Jun. 24
	23-23		10b (CAL)	11 Apr. 23
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	25-4		17	20 Oct. 25
	25-5		17	20 Oct. 25
	25-6		17	20 Oct. 25
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	25-9		17	20 Oct. 25
	25-10		17	20 Oct. 25
	25-11		17	20 Oct. 25
	25-12		17	20 Oct. 25
	25-13	*	19	12 Jun. 26
	25-14		17	20 Oct. 25
	25-15		17	20 Oct. 25
	25-16		17	20 Oct. 25
	25-17		17	20 Oct. 25
	25-18		17	20 Oct. 25
	25-19		17	20 Oct. 25
	25-20		17	20 Oct. 25
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	25-22		17	20 Oct. 25
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	26-4		12	27 Jun. 24
	26-5		9	09 Jun. 22
	26-6	*	19	12 Jun. 26
	26-7	*	19	12 Jun. 26
	26-8	*	19	12 Jun. 26
	26-9	*	19	12 Jun. 26
	26-10	*	19	12 Jun. 26
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	27-3	*	19	12 Jun. 26
	27-4	*	19	12 Jun. 26
	27-5	*	19	12 Jun. 26
	27-6	*	19	12 Jun. 26
	27-7	*	19	12 Jun. 26
	27-8	*	19	12 Jun. 26
	27-9	*	19	12 Jun. 26
	27-10	*	19	12 Jun. 26
	27-11	*	19	12 Jun. 26
	27-12	*	19	12 Jun. 26
	27-13	*	19	12 Jun. 26
	27-14	*	19	12 Jun. 26
	27-15	*	19	12 Jun. 26
	27-16	*	19	12 Jun. 26
	27-17	*	19	12 Jun. 26
	27-18	*	19	12 Jun. 26
	27-19	*	19	12 Jun. 26
	27-20	*	19	12 Jun. 26
	27-21	*	19	12 Jun. 26
	27-22	*	19	12 Jun. 26
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ATA 28 – FUEL SYSTEM				
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		28-2	* 19	12 Jun. 26
		28-3	12	27 Jun. 24
		28-4	* 19	12 Jun. 26
		28-5	9	09 Jun. 22
		28-6	* 19	12 Jun. 26
		28-7	* 19	12 Jun. 26
		28-8	12	27 Jun. 24
		28-9	* 19	12 Jun. 26
		28-10	12	27 Jun. 24
		28-11	12	27 Jun. 24
		28-12	10b (CAL)	11 Apr. 23
		28-13	10b (CAL)	11 Apr. 23
		28-14	10b (CAL)	11 Apr. 23
		28-15	12	27 Jun. 24
		28-16	* 19	12 Jun. 26
		28-17	12a (CAL)	07 Apr. 25
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		29-4		9	09 Jun. 22
		29-5		9	09 Jun. 22
		29-6		9	09 Jun. 22
		29-7		9	09 Jun. 22
		29-8		9	09 Jun. 22
		29-9		9	09 Jun. 22
		29-10	*	19	12 Jun. 26
		29-11	*	19	12 Jun. 26
		29-12	*	19	12 Jun. 26
ATA 30 – ICE AND RAIN PROTECTION					
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		TOC.30-2		9	09 Jun. 22
		TOC.30-3		9	09 Jun. 22
		TOC.30-4		9	09 Jun. 22
		30-1	*	19	12 Jun. 26
		30-2	*	19	12 Jun. 26
		30-3		9	09 Jun. 22
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		30-7	*	19	12 Jun. 26

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	30-13	*	19	12 Jun. 26
	30-14		9	09 Jun. 22
	30-15	*	19	12 Jun. 26
	30-16		9	09 Jun. 22
	30-17		9	09 Jun. 22
	30-18		9	09 Jun. 22
	30-19		9	09 Jun. 22
	30-20		9	09 Jun. 22
	30-21	*	19	12 Jun. 26
	30-22		9	09 Jun. 22
	30-23		9	09 Jun. 22
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	31-1	*	19	12 Jun. 26
	31-2	*	19	12 Jun. 26
	31-3	*	19	12 Jun. 26
	31-4		9	09 Jun. 22
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	31-9	*	19	12 Jun. 26
	31-10		17	20 Oct. 25
	31-11		17	20 Oct. 25
	31-12		17	20 Oct. 25
	31-13		17	20 Oct. 25
	31-14		17	20 Oct. 25
	31-15		17	20 Oct. 25
	31-16		17	20 Oct. 25
	31-17		17	20 Oct. 25
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		31-33	17	20 Oct. 25
	*	31-34	19	12 Jun. 26
		31-35	17	20 Oct. 25
		31-36	17	20 Oct. 25
		31-37	17	20 Oct. 25
		31-38	17	20 Oct. 25
		31-39	17	20 Oct. 25
		31-40	17	20 Oct. 25
		31-41	17	20 Oct. 25
		31-42	17	20 Oct. 25
		31-43	17	20 Oct. 25
		31-44	17	20 Oct. 25
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	32-9		17	20 Oct. 25
	32-10	*	19	12 Jun. 26
ATA 33 – LIGHTS				
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	33-8		9	09 Jun. 22
	33-9		12	27 Jun. 24
	33-10	*	19	12 Jun. 26
	33-11		9	09 Jun. 22
	33-12	*	19	12 Jun. 26
	33-13		9	09 Jun. 22
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	33-15	*	19	12 Jun. 26
	33-16	*	19	12 Jun. 26
	33-17		9	09 Jun. 22
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ATA 34 – NAVIGATION				
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		34-8	*	19	12 Jun. 26
		34-9		9	09 Jun. 22
		34-10		9	09 Jun. 22
		34-11		9	09 Jun. 22
		34-12	*	19	12 Jun. 26
		34-13	*	19	12 Jun. 26
		34-14	*	19	12 Jun. 26
		34-15	*	19	12 Jun. 26
		34-16	*	19	12 Jun. 26
		34-17	*	19	12 Jun. 26
		34-18	*	19	12 Jun. 26
		34-19	*	19	12 Jun. 26
		34-20	*	19	12 Jun. 26
		34-21	*	19	12 Jun. 26
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		34-25	*	19	12 Jun. 26
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	35-8		12	27 Jun. 24
	35-9	*	19	12 Jun. 26
	35-10	*	19	12 Jun. 26
ATA 36 – PNEUMATIC SYSTEM				
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	36-1		9	09 Jun. 22
	36-2	*	19	12 Jun. 26
	36-3		12	27 Jun. 24
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	TOC.42-2		9	09 Jun. 22
	42-1		12	27 Jun. 24
	42-2		9	09 Jun. 22
ATA 46 – INFORMATION SYSTEMS				
	TOC.46-1		10h (CAL)	15 Aug. 24
	TOC.46-2		10h (CAL)	15 Aug. 24
	46-1		10h (CAL)	15 Aug. 24
	46-2		10h (CAL)	15 Aug. 24
	46-3		10h (CAL)	15 Aug. 24
	46-4		10h (CAL)	15 Aug. 24
ATA 52 – DOORS				
	TOC.52-1		9	09 Jun. 22
	TOC.52-2	*	19	12 Jun. 26
	52-1		14	30 Apr. 25
	52-2	*	19	12 Jun. 26
	52-3		9	09 Jun. 22
	52-4		9	09 Jun. 22
	52-5		10e (CAL)	03 Jul. 23
	52-6		17	20 Oct. 25
	52-7		17	20 Oct. 25
	52-8		9	09 Jun. 22
	52-9		17	20 Oct. 25
	52-10		12	27 Jun. 24
	52-11		9	09 Jun. 22
	52-12		9	09 Jun. 22
	52-13	*	19	12 Jun. 26
	52-14	*	19	12 Jun. 26
	52-15		9	09 Jun. 22
	52-16		12	27 Jun. 24

Approved by TTCAA	
Signature:	Date 19 October, 2026

DESCRIPTION	PAGE		REVISION NO.	DATE
SECTION 1 – MEL OPERATIONAL & MAINTENANCE PROCEDURES - ATA SECTIONS				
ATA 56 – WINDOWS				
	TOC.56-1		9	09 Jun. 22
	TOC.56-2		9	09 Jun. 22
	56-1		12	27 Jun. 24
	56-2		9	09 Jun. 22
	56-3		9	09 Jun. 22
	56-4		9	09 Jun. 22
ATA 61 – PROPELLERS				
	TOC.61-1		12	27 Jun. 24
	TOC.61-2		12	27 Jun. 24
	61-1		9	09 Jun. 22
	61-2	*	19	12 Jun. 26
	61-3	*	19	12 Jun. 26
	61-4		12	27 Jun. 24
	61-5	*	19	12 Jun. 26
	61-6		9	09 Jun. 22
	61-7	*	19	12 Jun. 26
	61-8		17	20 Oct. 25
	61-9		12	27 Jun. 24
	61-10		12	27 Jun. 24
	61-11		12	27 Jun. 24
	61-12		12	27 Jun. 24
	61-13		12	27 Jun. 24
	61-14		12	27 Jun. 24

Approved by TTCAA	
Signature:	Date 19 October, 2026

DESCRIPTION	**PAGE**		**REVISION NO.**	**DATE**
SECTION 1 – MEL OPERATIONAL & MAINTENANCE PROCEDURES - ATA SECTIONS				
ATA 73 – ENGINE FUEL AND CONTROL				
	TOC.73-1	*	19	12 Jun. 26
	TOC.73-2	*	19	12 Jun. 26
	73-1		9	09 Jun. 22
	73-2		12	27 Jun. 24
	73-3		9	09 Jun. 22
	73-4	*	19	12 Jun. 26
	73-5	*	19	12 Jun. 26
	73-6		10b (CAL)	11 Apr. 23
	73-7		10b (CAL)	11 Apr. 23
	73-8		10b (CAL)	11 Apr. 23
	73-9		10b (CAL)	11 Apr. 23
	73-10		12	27 Jun. 24
	73-11		10b (CAL)	11 Apr. 23
	73-12		10b (CAL)	11 Apr. 23
ATA 74 - IGNITION				
	TOC.74-1		9	09 Jun. 22
	TOC.74-2		9	09 Jun. 22
	74-1	*	19	12 Jun. 26
	74-2	*	19	12 Jun. 26
ATA 75 - AIR				
	TOC.75-1		9	09 Jun. 22
	TOC.75-2		9	09 Jun. 22
	75-1		9	09 Jun. 22
	75-2		9	09 Jun. 22

Approved by TTCAA	
Signature:	Date 19 October, 2026

DESCRIPTION	PAGE		REVISION NO.	DATE
SECTION 1 – MEL OPERATIONAL & MAINTENANCE PROCEDURES - ATA SECTIONS				
ATA 77 – ENGINE INDICATING				
	TOC.77-1	*	19	12 Jun. 26
	TOC.77-2	*	19	12 Jun. 26
	77-1	*	19	12 Jun. 26
	77-2	*	19	12 Jun. 26
	77-3	*	19	12 Jun. 26
	77-4	*	19	12 Jun. 26
	77-5	*	19	12 Jun. 26
	77-6	*	19	12 Jun. 26
	77-7		9	09 Jun. 22
	77-8		12	27 Jun. 24
	77-9		9	09 Jun. 22
	77-10		9	09 Jun. 22
ATA 79 – OIL				
	TOC.79-1		9	09 Jun. 22
	TOC.79-2		9	09 Jun. 22
	79-1		9	09 Jun. 22
	79-2		9	09 Jun. 22

1. GENERAL

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10	Station Engineer – BGI		X	
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2. Location of Manual and Copy Requirements

The Chief Pilot controls and maintains the ATR Minimum Equipment List (MEL) Doc. No.: BW-ATR-MEL-0039 and the Distribution / Access for the ATR Minimum Equipment List Manual. The Technical Library will make available as indicated in paragraph 1, a copy of the ATR Minimum Equipment List (MEL) Doc. No.: BW-ATR-MEL-0039, or appropriate portions, to any personnel performing maintenance, flight or ground operations on Caribbean Airlines aircraft. The Chief Pilot identifies the required number of copies to meet distribution requirements.

(1) Each Department and Line Station receiving a manual will be responsible for its security, maintenance and currency. The person revising the manual will follow the instructions included with the revision. The Company's publication platforms will be updated as changes are issued.

(2) Request for distribution of the ATR Minimum Equipment List (MEL) Doc. No.: BW-ATR-MEL-0039 or additional copies of this manual should be made by written request through the Chief Pilot.

1.01 GOALS

The European air operations regulations (Paragraph 1.3.2 Annex II of regulation (EU) 2018/1139) requires that all equipment installed on an aircraft required for type certification or by operating rules shall be operative.

The following constitutes the master basis, approved by European air operations regulations, for Caribbean Airlines.

However, the Requirements also permit the use of a **Minimum Equipment List (MEL)** where compliance with certain equipment requirements is not necessary in the interests of safety under all operating conditions. Experience has shown that with various levels of redundancy designed into aircraft, operation of every system or installed component may not be necessary when the remaining operative equipment can provide an acceptable level of safety.

The **EASA MASTER MINIMUM EQUIPMENT LIST (MMEL)**, is supplied by the manufacturer ATR (the type certificate holder) TO IMPROVE AIRCRAFT UTILIZATION AND THEREBY PROVIDE MORE CONVENIENT AND ECONOMIC AIR TRANSPORTATION FOR THE PUBLIC. THE EASA MMEL INCLUDES THOSE ITEMS OF EQUIPMENT RELATED TO AIRWORTHINESS AND OPERATING REQUIREMENTS AND OTHER ITEMS OF EQUIPMENT WHICH THE EASA FINDS MAY BE INOPERATIVE AND YET **MAINTAIN AN ACCEPTABLE LEVEL OF SAFETY** BY APPROPRIATE CONDITIONS AND LIMITATIONS.

The MMEL is the basis for development of individual operator's MEL, which take into consideration the operator's particular aircraft equipment configuration and operational conditions.

An OPERATOR'S MEL may differ in format form the MMEL, **BUT CANNOT BE LESS RESTRICTIVE THAN THE MMEL**. The individual operator's MEL, **WHEN APPROVED**, permits operation of the aircraft with inoperative equipment.

The MMEL MUST NOT BE USED AS AN OPERATOR'S MEL because it does not take into account:

- Operator-specific requirements, definitions, operations or regulations
- Specific flight operations
- Equipment added or modified by the operator after manufacturer delivery (e.g. Supplemental Type Certificate)

1.02 PRINCIPLE

For the sake of brevity, the MEL does not include:

- Obviously required items such as wings, rudder, flaps engines, landing gear…
- Items which do not affect the airworthiness of the aircraft such as galley equipment, entertainment systems…
- Passenger convenience items…

HOWEVER, IT IS IMPORTANT TO REMEMBER THAT ALL EQUIPMENT RELATED TO THE AIRWORTHINESS AND THE OPERATING REQUIREMENTS OF THE AIRCRAFT NOT LISTED IN THE MMEL MUST BE OPERATIVE FOR EACH FLIGHT.

NOTE: 1) Refer to AFM for dispatch without certain structural components (Refer to AFM - DEVIATION GUIDE CDL).

2) Refer to AFM for dispatch with several systems inoperative (Refer to AFM - DEVIATION GUIDE SUPPLEMENTS).

Suitable conditions and limitations in the form of placards, maintenance procedures, crew operating procedures and other restrictions as necessary are specified in the MEL to ensure that an acceptable level of safety is maintained.

The MEL is intended to permit operation with inoperative items of equipment for a period of time until repairs can be accomplished. It is important that repairs be accomplished at the earliest opportunity. In order to maintain an acceptable level of safety and reliability the MMEL establishes limitations on the duration of and conditions for operation with inoperative equipment. The MEL provides for release of the aircraft for flight with inoperative equipment. Caribbean Airlines will establish a controlled and sound rectification program including the parts, personnel, facilities, procedures and schedules to ensure timely rectification.

When an item of equipment is discovered to be inoperative, it is reported by making an entry in the Aircraft Maintenance Record/Logbook as prescribed by European air operations regulations. The item is then either rectified or may be deferred per the MEL or other approval means acceptable to the TTCAA prior to further operation. MEL conditions and limitations do not relieve Caribbean Airlines from determining that the aircraft is in condition for safe operation with items of equipment inoperative. The flight planning shall take into account the inoperative equipment/function and the associated operational or performance penalties for the flight under MEL. An operational flight plan shall be completed for each intended flight based on considerations of aircraft performance, other operating limitations and relevant expected conditions on the route (including diversion route(s)) to be followed and at the aerodromes/ operating sites concerned.

Caribbean Airlines is responsible for exercising the necessary operational control to ensure that an acceptable level of safety is maintained and to evaluate the new operational capability of the aircraft.

The exposure to additional failures during continued operation with inoperative systems or components must also be considered. Wherever possible account has been taken in this MEL of multiple inoperative items. However, it is unlikely that all possible combinations of this nature have been accounted for. Therefore, when operating with multiple inoperative items, the inter-relationships between those items and the effect on aircraft operation and crew workload must be considered.

WHEN USING THE MEL, COMPLIANCE WITH THE STATED INTENT OF THE PREAMBLE, DEFINTIONS AND THE CONDITIONS AND LIMITATIONS SPECIFIED IN THE MEL IS REQUIRED.

2.1 MMEL REVISION

ATR provides a MMEL issue and indicates the aircraft configuration at aircraft delivery. ATR regularly issues subsequent revisions to the MMEL that indicates the configuration changes due to embodied modifications approved by ATR.

The List of Issues (LI) is a preliminary page on all publications and provides the following data:

- Issue dates - Date of publication in format MMM YY
- Reasons for issues - Approval information with the revision number in two-digit format.

For each new revision issue, Reason For Update (RFU) is a short description of the reason and/or type of changes for the related Data Modules (DMs). The issue date above the table corresponds to the last revision issue.

The List of Effective Data Module (LEDM) provides the container code, revision date, and aircraft applicability. For each DM at least one of the following statuses is applicable:

- Blank/Empty - No change in the DM, or
- **R** - Revised DM, or
- **N** - New DM, or
- **D** - Deleted DM, or
- **A** - Change of applicability only, or
- **R-A** - Revised DM and change of applicability, or
- **MT** - Moved To, or
- **MF** - Moved From.

The List Of Modifications (LOM) provides the aircraft configuration and applicable modifications with their number, title, and individual effectivity per Manufacturer Serial Number (MSN) or Registration Number. In the case of new modifications or changes in applicability, revision bars appear.

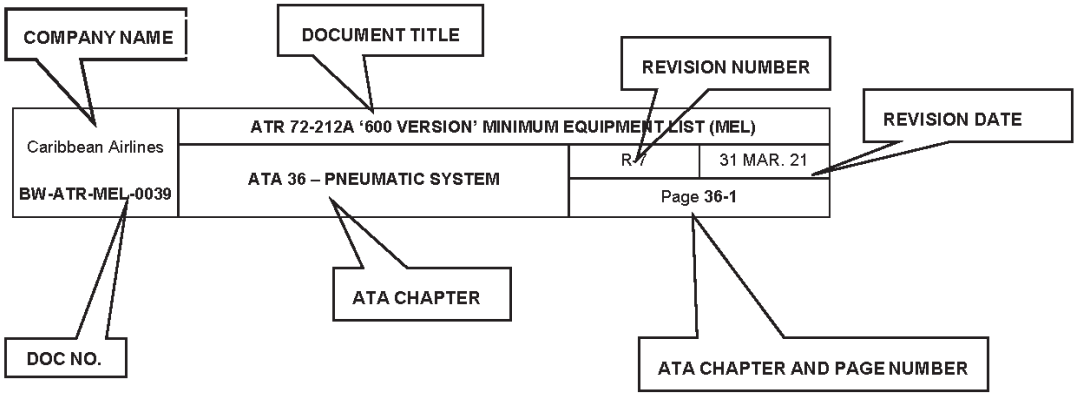
The Cross Reference Table (CRT) provides the list of applicable aircraft with the corresponding Fleet Serial Number (FSN), MSN, registration number, and model. In the case of new MSN or changes in FSN, Registration Number, or model, revision bars appear.

2.2 PRESENTATION OF THE MEL

The MEL is divided into Sections:

Section 0:	This Section includes general break down of the MEL.
PREAMBLE:	This Section includes general information and gives indications about how to read and understand the MEL.
ALERT MESSAGE LIST:	This Section is dedicated to "600 version" only,
MEL OPERATIONAL & MAINTENANCE PROCEDURES:	This Section contains the MEL items and their Operational and Maintenance Procedures classified by ATA Chapter.

Each content is organized with information related to the update revision.



2.3 HOW TO READ THE MEL OPERATIONAL & MAINTENANCE PROCEDURES PARTS

This section identified the systems per ATA classification with its associated MEL items and operational procedures (if required).

Examples:

Caribbean Airlines	OM : MEL ATR 72-212A MINIMUM EQUIPMENT LIST	ATR-600	
	SECTION 0 PREAMBLE	R-4	31 MAR. 20
		Page **PRE.0.9-1**	

① **22-36-02 AUTOPILOT DISCONNECT AURAL ALERT**

② **22-36-02A (A)**

③ RI	④ NI	⑤ NR	⑥ PLACARD	⑦ O	M
B	1	0	YES	NO	NO

May be inoperative provided that autopilot function is not used.

ITEM

- OR -

22-36-02B (B)

RI	NI	NR	PLACARD	O	M
B	1	0	YES	YES	YES

May be inoperative provided that both AP OFF lights are checked operative prior to each departure.

OPS PROC **OPERATIONAL PROCEDURES** ⑨

■ **On Ground, aircraft powered (EXT PWR or Hotel Mode)**

► AVIONICS VENT EXHAUST MODE..... NORM
► STICK PUSHER pb..... OFF
► AIR BLEEDS..... OFF

MP PROC **MAINTENANCE PROCEDURES**

MP ATR-A-36-11-60-00ZZZ-560Z-A.

21-10-01 FLOW SELECTOR

RI	NI	NR	PLACARD	O	M
C	1	0	YES	NO	NO

May be inoperative provided that one mode (NORM or HIGH) is operative and used.

⑧ ***NOTE:*** *Some smoke procedures request to select air flow HIGH. Nevertheless associated procedures efficiency has been demonstrated with air flow NORM.*

1) Item Number and Title

Identifies the system per ATA and its item, it indicates the equipment, systems, components for which dispatch conditions apply. An **"if installed"** annotation indicates that the listed item is not applicable to all models or configurations. It does not imply that the aircraft may be operated, in accordance with the MEL, with the item removed, unless clearly indicated in the associated Dispatch conditions.

2) Dispatch Condition

Gives remarks or conditions linked to a MEL dispatch. It indicates the suitable conditions and limitations in the form of placard, maintenance procedures, operational procedures and dispatch conditions as necessary to ensure that an acceptable level of safety is maintained.

Each dispatch condition has one or several requirements, called provisos. Provisos are alphabetically numbered a), b), c)...

CAL must comply with all the provisos of the selected dispatch condition.

Some MEL items may have several dispatch conditions. Each dispatch condition offers a different option to dispatch the aircraft. A reference identifies each dispatch condition.

EXAMPLE	**Dispatch condition reference 22-36-02B:** **- 22-36-02 corresponds to the item number, and** **- The letter "B" identifies the dispatch condition within the item 22-36-02.**

When a MEL item has several dispatch conditions, CAL must select and apply only one dispatch condition at a time.

An optional operational title may be provided in the MEL for some of the dispatch condition references. The aim of an operational title is to help CAL to better understand the operational differences between multiple dispatch conditions. In the frame of the MEL customization, this operational title may be modified at CAL's convenience provided that the customized operational title still ensure the correct identification, the content and the intent of the associated dispatch condition.

When necessary, CAL may successively apply other dispatch conditions of the same item provided that CAL complies with the MEL requirements and provided that the starting point of the repair interval countdown is not modified.

Any reference to another MEL item implies that the dispatch conditions of the referenced item are applicable in addition to those of the current item. Where multiple repair intervals apply, the most restrictive repair interval shall be used.

3) RI "Rectification Interval" Column

For each applicable item, the Rectification Interval column indicates the rectification interval category.

Each category is defined according to the following definitions:

CATEGROY A

No standard interval is specified, however, items in this category shall be rectified in accordance with the remark stated in the MEL.

Where a time period is specified in calendar days it shall start at 00:01 on the calendar day following the day of discovery. Where a time period is specified in number of flight or flight hours, it shall start at the beginning of the next flight that follows the discovery.

CATEGORY B

Items in this category shall be rectified within three (3) consecutive calendar days, excluding the day of discovery.

For example: if it was recorded at 10 am on January 26th, the three-day interval begins at midnight on January 26th, and ends at midnight the January 29th.

CATEGORY C

Items in this category shall be rectified within ten (10) consecutive calendar days (240 hours), excluding the day of discovery.

CATEGORY D

Items in this category shall be rectified within one hundred and twenty (120) consecutive calendar day but may only be deferred for a maximum of thirty (30) days. The item may then be deferred for a further thirty (30) days up to a maximum of one hundred and twenty (120) days.

DASH SYMBOL "-"

A dash symbol may appear to redirect to another referenced item which provides the rectification interval category.

NOTE: A flight should not be commenced if the scheduled arrival time is after the expiry of the rectification interval.

Rectification Interval Extension Policy

CAL may use a procedure for the extension of the applicable rectification intervals B, C, D for the same duration as specified in the MEL and according to the TTCAA provided:

- A description of specific duties and responsibilities for controlling extensions is established by CAL and accepted by the TTCAA, and
- CAL only grants a one-time extension of the applicable rectification interval, and

3) RI "Rectification Interval" Column (Cont'd)

The TTCAA is notified of any extension granted within a timescale acceptable to the TTCAA, and

Rectification is accomplished at the earliest opportunity.

When accepted by the TTCAA according to the conditions above, the application of the rectification interval policy is not a violation for the MEL to be not less restrictive than the MMEL.

4) NI "Number Installed" Column

Shows the number of installed equipment, components, systems or functions, it matches with the type-certificated configuration of the aircraft.

A Dash "-" symbol indicates a variable quantity installed on the aircraft.

5) NR "Number Required" Column

Specifies the minimum quantity of equipment, components, systems or functions which must be operative for a MEL dispatch, under the conditions listed in the dispatch conditions.

A Dash "-" symbol indicates a variable quantity, minimal required for dispatch of the aircraft.

6) Placard Column (YES / NO)

- YES: Each inoperative item must be placarded, as applicable, to inform and remind the crew members and maintenance personnel of the item's condition.

  Note: ***To the extent practical, placards should be located*** *adjacent to the control or indicator for the item affected. However, unless otherwise specified, placard wording and location will be determined by the CAL.*

- NO: No Placard is required.

7) Operational and Maintenance Procedures

An inoperative equipment may refer to an (O) Operational and/or Maintenance procedure (M) to ensure an acceptable level of safety.

It is CAL responsibility to ensure that all the (O) and/or (M) refer to an appropriate operational and/or maintenance procedure, or to a procedure established by CAL and accepted by the TTCAA. Then CAL must ensure that these procedures are appropriately applied.

Both (O) and (M) identifiers used singularly or in conjunction with each other, require that appropriate procedures be established, published and complied with if flight is accomplished with the item inoperative.

7) Operational and Maintenance Procedures (Cont'd)

a) "O" Column

The "YES" in Operational Column "O" identifies an Operational procedure when operating with the associated inoperative item. Operational procedures may require crew actions, limitations, performance penalties or crew awareness information to be taken into account.

These procedures are normally carried out by qualified flight or cabin crew, but may be accomplished by other qualified, approved personnel. However, the flight crew must always be informed of the applicable operational procedure prior to each flight under associated MEL item.

When flight situation occurs, the operational procedures must always be applied under the associated item (unless its periodicity of application is clearly specified in remarks and conditions).

The operational procedure is described just after the associated Dispatch Condition of the MEL item and is identified with the same number and title.

b) "M" Column

The "YES" in Maintenance Column (M) identifies a maintenance procedure to be performed. These procedures are normally carried out by maintenance personnel, but some elementary maintenance tasks may be accomplished by crew members or other qualified, approved personnel. Flight crew can accomplish some of these elementary (M) procedures, but it remains the responsibility of CAL and the TTCAA approval.

However, it is recommended that maintenance personnel accomplish procedures that require specialized knowledge, skills, or the use of tools and test equipment. The maintenance procedures should normally be performed once prior to the first flight under the associated item and, if applicable, must be repeated at the interval specified in the Remarks.

8) NOTES

There are three levels of notes:

- Notes at the level of the item: these notes appear below the item title and before the dispatch condition of this item. These notes apply to all the dispatch conditions of the item.
- Notes at the level of the dispatch condition: these notes appear at the end of the dispatch condition. These notes apply only to this dispatch condition.
- Notes at the level of the requirement: these notes appear within a proviso. These notes apply only to this requirement.

9) References

References are introduced to assist CAL in complying with MEL requirements. However, as indicated in the PREAMBLE, **CAL is still responsible for determining the applicable interrelationships and associated requirements.**

References are used in case of:

- Another source of information must be reviewed and associated restrictions, and/or procedures must be applied (e.g. "Refer to AFM…").
- Another ATA Chapter of the MEL must be reviewed, in order to determine the applicable item. The rectification interval category of the most restrictive must be applied.

2.4 HOW TO READ THE MEL MAINTENANCE PROCEDURES PART

This section includes the maintenance procedures associated to the corresponding MEL item.

Below, is an example with indications about how to read and understand the MEL Maintenance Part.

Caribbean Airlines	OM : MEL ATR 72-212A MINIMUM EQUIPMENT LIST	ATR-600	
	SECTION 0 – RECORD OF REVISIONS	R-4	31 MAR. 20
		Page **ROR 0.2-1**	

21-23-01 **OVERBOARD VALVE**

21-23-01A **Automatic Mode (A)**

RI	NI	NR	PLACARD	O	M
C	1	0	YES	YES	NO

Automatic mode may be inoperative provided that:
(a) Manual mode is checked operative before each departure, and
(b) Flight is conducted in order to maintain △P ≤ 1 PSI.

ITEM - OR -

21-23-01B **(B)**

RI	NI	NR	PLACARD	O	M
C	1	0	YES	NO	YES

Link to Maintenance Proc

May be inoperative provided that:
(a) It is deactivated, and
(b) Extended overwater flight is prohibited, and
(c) Flight level is limited to FL 170, and
(d) FLOW selector is considered inoperative, Refer to item 21-10-01 Flow Selector

Refer to item 21-10-01 Flow Selector
Maintenance procedure title

MAINT PROC

MAINTENANCE PROCEDURES

► Overboard Ventilation Valve De-energization

Refer to MP ATR-A-21-23-20-00ZZZ-560Z-A, setting the valve open in intermediate position.

NOTE

► *Avoid leaving avionics selected ON for a long period of time.*

2.5 DEFINITIONS

This list is not exhaustive. Additional definitions could be added by CAL in the MEL.

As required by regulation	Means that the listed item of equipment is subject to provisos (restrictive or permissive) expressed in the applicable legislation (e.g. regulation Air operations, Single European Sky legislation or the applicable airspace requirements). When the equipment is not required, it may be inoperative for the time specified by its rectification interval category.
Calendar Days	Some MEL items are assigned a time interval in "days" in the remark. "Days" must be considered as "calendar days", not including the calendar day that the failure was recorded.
Cargo Configuration	Refer to cargo operations when aircraft is operated with an ATR cargo installation. In case of STC (Supplemental Type Certificate) installation, CAL shall adapt its MEL with STC supplement.
Considered Inoperative	Must be treated as "inoperative". Therefore the "considered inoperative" MEL item must also be recorded in the Technical log and the associated dispatch exceptions must be applied, in addition to any applicable (O) and (M) procedure. The shorter rectification interval, between the initial inoperative item and the "considered inoperative" item, should be applied.
Day of Discovery	The calendar day that the failure has been recorded in the aircraft maintenance record/logbook.
ETOPS	Refers to operation according to "Extended Range operations of twin engine airplane" requirements. ETOPS restrictions in this MEL are applicable to ETOPS certified aircraft; Refer to *AFM*.

2.5 DEFINITIONS (Cont'd)

Extended Overwater Flight	A flight operated over water at a distance of more than 93 km (50 Nm), or 30 min at normal cruising speed, whichever is the lesser, away from land suitable for making an emergency landing, *Refer to ICAO Annex 6 parts II.*
Ferry Flight	Non-revenue flight with necessary flight crew member to conduct a ferry flight.
Flight	Is defined as the period of time that begins the moment at which an aircraft begins to move by its own means in preparation for takeoff, continues during and the applicable flight phases, and ends when the aircraft lands and comes to a complete stop in its parking area.
Flight Day	A 24-hour period from midnight to midnight based on either UTC or local time, during which at least one flight is initiated for the affected aircraft.
Icing Conditions	Visible moisture in any form is present, and OAT on ground and for takeoff is at or below 5° C or when TAT in flight is at or below 7° C. *Refer to AFM-LIM-04-04 ICING CONDITIONS.*
Inflammable or Combustible Material	Material that can catch fire and burn. Where loading of inflammable or combustible material is prohibited, no material may be loaded except the following: cargo handling equipment, flyaway kits (excluding e.g. cans of hydraulic fluid, cleaning solvents, batteries, chemical generators, etc.) and inflight service material (return catering-only closed catering trolleys/boxes, no newspaper, no alcohol or duty free goods).
Inoperative	Any time a system and/or component malfunctions to the extent that it does not accomplish its intended purpose and/or is not consistently functioning within its designed operating limit(s) or tolerance(s).
VMC	Atmospheric environment allows the flight to proceed under the Visual Meteorological Conditions applicable to the flight.

2.6 CAT 2 AUTOMATIC APPROACH

The required equipment are listed in the AFM.

2.7 GLOSSARY OF STANDARD NOMENCLATURE

A

AAS .....Anti-icing Advisory System
ABN ..... Abnormal
ABV ..... Above
A/C ..... Aircraft
AC .....Alternating Current
ACARS..... Aircraft Communication Addressing and Reporting System
ACAS.....Airborne Collision Avoidance System
AC BTC ..... AC Bus Tie Contactor
AC BTR ..... AC Bus Tie Relay
ACCU.....Accumulator
AC EBTC .....AC Emer Bus Transfer Contactor
ACM.....Air Cycle Machine
ACN.....Aircraft Classification Number
ACP ..... Audio Control Panel
ACR.....Aircraft Classification Rating
ACS ..... Air Conditioning Systems
ACW ..... Alternating Current Wild Frequency
ADC ..... Air Data Computer
ADF..... Automatic Direction Finding
ADI ..... Attitude Director Indicator
ADN .....Avionic Data Network
ADS .....Air Data System
ADS-B..... Automatic Dependent Surveillance-Broadcast
ADU ..... Advisory Display Unit
AFCS ..... Automatic Flight Control System
AFDX ..... Avionics Full Duplex
A/FEATH..... Auto Feathering
AFFF.....Aqueous Film Forming Foam
AFM ..... Airplane Flight Manual
AFR.....Airframe
AFU.....Auto Feather Unit
AGB ..... Accessory Gear Box
AGL ..... Above Ground Level
AH .....Ampere – Hours
AHRS..... Attitude and Heading Reference System
AHRU.....Attitude and Heading Reference Unit
AIDS ..... Aircraft Integrated Data System
AIL ..... Aileron
ALD..... Actual Landing Distance

2.7 GLOSSARY OF STANDARD NOMENCLATURE (Cont'd)

A

ALT ..... Attitude
ALTM ..... Altimeter
ALTN ..... Alternate
ANN ..... Annunciator
AOA ..... Angle of Attack
AP ..... Autopilot
APC ..... Active Phase Control
APM ..... Aircraft Performance Monitoring
APP ..... Approach
APS ..... Ambient Pressure Sensor
ARINC ..... Aeronautical Radio Incorporated
ARM ..... Armed
A/S ..... Antiskid
ASAP ..... As Soon As Possible
ASCB ..... Avionics Standard Communication Bus
ASD ..... Accelerate Stop Distance
ASI ..... Air Speed Indicator
ASTR ..... AC Stand by Bus Transfer Relay
ASYM ..... Asymmetry
ATC ..... Air Traffic Control
ATE ..... Automatic Test Equipment
ATPCS ..... Automatic Takeoff Power Control System
ATT ..... Attitude
ATTND ..... Attendant
AUTO ..... Automatic
AUX ..... Auxiliary
AVAIL ..... Available
AVS ..... AVionic Suite
AZ ..... Azimuth
AZWF ..... Actual Zero Fuel Weight

B

BARO ..... Barometric
BAT ..... Battery
BC ..... Black Course
BCP ..... Best Computed Position
BITE ..... Built in Test Equipment
BLW ..... Below
BPCU ..... Bus Power Control Unit
BPU ..... Battery Protection Unit

2.7 GLOSSARY OF STANDARD NOMENCLATURE (Cont'd)

B

BRG ..... Bearing
BRK ..... Brake
BSPS ..... Bleed Switching Pressure Sensor
BSC ..... Battery Start Contactor
BTC ..... Bus Tie Contactor
BTR ..... Bus Tie Relay
BXR ..... Battery Transfer Relay

C

CAB ..... Cabin
CAC ..... Crew Alerting Computer
CAC ..... Core Avionic Cabinet
CAN ..... Controller Area Network
CAP ..... Crew Alerting Panel
CAPT ..... Captain
CAS ..... Calibrated Airspeed
CAT ..... Category
C/B ..... Circuit Breaker
CBV ..... Cross Bleed Valve
CCAS ..... Centralized Crew Alerting System
CCW ..... Counter Clockwise
CD ..... Coefficient of Drag
CDCU ..... Cockpit Door Control Unit
CDI ..... Course Deviation Indicator
CDL ..... Configuration Deviation List
CDLS ..... Cockpit Door Locking System
CDS ..... Cockpit Display System
CDTS ..... Compressor Discharge Temperature Sensor
CDU ..... Cabin Display Unit
CFC ..... Constant Frequency Contactor
CFR ..... Code of Federal Regulations
CG ..... Center of Gravity
CHAN ..... Channel
CHC ..... Charge Contactor
CHG ..... Charge
CKPT ..... Cockpit
C/L ..... Checklist
CL ..... Condition Lever or Coefficient of Lift
CLA ..... Condition Lever Angle
CLB ..... Climb
CLR ..... Clear
CM ..... Crew Member

2.7 GLOSSARY OF STANDARD NOMENCLATURE (Cont'd)

C

CNTNR ..... Container
CNTR ..... Contactor
CNTRS ..... Contrast
COM ..... Communication
COMPT ..... Compartment
COMPTR ..... Computer
CONFIG ..... Configuration
CONT ..... Continuous
CORRECT ..... Correction
CPA ..... Closest Point of Approach
CPCS ..... Cabin Pressure Control System
CPL ..... Autopilot Coupling
CPM ..... Core Processing Module
CRC ..... Continuous Repetitive Chime
CRG ..... Cargo
CRS ..... Course
CRT ..... Cathode Ray Tube
CRZ ..... Cruise
CTL ..... Control
CVR ..... Cockpit Voice Recorder
CW ..... Clockwise

D

DADC ..... Digital Air Data Computer
DADS ..... Digital Air Data System
DC ..... Direct Current
DCA ..... Data Concentration Application
DEC ..... Decrease
DELTA P ..... Differential Pressure
DEV ..... Deviation
DFDR ..... Digital Flight Data Recorder
DGAC ..... Direction Générale de l'Aviation Civile
DGR ..... Degraded
DH ..... Decision Height
DIFF ..... Differential
DIM ..... Light Dimmer
DISC ..... Disconnect
DISCH ..... Discharge
DIST ..... Distance
DLK ..... Data Link

2.7 GLOSSARY OF STANDARD NOMENCLATURE (Cont'd)

D

DM ..... Data Module
DME ..... Distance Measuring Equipment
DN ..... Down
DSPL ..... Display
DU ..... Display Unit

E

EADI ..... Electronic Attitude Director Indicator
EASA ..... European Aviation Safety Agency
EBCC ..... Emergency Battery Charge Contactor
EBTC ..... Emer Bus Transfer Contactor
ECL ..... Electronic Checklist
ECS ..... Environmental Control System
ECU ..... Electronic Control Unit
E/E ..... Electrical/Electronic
EEC ..... Electronic Engine Control
EFB ..... Electronic Flight Bag
EFCP ..... EFIS Control Panel
EFIS ..... Electronic Flight Instrument System
EFOB ..... Estimated Fuel on Board
EFVS ..... Enhanced Flight Vision System
EGHR ..... External Ground Handling Relay
EGPWS ..... Enhanced Ground Proximity Warning System
EHSI ..... Electronic Horizontal Situation Indicator
EHV ..... Electro Hydraulic Valve
ELEC ..... Electrical
ELEV ..... Elevator
ELV ..... Elevation
EMER ..... Emergency
ENG ..... Engine
EPC ..... External Power Contactor
EPE ..... Estimated Position Error
EQPT ..... Equipment
ERACKV ..... Emergency Ram Air Check Valve
ERAV ..... Emergency Ram Air Valve
ERIU ..... Engine Rating Interface Unit
ERP ..... Eye Reference Point
ESS ..... Essential

2.7 GLOSSARY OF STANDARD NOMENCLATURE (Cont'd)

E

ET ..... Elapsed Time
ETA ..... Estimated Time of Arrival
ETE ..... Estimated Time En route
ETOPS .....Extended-Range Twin Engine Operations Performance Standard
EVS ..... Enhance Vision Sensor
EWD ..... Engine and Warning Display
EXC ..... External Power/Service Bus Contactor
EXH ..... Exhaust
EXT ..... External

F

FAA..... Federal Aviation Administration
FAF ..... Final Approach Fix
FAIL ..... Failed, Failure
FAR..... Federal Aviation Regulations
FCOC ..... Fuel Cooled Oil Cooler
FCOM ..... Flight Crew Operating Manual
FCU ..... Fuel Control Unit
FCV ..... Flow Control Valve
FD ..... Flight Director
FDA ..... Flight Data Application
FDAU ..... Flight Data Acquisition Unit
FDE..... Fault Detection and Exclusion
FDEP ..... Flight Data Entry Panel
FEATH..... Feathered
FF ..... Fuel Flow
FGCP ..... Flight Guidance Control Panel
FGS ..... Flight Guidance System
FI ..... Flight Idle
FL ..... Flight Level
FLT..... Flight
FMA..... Flight Mode Annunciators
FMS..... Flight Management System
F/O ..... First Officer
FOB..... Fuel on Board
FODU ..... Feeder Overheat Detection Unit
FOS ..... Flight Operations Software
FOV ..... Field of View
FPA ..... Flight Path Angle
FPLN ..... Flight Plan
FPV ..... Flight Path Vector

2.7 GLOSSARY OF STANDARD NOMENCLATURE (Cont'd)

F

FQI ..... Fuel Quantity Indication
FSN ..... Fleet Serial Number
FSV ..... Flow Sensor Venturi
FTO ..... Final TakeOff
FTR ..... Feather
FU ..... Fuel Used
FWA ..... Flight Warning Application
FWD ..... Forward
FWS ..... Flight Warning System

G

GA ..... Go Around
GAL ..... Galley
GC ..... Generator Contactor
GCR ..... Ground Clutter Reduction
GCU ..... Generator Control Unit
GDM ..... Group of Data Module
GEN ..... Generator
GI ..... Ground Idle
GMT ..... Greenwhich Mean Time
GND ..... Ground
GNSS ..... Global Navigation Satellite System
GPS ..... Global Positioning System
GPU ..... Ground Power Unit
GPWS ..... Ground Proximity Warning System
G/S ..... Glide Slope
GS ..... Ground Speed
GSE ..... Ground Support Equipment
GW ..... Gross Weight
GXS ..... ACW Generator/Service Bus Contactor

H

HBOV ..... Handling Bleed Overboard Valve
HBV ..... Handling Bleed Valve
HD ..... Head Down
HDG ..... Heading
HDLG ..... Handling
HDOP ..... Horizontal Dilution Of Precision
HEBB ..... Hot Emergency Battery Bus

2.7 GLOSSARY OF STANDARD NOMENCLATURE (Cont'd)

H

HEBTC ..... Hot Emer Battery Transfer Contactor
HF ..... High Frequency
HFOM ..... Horizontal Figure Of Merit
HI ..... High
HIL ..... Horizontal Integrity Limit
HLD ..... Hold
HMBTC ..... Hot Main Battery Transfer Contactor
HMD ..... Head Mounted Display
HMU ..... Hydromechanical Unit
HOBV ..... Handling Overboard Valve
HP ..... High Pressure
HPV ..... High Pressure Valve
HSI ..... Horizontal Situation Indicator
HTG ..... Heating
HU ..... Head Up
HYD ..... Hydraulic

I

IAF ..... Initial Approach Fix
IAS ..... Indicated Air Speed
IASC ..... Integrated Air System Control
ICAO ..... International Civil Aviation Organisation
ICP ..... Index Control Panel
IDT ..... Ident
IEP ..... Ice Evidence Probe
IESI ..... Integrated Electronic Standby Instrument
I.F ..... Intermediate Frequency
IFR ..... Instrument Flight Rules
IGN ..... Ignition
ILS ..... Instrument Landing System
IMC ..... Instrumental Meteorological Conditions
IMU ..... Inertial Measurement Unit
IN ..... Inertial Navigation
INC ..... Increase
IND ..... Indicator
IN/HG ..... Inches of Mercury
INHIB ..... Inhibit
INOP ..... Inoperative
INS ..... Inertial Navigation System
INST ..... Instrument

2.7 GLOSSARY OF STANDARD NOMENCLATURE (Cont'd)

I

INT ..... Interphone
INU .....Inertial Navigation Unit
INV ..... Inverter
IOM ..... Input Output Module
IPD .....Inter-Pupillary Distance
IRS .....Inertial Reference System
ISA .....International Standard Atmosphere
ISOL..... Isolation
ISPSS.....In Seat Power Supply System
ISV ..... Isolation Shut-Off Valve
ITT ..... Inter Turbine Temperature

J

JAA.....Joint Aviation Authorities
JAR..... Joint Aviation Requirements
JIC .....Job Instruction Card

L

LAT..... Lateral
LAV .....Lavatory
LBA..... Lowest Blade Angle
LDG .....Landing
LDTA ..... Landing Distance at Time of Arrival
LEDM..... List of Effective Data Module
L/G .....Landing Gear
LH .....Left Hand
LIM ..... Limitation
LMG.....Left Main Gear
LMU .....Light Managing Unit
LNAV .....Lateral Navigation
LO .....Low
LOC ..... Localizer
LOM.....List of MODs
LOS.....Line Of Sight
LP .....Low Pressure

2.7 GLOSSARY OF STANDARD NOMENCLATURE (Cont'd)

L

LRU ..... Line Replaceable Unit
LSK ..... Line Select Key
LT ..... Light
LVL ..... Level

M

MAC ..... Mean Aerodynamic Chord
MAN ..... Manual
MAP ..... Ground Mapping
MAX ..... Maximum
MBCC ..... Main Battery Charge Contactor
MBTC ..... Main Bus Transfer Contactor
MC ..... Master Caution
MCDU ..... Multi Function Control Display Unit
MCP ..... Multi Function Control Panel
MCT ..... Maxi Continuous Thrust
MDA ..... Minimum Descent Altitude
MEA ..... Minimum En route Altitude
MECH ..... Mechanic
MEL ..... Minimum Equipment List
MFC ..... Multi Function Computer
MFCU ..... Mechanical Fuel Control Unit
MFD ..... Multi Function Display
MFW ..... Minimum Flight Weight
MGT ..... Management
MIC ..... Microphone
MIN ..... Minimum
MIP ..... Main Instruments Panel
MISC ..... Miscellaneous
MIXTS ..... Mix Temperature Sensor
MKR ..... Marker
MLG ..... Main Landing Gear
MLS ..... Microwave Landing System
MLW ..... Maximum Landing Weight
MMEL ..... Master Minimum Equipment List
MMO ..... Maximum Operating Mach
MMPU ..... Multi Media Player Unit

2.7 GLOSSARY OF STANDARD NOMENCLATURE (Cont'd)

M

MOD ..... Modification
MP ..... Maintenance Procedure
MPC ..... Multi Purpose Computer
MPU ..... Micro Processor Unit
MRT ..... Manual Release Tool
MRW ..... Maximum Ramp Weight
MSA ..... Minimum Safe Altitude/Minimum Sector Altitude
MSG ..... Messages
MSN ..... Manufacture Serial Number
MSIS ..... Minimum Severe Icing Speed
MSN ..... Manufacturer Serial Number
MTOW ..... Maximum TakeOff Weight
MTW ..... Maximum Taxi Weight
MW ..... Master Warning
MZFW ..... Maximum Zero Fuel Weight

N

NAC ..... Nacelle
NACA ..... National Advisory Committee for Aeronautics
NAMS ..... New Air Management System
NAV ..... Navigation
NAVAID ..... Navigation Aid
ND ..... Navigation Display
NDB ..... Non Directional Beacon or Navigation Database (GPS)
NEG ..... Negative
NG ..... Nose Gear
NH ..... High Pressure Spool Rotation Speed
NIL ..... Nothing, No Object
NL ..... Low Pressure Spool Rotation Speed
NLG ..... Nose Landing Gear
NM ..... Nautical Mile
NORM ..... Normal
NOTAM ..... Notice To Airmen
NP ..... Propeller Rotation Speed
NPU ..... Navigation Processor Unit
NUC ..... Non Uniformity Correction
NVM ..... Non Volatile Memory
N/W ..... Nose Wheel
NWS ..... Nose Wheel Steering

2.7 GLOSSARY OF STANDARD NOMENCLATURE (Cont'd)

O

OAT ..... Outside Air Temperature
OBS ..... Omni Bearing Selector
OSD ..... Operational Suitability Data
OT ..... Other Traffic
OTF ..... Optical Tracker Fixed Unit
OUTB ..... Outboard
OVBD ..... Overboard
OVERTEMP ..... Overtemperature
OVHD ..... Overhead
OVHP ..... Overhead Panel
OVHT ..... Overheat
OVLY ..... Overlay
OVRD ..... Override
OVSPD ..... Overspeed
OXY ..... Oxygen

P

PA ..... Passenger Address
PAX ..... Passenger
PBE ..... Protective Breathing Equipment
PBN ..... Performance Based Navigation
PCN ..... Pavement Classification Number
PCR ..... Pavement Classification Rating
PCU ..... Propeller Control Unit
PDPS ..... Pack Discharge Pressure Sensor
PDTS ..... Pack Discharge Temperature Sensor
PEBMV ..... Parking/Emergency Braking Metering Valve
PEC ..... Propeller Electronic Control
PED ..... Portable Electronic Device
PF ..... Pilot Flying
PFC ..... Porous Friction Course
PFD ..... Primary Flight Display
PFTS ..... Power Feeder Thermal Sensor
PIPS ..... Pressure Inlet Pressure Sensor
PITS ..... Pack Inlet Temperature Sensor
PIU ..... Propeller Interface Unit
PL ..... Power Lever
PLA ..... Power Lever Angle
PM ..... Pilot Monitoring
PNL ..... Panel
POS ..... Position

2.7 GLOSSARY OF STANDARD NOMENCLATURE (Cont'd)

P

PR ..... Pressure
PRESS ..... Pressurization
PRIM ..... Primary
PRKG ..... Parking
PROC ..... Procedure
PROP ..... Propeller
PROX SW ..... Proximity Switch
PRSOV ..... Pressure Regulating & Shut Off Valve
PRV ..... Pressure Regulating Valve
PSEU ..... Proximity Switch Electronic Unit
PSI ..... Pound per Square Inch
PSU ..... Passenger Service Unit
PSV ..... Propeller Servo Valve
PT ..... Point
PT ..... Proximity Traffic TCAS
PTB ..... Power Trim Box
PTS ..... Pack Temperature Sensor Channels
PTT ..... Push To Talk, Push To Test
PTW ..... Pitch Thumb Wheel
PVCU ..... Power and Video Control Unit
PVM ..... Propeller Valve Module
PWM ..... Pulse Width Modulation
PWR ..... Power

Q

QAR ..... Quick Access Recorder
QRC ..... Quick Release Control
QRH ..... Quick Reference Handbook
QRT ..... Quick Reference Table
QT ..... Quart
QTY ..... Quantity

2.7 GLOSSARY OF STANDARD NOMENCLATURE (Cont'd)

R

RA ..... Resolution Advisory TCAS or Radio Altimeter
RAD/ALT ..... Radio Altitude
RAD/INT ..... Radio/Interphone
RAIM ..... Receiver Autonomous Integrity Monitoring
RARV ..... Ram Air Regulating Valve
RCAU ..... Remote Control Audio Unit
RCDR ..... Recorder
RCF ..... Resident Compact Flash
RCL ..... Recall
RCP ..... Refuel Control Panel
RCT ..... Rain Echo Attenuation Compensation Technique
RCU ..... Releasable Centering Unit
RECIRC ..... Recirculation
REV ..... Reverse
RGA ..... Reverse Go-Around
RGA ..... Reduction Gear Box
RH ..... Right Hand
RLY ..... Relay
RMA ..... Radio Management Application
RMG ..... Right Main Gear
RMI ..... Radio Magnetic Indicator
RMS ..... Radio Management System
RNAV ..... Area Navigation
RNP ..... Required Navigation Performance
RPM ..... Revolution Per Minute
RQD ..... Required
RQRD ..... Required
RTO ..... Reserve TakeOff
RTOW ..... Regulatory TakeOff Weight
RUD ..... Rudder
RVDT ..... Rotary Variable Differential Transducer
RWS ..... Reactive Windshear
RWY ..... Runway
RWYCC ..... Runway Condition Code

S

SAT ..... Static Air Temperature
SB ..... Service Bulletin
SBAS ..... Satellite Based Augmentation System

2.7 GLOSSARY OF STANDARD NOMENCLATURE (Cont'd)

S

SBTC .....Stand By Bus Transfer Contactor
SBY.....Stand By
SC .....Single Chime, Starter Contactor
SCU .....Signal Conditioning Unit
SD .....System Display
SDTC .....Static Inverter Override Transfer Contactor
SEC..... Secondary
SELCAL..... Selective Calling
SFV..... Safety Valve
SG .....Symbol Generator
SGL..... Single
SGU ..... Symbol Generator Unit
SHP.....Shaft Horse Power
SID .....Standard Instrument Departure
SMA.....System Maintenance Advisory
SMK..... Smoke
SMKG ..... Smoking
SO .....Shut-Off
S.O......Shut-Off
SOV ..... Shut Off Valve
SPD.....Speed
SPLR ..... Spoiler
SPLY..... Supply
SPS..... Single-Point Performance Software
SSR.....Secondary Surveillance Radar
STAB .....Stabilizer
STAR .....Standard Arrival
STBY ..... Standby
STR..... Service Bus Transfer Relay
STRG.....Steering
SURV.....Surveillance
SVCE ..... Service
SVS..... Synthetic Vision System
SWM ..... Switch Module
SYNPHR..... Synchrophaser
SYS..... System

2.7 GLOSSARY OF STANDARD NOMENCLATURE (Cont'd)

T

T2CAS.....Traffic and Terrain Collision Avoidance System
T3CAS..... TAWS, TCAS and Transponder Combined Unit
TA .....Traffic Advisory TCAS
TAD.....Terrain Awareness Display
TAS .....True Air Speed
TAT ..... Total Air Temperature
TAWS..... Terrain Awareness Warning System
TBD..... To be Determined
TCAS.....Traffic Alert and Collision Avoidance System
TCF .....Terrain Clearance Floor
TCS ..... Touch Control Steering
TCV ..... Temperature Control Valve
T/D .....Time Delay
TD .....Type Design
TEMP ..... Temperature
TGT ..... Target
TIC .....Turbine Inlet Control
TK .....Tank
TLU .....Travel Limiting Unit (Rudder)
TM .....Torque Motor
T.O. ..... TakeOff
TOC.....Top Of Climb
TOD.....Top Of Descent
TOR.....TakeOff Run
TOW..... TakeOff Weight
TQ ..... Torque
TRU..... Transformer Rectifier Unit
TSCU..... Torque Signal Conditioning Unit
TTG .....Time To Go
TTOL..... Taxi, TakeOff and Landing

2.7 GLOSSARY OF STANDARD NOMENCLATURE (Cont'd)

U

UBC ..... Utility Bus Contactor
U/F ..... Underfloor
UHF ..... Ultra High Frequency
UNCPL ..... Uncouple
UNDV ..... Undervoltage
UNLK ..... Unlock
USB ..... Upper Sideband/Universal Serial Bus
UTLY ..... Utility

V

V1 ..... Decision Speed
V2 ..... TakeOff Safety Speed
VCCU ..... Video Cabin Command Unit
VCP ..... Virtual Control Panel
VENT ..... Ventilation
VENTS ..... Ventilated Temperature Sensor
VERT ..... Vertical
VFOM ..... Vertical Figure of Merit
VFR ..... Visual Flight Rules
VHF ..... Very High Frequency
VLOF ..... LiffOff Speed
VMC ..... Visual Meteorological Conditions
VMO ..... Maximum Operating Speed
VNAV ..... Vertical Navigation
VOR ..... VHF OMNI Directional Range
VPRL ..... Vertical Path Reference Line
VR ..... Rotation Speed
VS ..... Vertical Speed
VSI ..... Vertical Speed Indicator
VTF ..... Vector To Final
VU ..... Visual Unit

W

WARN ..... Warning
WAT ..... Weight Altitude Temperature
WBM ..... Weight and Balance Manual

2.7 GLOSSARY OF STANDARD NOMENCLATURE (Cont'd)

W

WE ..... Water Extractor
WOW ..... Weight On Wheel
WPT ..... Waypoint
W/S ..... Windshear
WS ..... Water Sprayer
WX ..... Weather Mode
WXR ..... Weather Radar

X

X ..... Cross
XFEED ..... Cross Feed
XFR ..... Transfer
XPDR ..... Transponder

Y

YD ..... Yaw Damper

Z

ZA ..... Aircraft Altitude
Z CAB ..... Cabin Altitude
ZCTH ..... Theoretical Cabin Altitude
ZFW ..... Zero Fuel Weight
ZP ..... Pressure Altitude
ZPRL ..... Zero Pitch Reference Line
ZRA ..... Radio Altimeter Altitude

HISTORICAL DATA

The following source publication have been used for the development of the ATR 72-212A '600 VERSION' MMEL:

- ATR 72-600 Scheduled Revision DIFF - Rev19.0 Issue Date 12 JUN 2026

ORGANIZATION

This MEL comprises

- The Caribbean Airlines ATR EASA APPROVED MMEL combined with Procedures.
  The MEL is divided into two Sections as follows:
  SECTION 0 - GENERAL
  Section 1 – MEL OPERATIONAL & MAINTENANCE PROCEDURES

REVISIONS

Two methods to update the Caribbean Airlines ATR 72-212A '600 VERSION' MEL:

- Normal Revisions
- Temporary Revisions

Normal Revisions

Based on ATR MMEL issued to cover non-urgent corrections and changes.

Filing Instructions Page

All revisions will include Filing Instructions, Record of Amendment Approval, List of Effective Pages, and Manual Organization & Administration.

Revision pages will be annotated to show the date of issue and the portion of the text which has been revised, as indicated by black vertical line adjacent to the changes, or arrows pointing outwards (←) indicating that text has been deleted.

Consult the List of Effective Pages. The List of Effective pages determines the correct content of the Manual. On the List of Effective Pages, a New page is identified by the character *'n'*, a Revised page by an *'*'* and a Deleted page by the character *'d'*. All Revisions shall be numbered in sequence.

Pages may be re-dated without revision bars. This is due to repagination of contents or date change, but there will not be a corresponding revision highlight.

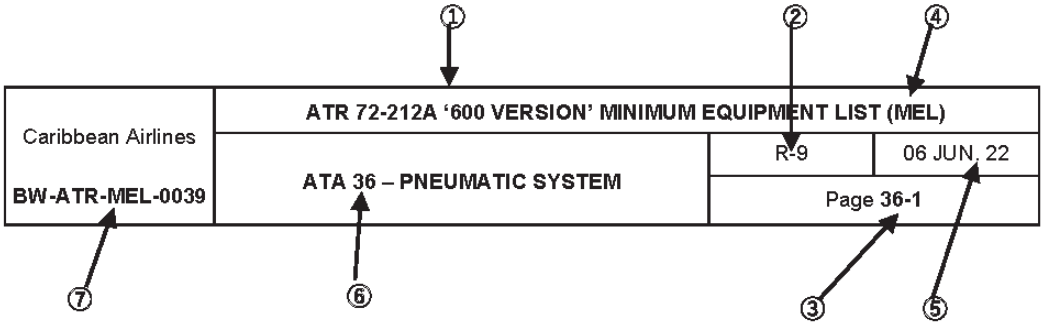
Temporary Revisions

Details of revisions which may be urgently required in the interest of flight safety, or which are supplementary to the MEL will be promulgated as Temporary Revision printed on yellow paper. They are accompanied by filing instructions, which will be filed at the front of the manual and will be incorporated into the manual at the next revision, which should not exceed a period of twelve (12) months between revisions.

ADMINSTRATION

The Fleet Manager – ATR is responsible for the administration of the Manual and any changes must be authorized by the Executive Manager – Flight Operations and / or Chief Pilot. The MEL and revisions thereto, requires TTCAA approval.

PAGINATION



① Section Title

② Revision Number

Revision numbers are synchronized with ATR MMEL revision numbers. Company initiated Temporary Revisions are indicated by an alphabet notation and (CAL) next to the current MMEL revision number e.g. 3a(CAL).

③ ATA Number (MEL ATA Items only) / Page Number

④ Aircraft Applicability

⑤ Revision Date

⑥ ATA Number (MEL/ECAM ATA Items only) Subchapter Title

⑦ CAL Document Number

DEFENTIONS OF NOTES USED IN MEL SECTION

- NOTE LEVEL 1

  Written in bold and located below a specific proviso, this NOTE is applicable only to that specific proviso associated with a given item.

INTENTIONALLY
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Modification Number	Title	Validity
00332	IGNITION – DUPLICATE THE IGNITION BOX	0968 / 0973 / 0989 / 0993 / 0997 / 1133 / 1318 / 1531 / 1534 / 1540
00659	FUEL – INSTALL FUEL TEMPERATURE INDICATOR IN THE TANKS	0968 / 0973 / 0989 / 0993 / 0997 / 1133 / 1318 / 1531 / 1534 / 1540
01830	FLIGHT CONTROLS – REPLACE ELEVATOR TRIM CONTROL SWITCH COVERS	0968 / 0973 / 0989 / 0993 / 0997 / 1133 / 1318 / 1531 / 1534 / 1540
02813	NAVIGATION - PROVIDE PROV.FOR COLLINS TCAS AND INSTALL 2ATC'S MODE S AND SEXTANT VSI/TCAS ON COLLINS RADIO NAV SYST	0968 / 0973 / 0989 / 0993 / 0997 / 1133 / 1318 / 1531 / 1534 / 1540
03262	ICE AND RAIN PROTECTION – ICE DETECTION – INSTALL AN ICING INDICATOR LIGHT	0968 / 0973 / 0989 / 0993 / 0997 / 1133 / 1318 / 1531 / 1534 / 1540
03530	FLIGHT CONTROLS – ELEVATOR INSTALL CLUTCH RECONDITIONING DEVICE (PROD. SOLUTION)	0968 / 0973 / 0989 / 0993 / 0997 / 1133 / 1318 / 1531 / 1534 / 1540
03663	GENERAL – INSTALL OVER BOARD EXHAUST FOR HBOV, SHROUD BLEED, FUEL FLOW TRANSMITTERS AND JET PUMPS	0968 / 0973 / 0989 / 0993 / 0997 / 1133 / 1318 / 1531 / 1534 / 1540
04116	ATR 42-72 – HYRDRAULIC POWER – RELOCATE GREEN PUMP CONTROL CIRCUIT BREAKER	0968 / 0973 / 0989 / 0993 / 0997 / 1133 / 1318 / 1531 / 1534 / 1540
04150	EQUIPMENT/FURNISHING – EMERGENCY EQUIPMENT – REPLACE FLIGHT COMPARTMENT ASBESTOS GLOVES	0968 / 0973 / 0989 / 0993 / 0997 / 1133 / 1318 / 1531 / 1534 / 1540

Modification Number	Title	Validity
04358	DOORS – SERVICE DOOR – REPLACE THE OPEN DEVICE RIGID MECHANISM	0968 / 0973 / 0989 / 0993 / 0997 / 1133 / 1318 / 1531 / 1534 / 1540
04371	PROPELLERS 14SF-11E PROPELLER CONTROL – INSTALL ELECTRONIC REGULATION ON ATR 72.200	0968 / 0973 / 0989 / 0993 / 0997 / 1133 / 1318 / 1531 / 1534 / 1540
04373	FLIGHT CONTROLS – AILERON AND TAB – INSTALL AILERONS FITTED WITH SPRING TABS WITH GUST LOCK	0968 / 0973 / 0989 / 0993 / 0997 / 1133 / 1318 / 1531 / 1534 / 1540
04411	OXYGEN – DISTRIBUTION – MODIFY THE SYSTEM FOR 25 % OF THE PASSENGERS (N.L.)	0968 / 0973 / 0989 / 0993 / 0997 / 1133 / 1318 / 1531 / 1534 / 1540
04457	PROPELLERS - INSTALL 568F 6 BLADE PROPELLERS (WITH ELECTRONIC REGULATION)	0968 / 0973 / 0989 / 0993 / 0997 / 1133 / 1318 / 1531 / 1534 / 1540
04523	EQUIPMENT/FURNISHINGS – MODIFY LIGHTING OF TYPE III EMERGENCY EXIT HANDLES	0968 / 0973 / 0989 / 0993 / 0997 / 1133 / 1318 / 1531 / 1534 / 1540
04539	EQUIPMENT/FURNISHINGS – FLIGHT COMPARTMENT – CHANGE FIRE STANDARD GLOVES	0968 / 0973 / 0989 / 0993 / 0997 / 1133 / 1318 / 1531 / 1534 / 1540
04686	FUEL - QUANTITY INDICATION - ADD LOW LEVEL DETECTION SYSTEM	0968 / 0973 / 0989 / 0993 / 0997 / 1133 / 1318 / 1531 / 1534 / 1540

Modification Number	Title	Validity
05018	COMMUNICATIONS - CABIN AND FLIGHT CREWCALL SYSTEM - ADD A SECOND NEW LOOK ATTENDANT HANDSET IN "EMER CALL" CONFIG	0968 / 0973 / 0989 / 0993 / 0997 / 1133 / 1318 / 1531 / 1534 / 1540
05040	LIGHTS - PAX COMPARTMENT EXTERIOR LIGHTING - REPLACE ESCAPE PATH MARKING SYSTEM	0968 / 0973 / 0989 / 0993 / 0997 / 1133 / 1318 / 1531 / 1534 / 1540
05339	FLIGHT CONTROLS - ELEVATOR SYSTEM - INSTALL TRIM SWITCH NEW SUPPORT	0968 / 0973 / 0989 / 0993 / 0997 / 1133 / 1318 / 1531 / 1534 / 1540
05390	AUDIO AND VIDEO MONITORING - INSTALL A VIDEO SURVEILLANCE SYSTEM	0968 / 0973 / 0989 / 0993 / 0997 / 1133 / 1318
05614	NAVIGATION - INSTALL MOD "S" TRANSPONDER WITH ADS/B CAPABILITY	0968 / 0973 / 0989 / 0993 / 0997 / 1133 / 1318 / 1531 / 1534 / 1540
05785	SYSTEM INTEGRATION & DISPLAY – INSTALL TWO ELECTRONIC FLIGHT BAGS IN COCKPIT	1133 / 1318
05908	ENGINE FUEL & CONTROL GENERAL - INSTALL BOOST FUNCTION	1531 / 1534 / 1540
05948	NAVIGATION- INSTALL NEW AVIONICS SUITE	0968 / 0973 / 0989 / 0993 / 0997 / 1133 / 1318 / 1531 / 1534 / 1540
05977	FUEL - INSTALL NEW FUEL GAUGING SYSTEM IN KG ON ATR72-212A	0968 / 0973 / 0989 / 0993 / 0997 / 1133 / 1318 / 1531 / 1534 / 1540

Modification Number	Title	Validity
06521	NAVIGATION - REPLACE AVIONICS SUITE SOFTWARE BY L2B2	0968 / 0973 / 0989 / 0993 / 0997 / 1133 / 1318 / 1531 / 1534 / 1540
06593	EQUIPMENT/FURNISHINGS - INSTALL GEVEN " PRESTIGE" SEATS (HYBRID BACKRESTS)	0968 / 0973 / 0989 / 0993 / 0997 / 1133 / 1318 / 1531 / 1534 / 1540
06977	INDICATING/RECORDING SYSTEMS – NAS SOFTWARE ENHANCEMENT STANDARD 2)	0968 / 0973 / 0989 / 0993 / 0997 / 1133 / 1318 / 1531 / 1534 / 1540
07144	NAVIGATION - REPLACE FGCP BY ONE WITH VNAV P/B	1318 / 1531 / 1534 / 1540
07574	EQUIPMENT/FURNISHINGS - INSTALL CABIN VIDEO SURVEILLANCE SYSTEM	1531 / 1534 / 1540
07580	COMMUNICATIONS - VHF SELCAL INST & ACTV	1531 / 1534 / 1540
10367	ELECTRICAL POWER - MODIFY BATTERY SWITCH ELECTRICAL INTEGRATION	0968 / 0973 / 0989 / 0993 / 0997 / 1133 / 1318 / 1531 / 1534 / 1540
10453	FLIGHT CONTROLS – ATR72 – REMOVE TLU COMMAND – MONITORING COMMON POINT	0989

FSN	MSN	REGISTRATION	MODEL
0051	0968	9Y-TTA	72-212A
0052	0973	9Y-TTB	72-212A
0053	0989	9Y-TTC	72-212A
0054	0993	9Y-TTD	72-212A
0055	0997	9Y-TTE	72-212A
0056	1133	9Y-TTF	72-212A
0059	1318	9Y-TTI	72-212A
0060	1531	9Y-TTG	72-212A
0061	1534	9Y-TTH	72-212A
0062	1540	9Y-TTJ	72-212A

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1. DEFINITION

Non-essential Equipment & Furnishings (NEF) are those items installed on the aircraft as part of the original type certification, supplemental type certificate, or other form of alteration that have no effect on the safe operation of flight and would not be required by the applicable certification rules of operational rules. They are those items that, if inoperative, damaged, or missing, have no effect on the aircraft's ability to be operated safely under all operational conditions.

These non-essential items may be installed in areas including, but not limited to, the passenger compartment, flight deck area, service areas, cargo areas, crew rest areas, lavatories, and galley areas. NEF items are not items already identified in the MEL or CDL of the applicable aircraft. They do not include items that are functionally required to meet the certification rule or for compliance with any operational rule. The NEF process shall not provide for deferral of items within serviceable limits identified in the manufacturer's maintenance manual or approved maintenance program such as wear limits, fuel/hydraulic leaks rates, oil consumption, etc. Cosmetic items that are fully serviceable but worn or soiled may be deferred under the NEF process.

2. NEF DEFERRAL PROCEDURE

1. A list of NEF items is provided at MEL item 25-36-1 for convenience of use. Due to the wide varieties of aircraft systems, a complete list of all NEF items is not possible. Items not included in this list but satisfy the NEF definition may also be deferred by this NEF deferral procedure. These items can be added to the NEF list when approved by the TTCAA. The NEF item evaluation will follow NEF flowchart.

   NOTE: MCC WILL REPORT THE UPDATE OF NEF LIST TO FLIGHT OPERATIONS TO UPDATE THE SAID LIST IN MEL/CDL.

2. Repair intervals of items listed in the NEF list shall be repaired:
   1) Within the repair intervals stipulated by relevant maintenance manuals, e.g., SRM, AMM, etc. or

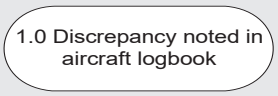
   2) The next maintenance opportunity but not exceeding deferrals of thirty (30) day intervals but not more than one hundred and twenty (120) days in total.

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3. NEF DECISION TREE DISCUSSION

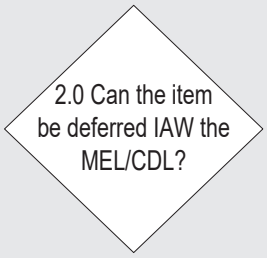
1.0 Discrepancy noted in aircraft logbook

The inoperative, damaged or missing item must be identified and documented in the aircraft logbook (or other approved location) by:



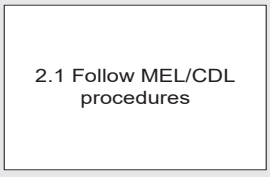
1. Flight Crew; or
2. Company maintenance personnel; or
3. Personnel authorized and approved to perform such functions as outlined in the maintenance program.

2.0 Can the item be deferred in accordance with (IAW) the MEL, CDL, or the MEL?



1. If the inoperative, damaged, or missing item is listed in the MEL or CDL, then the deferral procedures for that item must be followed. If the item is a subcomponent of a primary system identified in the MEL/CDL, where no previous relief was authorized, the subcomponent **may not be deferred** in accordance with the NEF procedures outlined in Chapter 25 of the MEL.

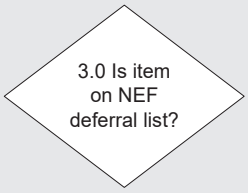
2.1 Follow MEL/CDL Procedures



1. If the item is identified in another part of the MEL/CDL, then the procedures approved for the deferral of such item must be followed.

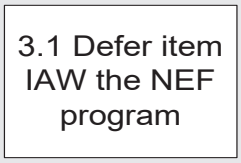
3. NEF DECISION TREE DISCUSSION (Cont'd)

3.0 Is item on the NEF Deferral List?



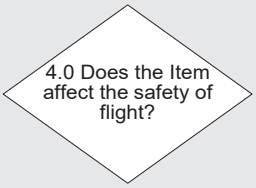
1. Is the item on the NEF list? If yes, then follow the NEF deferral procedures in step 3.1. (Items that are not previously on the NEF list should proceed to step 4.0.)

3.1 Defer item IAW the NEF Deferral Program



1. If the item is identified in the NEF deferral list, then the procedures approved for the deferral of such item shall be followed.

4.0 Does the item affect the safety of flight?



1. Is it obvious from a maintenance or operational perspective that the item, in and of itself, could have an adverse effect on the safe conduct of flight? If there is an obvious safety-of-flight issue, then the inoperative, damaged, or missing item may not be deferred and step 4.1 shall be followed.

4.1 Repair item prior to flight



1. The item may not be deferred and must be repaired prior to flight.

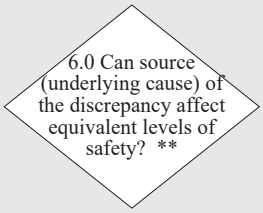
3. NEF DECISION TREE DISCUSSION (Cont'd)

5.0 Can source (underlying cause) of the discrepancy be identified? (If applicable)



1. Can the source of the discrepancy be identified? This step may or may not apply to the item under consideration. If the source can be identified, then proceed to step 6.0, otherwise proceed to step 4.1.

6.0 Can source (underlying cause) of the discrepancy affected equivalent levels of safety?



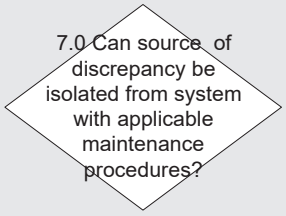
1. If the source (underlying cause) of the discrepancy affect equivalent levels of safety, then it must be determined if it can be isolated from all other systems so as to alleviate any safety concern.

***** NOTE: IN MAKING THIS DETERMINATION, VERY CLOSE COORDINATION BETWEEN THE FLIGHT CREW, MAINTENANCE AND OPERATIONS PERSONNEL MAY BE REQUIRED.***

2. If, after review, the source of the discrepancy could be considered a safety-of-flight concern, the item must be repaired prior to flight (step 4.1). If the source of the discrepancy is not a safety-of-flight concern then defer the item in accordance with the approved NEF procedures in step 8.0. If it cannot be determined, or is uncertain, that the source of the discrepancy is a safety-of-flight concern then proceed to 7.0.

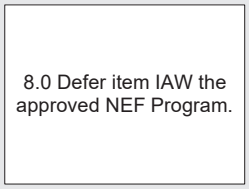
3. NEF DECISION TREE DISCUSSION (Cont'd)

7.0 Can source (underlying cause) of discrepancy be isolated from the system with applicable maintenance procedures?



1. If applicable, the source (underlying cause) of the discrepancy must be isolated from all other systems so as to alleviate the safety-of-flight concern.
2. If the item cannot be safely isolated then the item must be repaired prior to flight (step 4.1).
3. If isolated, the isolation of the source must pass the entire test identified in the evaluative process (steps 4.0-7.0) for the item.
4. If source can be isolated then proceed to step 8.0.

8.0 Defer Item IAW the approved NEF program.



1. Defer the item if, after completing the previous 8 steps, the item can be deferred IAW the NEF program.

NOTE: ***BEFORE AN ITEM CAN BE DEFERRED AS AN NEF ITEM, THE NEF PROGRAM EVALUATION PROCESS FOR DETERMINING SHALL BE FOLLOWED IF AN ITEM CAN BE CONSIDERED A NEF. ALTHOUGH NEF ITEMS ARE NOT SAFETY-OF-FLIGHT ITEMS, THEY HAVE NOT BEEN EVALUATED THROUGH THE NORMAL AEG REVIEW PROCESS AND THEREFORE REQUIRE THE CONCURRENCE OF THE FLIGHT CREW, MAINTENANCE, AND OPERATIONAL PERSONNEL, IF APPLICABLE. NEF ITEMS ARE NOT DEFERRED UNDER THE AUTHORITY OF AN AIRFRAME AND POWERPLANT CERTIFICATE BUT RATHER THE ITEM IS DEFERRED UNDER THE NEF PROGRAM.***

3. NEF DECISION TREE DISCUSSION (Cont'd)

8.0 Defer Item IAW the approved NEF program. (Cont'd)

The evaluation process should determine items such as:

Is the item required for the operational rules in which the aircraft is operated?

a. Does it create the potential for fire/smoke or other hazardous conditions?

b. Could it have an adverse effect on other required systems or components?

c. Does its condition potentially affect the safety of passengers, crew, or service personnel?

d. Could it have a negative impact on emergency or abnormal procedures?

f. Does it create additional workload for the crew at critical times of flight or flight preparation?

g. Crewmembers may need to evaluate the deferred NEF on a flight-by-flight basis.

NOTE: THE ABOVE EVALUATION PROCESS MUST BE ACCOMPLISHED FOR THE INOPERATIVE, DAMAGED, OR MISSING ITEMS AT ITS FACE VALUE, AND ALSO FOR THE UNDERLYING CAUSE OF THE DISCREPANCY.

9.0 Update NEF Deferral list as required.

9.0 Update NEF Deferral List as required.

1. **Caribbean Airlines** will continually add items to the NEF list as they see fit.

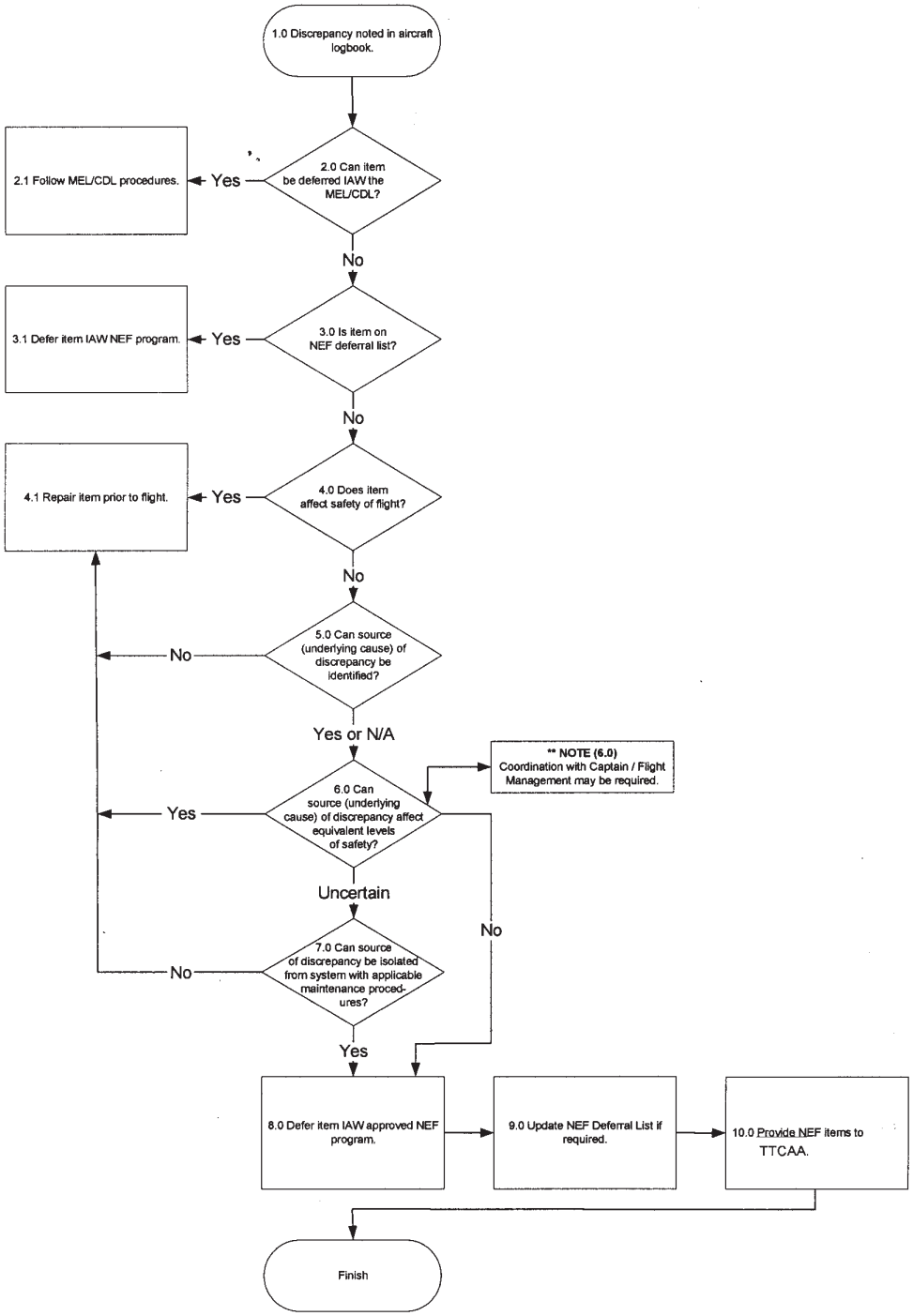
10.0 Provide NEF Items to TTCAA.

10.0 Provide NEF items to TTCAA.

1. **Caribbean Airlines** will provide the items placed in the NEF program to the TTCAA for review. This review is not to be conceived as a requirement to approve the NEF items. It is merely a means to provide oversight to ensure the program is effective.

MEL CHAPTER 25 EQUIPMENT AND FURNISHINGS
NONESSENTIAL EQUIPMENT AND FURNISHING (NEF)

FIGURE 1: NEF DECISION TREE



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ATR 72-600 is the commercial designation of the ATR 72-212A '600 VERSION' fitted with Mod. 05948 NAVIGATION - INSTALL NEW AVIONICS SUITE.

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1. INTRODUCTION

1.01 GENERAL

The main purpose of the Alert Message List is to provide a quick means to cross reference an alert message with the possible associated MEL item. The "AML.Alert Message List" section shall therefore be used for guidance purpose only.

This Alert Message List is intended to assist CAL in defining if the condition presented by the alert message is a potential deferrable one or not prior to performing troubleshooting. It is not the intent of this list to replace or disregard proper troubleshooting of the associated alert message.

When an alert message is displayed, its cause must be investigated by using the appropriate means or associated maintenance procedures. Only after proper identification of the failed component a MEL relief may be used. Once this MEL relief is used, the Alert Message List is no longer applicable to this related item.

If CAL chooses not to consider MEL relief, then maintenance must be accomplished to rectify any system and/or component failure before aircraft dispatch.

The flight crew is responsible for checking that the condition of the aircraft systems/components complies with the MEL requirements. Prior to operation with any item inoperative acceptance by the crew is required in accordance with the continuing airworthiness management procedures. If not, the dispatch of the aircraft is not permitted.

NOTE: *This cross reference table shall be used for guidance purpose only as it does not cover all the combinations of systems and/or component failures that may cause Alert message(s).*

1.02 READING THE TABLE

1) Class of Alert message section: Warning, Caution, Advisory
   - **WARNING** messages: Refer to WARNING MESSAGE
   - **CAUTION** messages: Refer to CAUTION MESSAGE
2) EWD message: this column indicates the alert message title that appears on the Alert Window on the EWD.
3) Meaning / Assessment conditions: this column provides additional information when required. When a failed component is identified as a root cause, it either leads to a potential dispatch or a no dispatch case (NO GO) following the nature of the failed component.

1.02 READING THE TABLE (Cont'd)

4) Condition of dispatch:

- **NO DISPATCH:** Since the associated EWD alert message can be triggered by one (or more) system, equipment, component, or functional conditions or failures, a MEL item relief cannot be directly linked to these EWD alert messages. CAL must perform troubleshooting to identify the root cause(s) that triggered the EWD alert message.

  Dispatch may be possible for the identified inoperative system(s), equipment, component(s), or function(s) if a MEL item exists for the identified inoperative condition.

  If not, or if CAL decides not to consider MEL relief, maintenance or repair must be completed to resolve any system, equipment, or component failure before the aircraft is released to service.
- **XX-XX-XX:** Potential Dispatch. Troubleshooting is necessary to identify the specific conditions that triggered the EWD alert message. When identified, a dispatch may be possible under the reference MEL item.
- **Apply EWD procedure:** some of the FWS messages can be resolved by applying the associated procedure. It is up to the crew to assess situation after application of the procedure.

ENGINE		
EWD Message	**Meaning / Assessment conditions**	**Condition of Dispatch**
ENG 1 FIRE		NO DISPATCH
ENG 1 OIL PRESS		NO DISPATCH
ENG 1 OUT		NO DISPATCH
ENG 1+2 OUT		NO DISPATCH
NAC OVHT		Apply EWD Procedure
ENG 2 FIRE		NO DISPATCH
ENG 2 OIL PRESS		NO DISPATCH
ENG 2 OUT		NO DISPATCH
PROP BRK		61-50-01

ELEC		
EWD Message	**Meaning / Assessment conditions**	**Condition of Dispatch**
ELEC DC GEN 1+2		NO DISPATCH

FLIGHT CONTROL		
EWD Message	**Meaning / Assessment conditions**	**Condition of Dispatch**
FLAPS UNLK		NO DISPATCH
PITCH DISC	Pitch disconnect occurred in flight	NO DISPATCH
	Pitch disconnect occurred on ground	Apply EWD Procedure
PITCH RECONNCT	If message remains after 10s after reconnection attempt Refer to 27.37-02 Elevators Clutch Reconditioning Device	NO DISPATCH

LANDING GEAR		
EWD Message	**Meaning / Assessment conditions**	**Condition of Dispatch**
LDG GEAR NOT DN		NO DISPATCH

FIRE PROTECTION		
EWD Message	**Meaning / Assessment conditions**	**Condition of Dispatch**
FWD SMK		NO DISPATCH
AFT SMK		NO DISPATCH
ELEC SMK		NO DISPATCH

AIR		
EWD Message	**Meaning / Assessment conditions**	**Condition of Dispatch**
EXCESS CAB ALT		NO DISPATCH
EXCESS CAB △P		NO DISPATCH
NEGATIVE CAB △P		NO DISPATCH

MISCELLANEOUS		
EWD Message	**Meaning / Assessment Conditions**	**Condition of Dispatch**
CONFIG		NO DISPATCH

ENGINE		
EWD Message	**Meaning / Assessment Conditions**	**Condition of Dispatch**
ENG 1 LO PITCH		NO DISPATCH
ENG 1 OIL OVHT		NO DISPATCH
ENG 1 OVER LIM		NO DISPATCH
ENG 1 PEC		NO DISPATCH
ENG 1 PEC SG CH		61-21-02
ENG 1 PROP LIM		NO DISPATCH
ENG 2 LO PITCH		NO DISPATCH
ENG 2 OIL OVHT		NO DISPATCH
ENG 2 OVER LIM		NO DISPATCH
ENG 2 PEC		NO DISPATCH
ENG 2 PEC SG CH		61-21-02
ENG 2 PROP LIM		NO DISPATCH
ENG BOOST		73-23-05
ENG EEC		73-23-02-01
ENG EEC 1+2		NO DISPATCH
ENG PROP NP		61-21-06
ENG START	NH <20%, 10 s after starter pushbutton ON	NO DISPATCH
	Start light ON past 45% NH or GCU failure during start	Apply EWD Procedure
ENG START ITT	If Engine ITT parameters outside AFM starting limits (Refer to: AFM–LIM-5-70-1-1 ENGINES)	NO DISPATCH
	If Engine ITT parameters within AFM starting limits (Refer to: AFM-LIM-5-70-1-1 ENGINES)	Apply EWD Procedure
ENG START NO NH	NH indication issue If Engine starting successful	77-11-01-02
ENG X START		Apply EWD Procedure

ENGINE (Cont'd)		
EWD Message	**Meaning / Assessment Conditions**	**Condition of Dispatch**
ENG 1 OIL TEMP H	If Engine OIL TEMP parameters outside AFM limits (Refer to: AFM-LIM-5-70-1-1 ENGINES)	NO DISPATCH
	If Engine OIL TEMP parameters within AFM limits (Refer to: AFM-LIM-5-70-1-1 ENGINES)	Apply EWD Procedure
ENG 1 OIL TEMP L	If Engine OIL TEMP parameters outside AFM limits (Refer to: AFM-LIM-5-70-1-1 ENGINES)	NO DISPATCH
	If Engine OIL TEMP parameters within AFM limits (Refer to: AFM-LIM-5-70-1-1 ENGINES)	Apply EWD Procedure
IDLE GATE		NO DISPATCH
LOOP 1A		26-12-01
LOOP 1B		26-12-01
LOOP 2A		26-12-01
LOOP 2B		26-12-01
SINGLE ENGINE		NO DISPATCH
ENG 2 OIL TEMP H	If Engine OIL TEMP parameters outside AFM limits (Refer to: AFM-LIM-5-70-1-1 ENGINES)	NO DISPATCH
	If Engine OIL TEMP parameters within AFM limits (Refer to: AFM-LIM-5-70-1-1 ENGINES)	Apply EWD Procedure
ENG 2 OIL TEMP L	If Engine OIL TEMP parameters outside AFM limits (Refer to: AFM-LIM-5-70-1-1 ENGINES)	NO DISPATCH
	If Engine OIL TEMP parameters within AFM limits (Refer to: AFM-LIM-5-70-1-1 ENGINES)	Apply EWD Procedure

FUEL		
EWD Message	**Meaning / Assessment Conditions**	**Condition of Dispatch**
ENG FUEL CLOG		NO DISPATCH
FUEL 1 TEMP HI	If FUEL TEMP parameter outside AFM limits (Refer to AFM -LIM-5-70-4-2 Fuel Temperature)	NO DISPATCH
	If FUEL TEMP parameter within AFM limits (Refer to AFM - LIM-5-70-4-2 Fuel Temperature)	Apply EWD Procedure
FUEL 1 TEMP LO	If FUEL TEMP parameter outside AFM limits (Refer to AFM - LIM-5-70-4-2 Fuel Temperature)	NO DISPATCH
	If FUEL TEMP parameter within AFM limits (Refer to AFM - LIM-5-70-4-2 Fuel Temperature)	Apply EWD Procedure
FUEL 2 TEMP HI	If FUEL TEMP parameter outside AFM limits (Refer to AFM - LIM-5-70-4-2 Fuel Temperature)	NO DISPATCH
	If FUEL TEMP parameter within AFM limits (Refer to AFM - LIM-5-70-4-2 Fuel Temperature)	Apply EWD Procedure
FUEL 2 TEMP LO	If FUEL TEMP parameter outside AFM limits (Refer to AFM - LIM-5-70-4-2 Fuel Temperature)	NO DISPATCH
	If FUEL TEMP parameter within AFM limits (Refer to AFM - LIM-5-70-4-2 Fuel Temperature)	Apply EWD Procedure
FUEL FEED LO PR		NO DISPATCH
FUEL LO LVL	Fuel tank quantity < 160 kg	NO DISPATCH
	Fuel tank quantity ≥ 160 kg	NO DISPATCH
FUEL MISMATCH	Other case	NO DISPATCH
	Fuel quantity indication	28-42-01
FUEL UNBALANCED	Other case	Apply EWD Procedure
	Fuel quantity indication inoperative	28-42-01
	Fuel leak confirmed	NO DISPATCH

ELEC		
EWD Message	**Meaning / Assessment Conditions**	**Condition of Dispatch**
ELEC AC BUS 1		NO DISPATCH
ELEC AC BUS 2		NO DISPATCH
ELEC AC STBY		NO DISPATCH
ELEC ACW 1+2		NO DISPATCH
ELEC ACW BUS 1		NO DISPATCH
ELEC ACW BUS 2		NO DISPATCH
ELEC ACW GEN 1		24-22-01
ELEC ACW GEN 2		24-22-01
ELEC BAT CHG		NO DISPATCH
ELEC BAT DISCH		NO DISPATCH
ELEC CHG LOSS		NO DISPATCH
ELEC DC BUS 1		NO DISPATCH
ELEC DC BUS 2		NO DISPATCH
ELEC DC EMER		NO DISPATCH
ELEC DC ESS		NO DISPATCH
ELEC DC GEN		24-30-01
ELEC DC SBY BUS		NO DISPATCH
ELEC DC SHED		NO DISPATCH
ELEC INV 1		NO DISPATCH
ELEC INV 2		24-21-01
ELEC STBY UNDV		NO DISPATCH

HYDRAULIC		
EWD Message	**Meaning / Assessment Conditions**	**Condition of Dispatch**
HYD AUX LO PR	Auxiliary hydraulic pump inoperative	29-11-02
	Other case	NO DISPATCH
HYD AUX OVHT	Auxiliary hydraulic pump inoperative	29-11-02
	Other case	NO DISPATCH
HYD BLUE LO LVL		NO DISPATCH
HYD BLUE LO PR	Main hydraulic pump inoperative	29-11-01
	Other case	NO DISPATCH
HYD BLUE OVHT	Main hydraulic pump inoperative	29-11-01
	Other case	NO DISPATCH
HYD GREEN LO PR	Main hydraulic pump inoperative	29-11-01
	Other case	NO DISPATCH
HYD GREEN LO LVL		NO DISPATCH
HYD GREEN OVHT	Main hydraulic pump inoperative	29-11-01
	Other case	NO DISPATCH
HYD PUMPS LOSS		NO DISPATCH
HYD SYS LOSS		NO DISPATCH

FLIGHT CONTROL		
EWD Message	**Meaning / Assessment Conditions**	**Condition of Dispatch**
AIL LOCK LIT	With GUST LOCK lever engaged	27-70-01-01
	With GUST LOCK lever disengaged	NO DISPATCH
FLAPS ASYM		NO DISPATCH
FLAPS JAM	Flaps Control System inoperative with flaps confirmed in retracted position	27-51-01
	Other case	NO DISPATCH
FLT CTL PUSHER		NO DISPATCH
FLT CTL RUD RCU		27-20-02
FLT CTL TLU		27-23-01
PITCH TRIM ASYM		NO DISPATCH

LANDING GEAR		
EWD Message	**Meaning / Assessment Conditions**	**Condition of Dispatch**
LDG GEAR	If any UNLOCK LIGHT INDICATION on one panel only	32-61-01
	If same UNLOCK LIGHT INDICATION on both panels	NO DISPATCH
WHEELS A-SKID		32-42-02
WHEELS BRK HOT		NO DISPATCH

DOORS		
EWD Message	**Meaning / Assessment Conditions**	**Condition of Dispatch**
DOOR FWD COMPT	When false alert confirmed	52-11-01
		NO DISPATCH
DOORS	When false alert confirmed	52-11-01
		NO DISPATCH

AIR		
EWD Message	**Meaning / Assessment Conditions**	**Condition of Dispatch**
AIR AUTO PRESS		21-31-01
		21-31-03
AIR BLEED 1		36-11-02-01
		36-11-02-02
		36-11-01
AIR BLEED 2		36-11-02-01
		36-11-02-02
		36-11-01
AIR BLEED 1 LEAK		NO DISPATCH

AIR (Cont'd)		
EWD Message	**Meaning / Assessment Conditions**	**Condition of Dispatch**
AIR BLEED 1 OVHT		36-11-02-01
		36-11-02-02
		36-11-01
AIR BLEED 2 LEAK		NO DISPATCH
AIR BLEED 2 OVHT		36-11-02-01
		36-11-02-02
		36-11-01
AIR DITCH		NO DISPATCH
AIR DUCT 1 OVHT	Pack AUTO Temperature Control inoperative	21-61-01
	If alert persists in manual mode: pack inoperative	21-51-01
AIR DUCT 2 OVHT	Pack AUTO Temperature Control inoperative	21-61-01
	If alert persists in manual mode: pack inoperative	21-51-01
AIR OVBD		21-23-01
AIR PACK	Pack inoperative	21-50-01
	Pack valve inoperative	21-51-01
AIR PACK 1+2		NO DISPATCH
AIR RECIR FAN 1		21-22-01
AIR RECIR FAN 2		21-22-01
AIR VENT EXH	Passenger configuration	21-23-03-01
AIR X VALVE		36-11-04
CAB △P		NO DISPATCH
CAB ALT		NO DISPATCH

ANTI-ICE		
EWD Message	**Meaning / Assessment Conditions**	**Condition of Dispatch**
A-ICING ALPHA		30-31-02
A-ICING ALPHAs		NO DISPATCH
A-ICING AUTO		30-11-06
A-ICING BLEED	Airframe De Icing Boots System inoperative	30-11-02
	Engine De Icing Boots System inoperative	30-21-01
A-ICING DETECT		30-80-01
A-ICING ENG 1	Engine De Icing Boots System inoperative	30-21-01
	De Ice valve inoperative	30-11-01
A-ICING ENG 2	Engine De Icing Boots System inoperative	30-21-01
	De Ice valve inoperative	30-11-01
A-ICING FRAME		30-11-02
A-ICING HORNS 1		30-53-01
A-ICING HORNS 2		30-53-01
A-ICING L WSHLD		30-60-01
A-ICING PIT CPT	Probe Heating System - Pitot inoperative	30-30-01
	Probe Heating System - Static inoperative	30-30-02
A-ICING PIT F/O	Probe Heating System - Pitot inoperative	30-30-01
	Probe Heating System - Static inoperative	30-30-02
A-ICING PIT SBY	Probe Heating System - Pitot inoperative	30-30-01
	Probe Heating System - Static inoperative	30-30-02
A-ICING PROP 1		30-61-01
A-ICING PROP 2		30-61-01
A-ICING R WSHLD		30-60-01
A-ICING SEL		NO DISPATCH
A-ICING TAT CPT		30-31-01
		30-31-04
A-ICING TAT F/O		30-31-01
		30-31-04
A-ICING WINDOWS		30-60-02

←

AUTOPILOT		
EWD Message	**Meaning / Assessment Conditions**	**Condition of Dispatch**
AILERON MISTRIM		NO DISPATCH
AP PITCH TRIM		NO DISPATCH
PITCH MISTRIM		NO DISPATCH
RUDDER MISTRIM		NO DISPATCH
YAW AUTO TRIM		NO DISPATCH

MPC		
EWD Message	**Meaning / Assessment Conditions**	**Condition of Dispatch**
APM FAULT		31-31-03-02

AVIONICS		
EWD Message	**Meaning / Assessment Conditions**	**Condition of Dispatch**
ADC		34-11-01
ADC 1+2		NO DISPATCH
ADS-B		34-52-04
AHRS		34-20-01
AHRS 1+2		NO DISPATCH
AHRS NOT ALIGN		NO DISPATCH
ALT DISAGREE		NO DISPATCH
ATT DISAGREE	AHRS inoperative	34-20-01
AUDIO SEL CAPT		NO DISPATCH
AUDIO SEL F/O		NO DISPATCH
AVNX CPM 1		NO DISPATCH
AVNX CPM 2		NO DISPATCH
AVNX CPM 1 + IOM 22		NO DISPATCH
AVNX CPM 1 + IOM 23		NO DISPATCH
AVNX CPM 2 + IOM 12		NO DISPATCH
AVNX CPM 2 + IOM 13		NO DISPATCH
AVNX IOM 12		NO DISPATCH
AVNX IOM 12 + 13		NO DISPATCH
AVNX IOM 12 + 22		NO DISPATCH
AVNX IOM 12 + 23		NO DISPATCH
AVNX IOM 13		NO DISPATCH
AVNX IOM 13 + 22		NO DISPATCH
AVNX IOM 13 + 23		NO DISPATCH
AVNX IOM 22		NO DISPATCH
AVNX IOM 22 + 23		NO DISPATCH
AVNX IOM 23		NO DISPATCH
AVNX SGL SWITCH		NO DISPATCH

AVIONICS (Cont'd)		
EWD Message	**Meaning / Assessment Conditions**	**Condition of Dispatch**
DME LOSS		34-50-02
DU TEMP HI	Refer to AFM.PRO.NNO.ABN.34.6-DU TEMP HI to try to reset alert.	Apply EWD Procedures
	If reset unsuccessful	31-60-01
DU DISCREPANCY	Refer to AFM.PRO.NNO.ABN.34.6-DU DISCREPANCY to try to reset alert.	Apply EWD Procedures
	If reset unsuccessful	31-60-01
DUs TEMP HI	Refer to AFM.PRO.NNO.ABN.34.6-SEVERAL DUs TEMP HI to try to reset alert	Apply EWD Procedures
	If reset unsuccessful	31-60-01
FWS SGL CH		NO DISPATCH
GPWS		34-48-01-02
HDG DISAGREE		NO DISPATCH
HF		23-11-01
HF EMITTING	In case of permanent message	NO DISPATCH
IAS DISAGREE		NO DISPATCH
RMS		23-52-01
RMS 1 + 2		NO DISPATCH
SGL DME		34-50-02
TCAS		34-57-01-01
TERRAIN		34-48-01-03
VHF		23-12-01-01
VHF 1 + 2		NO DISPATCH
VHF EMITTING	In case of permanent message	NO DISPATCH

AVIONICS (Cont'd)		
EWD Message	**Meaning / Assessment Conditions**	**Condition of Dispatch**
XPDR	All affected XPDR functions are lost	34-52-02
	All affected XPDR functions are lost	34-52-03
	All affected XPDR functions are lost	34-52-04
XPDR 1 + 2	All affected XPDR functions are lost	34-52-02
	All affected XPDR functions are lost	34-52-03
	All affected XPDR functions are lost	34-52-04

MISCELLANEOUS		
EWD Message	**Meaning / Assessment Conditions**	**Condition of Dispatch**
BAD CONF		NO DISPATCH
DATA RECORDER		31-32-01
DATALINK		23-24-01
FDAU		31-31-03-01
OXY		NO DISPATCH
RAD-ALT		34-42-01
RAD-ALT LOSS		34-42-01
VOICE RECORDER		23-71-01
AIR AFT DET FAN		26-19-01

MFC		
EWD Message	**Meaning / Assessment Conditions**	**Condition of Dispatch**
MFC 1A		NO DISPATCH
MFC 1A + 1B		NO DISPATCH
MFC 1A + 2A		NO DISPATCH
MFC 1A + 2B		NO DISPATCH
MFC 1B		NO DISPATCH
MFC 1B + 2A		NO DISPATCH
MFC 1B + 2B		NO DISPATCH
MFC 2A		NO DISPATCH
MFC 2A + 2B		NO DISPATCH
MFC 2B		NO DISPATCH

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ATA 30	**ICE AND RAIN PROTECTION**
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ATA 33	**LIGHTS**
ATA 34	**NAVIGATION**
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ATA 36	**PNEUMATIC SYSTEM**
ATA 42	**INTEGRATED MODULAR AVIONICS**
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ATA 73	**ENGINE FUEL AND CONTROL**
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ATA 75	**AIR**
ATA 77	**ENGINE INDICATING**
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21-10-01 FLOW SELECTOR

21-10-01A NORM Mode Inoperative (A)

RI	NI	NR	PLACARD	O	M
C	1	0	YES	YES	NO

May be inoperative provided that:
(a) HIGH mode is operative and used, and
(b) Takeoff with BLEED ON is prohibited.

←

OPERATIONAL (O) PROCEDURES

- **Before takeoff**
  - ► BLEED VALVES ..... OFF
- **After takeoff**
  - ► BLEED VALVES ..... ON

- OR -

21-10-01B HIGH Mode Inoperative (B)

RI	NI	NR	PLACARD	O	M
C	1	0	YES	NO	NO

May be inoperative provided that NORM mode is operative and used.

NOTE: *Some smoke procedures request to select air flow HIGH. Nevertheless associated procedures efficiency has been demonstrated with air flow NORM.*

21-22-01 RECIRCULATION FAN

RI	NI	NR	PLACARD	O	M
C	2	0	YES	NO	YES

One or both may be inoperative.

MAINTENANCE (M) PROCEDURES *(CAL)*
Affected recirculation fan switch to OFF.
Ensure affected recirculation fan switch PLACARDED.

21-23-01 OVERBOARD VALVE

21-23-01A Automatic Mode (A)

RI	NI	NR	PLACARD	O	M
C	1	0	YES	YES	NO

Automatic mode may be inoperative provided that:
(a) Manual mode is checked operative before each departure, and
(b) Flight is conducted in an unpressurized configuration with maximum operating altitude limited to 10 000 ft.

OPERATIONAL (O) PROCEDURES

- **On ground before departure**
  - ► OVBD VALVE SELECTOR SWITCH .....FULL CLOSE
  - ► OVBD VALVE .....CHECK FULLY CLOSED
  - ► OVBD VALVE SELECTOR SWITCH .....FULL OPEN
  - ► OVBD VALVE ..... CHECK FULLY OPEN
- **On ground, engine 1 not running**
  - ► OVBD VALVE SELECTOR SWITCH .....FULL OPEN
- **In flight or on ground, engine 1 running**
  - ► OVBD VALVE SELECTOR SWITCH .....FULL CLOSE
  - ■ **If EXHAUST MODE comes on amber and △P ≤ 1 PSI**
    - ► OVBD VALVE SELECTOR SWITCH .....FULL OPEN

CAUTION: ► Do not select OVBD VALVE OPEN if △P > 1 PSI.

Unpressurized Flight
Limitation
Maximum operating altitude ..... 10 000 ft
Normal Procedures
►OVBD VALVE .....FULL OPEN
►CAB PRESS MODE SEL..... MAN
►MAN RATE knob .....MAX INCREASE
►BLEED VALVE .....ON

NOTE: *Control aircraft vertical speed to take into account passenger comfort.*

- OR -

21-23-01 OVERBOARD VALVE

21-23-01B (B)

RI	NI	NR	PLACARD	O	M
C	1	0	YES	YES	YES

May be inoperative provided that:
(a) It is deactivated,
(b) Extended overwater flight is prohibited,
(c) Maximum operating altitude is limited to 17 000 ft, and
(d) FLOW selector is considered inoperative, **Refer to item 21-10-01 Flow Selector – HIGH mode inoperative (B).**

OPERATIONAL (O) PROCEDURES

■ **In case of Overboard Valve – Automatic Mode (A)**

Limitation

Flight is conducted in an unpressurized configuration.

Maximum operating altitude ..... 10 000 ft

Check procedure of the Overboard Valve in manual mode

► OVBD VALVE SELECTOR SWITCH..... FULL CLOSE
► OVBD VALVE..... CHECK FULLY CLOSED
► OVBD VALVE SELECTOR SWITCH..... FULL OPEN
► OVBD VALVE..... CHECK FULLY OPEN
► CAB PRESS MODE SEL.....MAN
► MAN RATE knob ..... MAX INCREASE

NOTE: *Adjust aircraft vertical speed to take into account passenger comfort.*

- OR -

■ **In case of Overboard Valve (B)**

- **After ENG 1 start**

► EXHAUST MODE.....OVBD

NOTE: ► *Avoid leaving avionics selected ON for a long period of time on ground.*

► *On ground with aircraft network energized (GPU connected), the mechanic ground call system is activated as extract fan is inoperative.*

MAINTENANCE (M) PROCEDURES

► Overboard Ventilation Valve De-energization.
Refer to MP ATR-A-21-23-20-00ZZZ-560Z-A, setting the valve open in intermediate position.

NOTE: ► *Avoid leaving avionics selected ON for a long period of time.*

21-23-02 UNDERFLOOR VALVE

RI	NI	NR	PLACARD	O	M
C	1	0	YES	YES	NO

May be inoperative in closed position provided that:
(a) Exhaust mode is selected OVBD, and
(b) Extended overwater flight is prohibited, and
(c) Maximum operating altitude is limited to 17 000 ft.

OPERATIONAL (O) PROCEDURES

► Select exhaust mode OVBD.

NOTE: ► *Avoid leaving avionics selected ON for a long period of time on ground.*
► On ground only with aircraft network energized (GPU connected), the mechanic ground call system is activated as extract fan is inoperative.

21-23-03-01 EXTRACT FAN – PASSENGER CONFIGURATION

RI	NI	NR	PLACARD	O	M
C	1	0	YES	YES	NO

May be inoperative provided that:
(a) Exhaust mode is selected OVBD, and
(b) Extended overwater flight is prohibited, and
(c) Maximum operating altitude is limited to 17 000 ft.

OPERATIONAL (O) PROCEDURES

► Select exhaust mode OVBD.

NOTE

► *Avoid leaving avionics selected ON for a long period of time on ground.*

► *On ground only with aircraft network energized (GPU connected), the mechanic ground call system is activated as extract fan is inoperative.*

21-30-01-01 PRESSURIZATION SYSTEM – PASSENGER CONFIGURATION

RI	NI	NR	PLACARD	O	M
C	1	0	YES	YES	YES

The pressurization system (automatic and manual modes) may be inoperative provided that:

(a) Flight is conducted in an unpressurized configuration with maximum operating altitude limited to 10 000 ft, and

(b) Outflow valves are secured in open position.

OPERATIONAL (O) PROCEDURES

■ **Unpressurized Flight**

Limitation

Maximum operating altitude ..... 10 000 ft

Normal Procedures

► OVBD VALVE .....FULL OPEN

► CAB PRESS MODE SEL ..... MAN

► MAN RATE knob.....MAX INCREASE

► BLEED VALVE.....ON

NOTE: *Control aircraft vertical speed to take into account passenger comfort.*

■ **If manual mode is operative**

► MAINTENANCE PROCEDURE .....APPLY

MAINTENANCE (M) PROCEDURES

► Electropneumatic Outflow Valve and Pneumatic Outflow Valve Blockage in Open Positon.

Refer to MP ATR-A-21-31-20-00ZZZ-560Z-A.

21-31-01 AUTOMATIC PRESSURIZATION CONTROL

RI	NI	NR	PLACARD	O	M
C	1	0	YES	YES	NO

May be inoperative **for a pressurized flight** provided that:
(a) Manual pressurization is operative, and
(b) CAB ALT, CAB RATE and △P indications are operative.

OPERATIONAL (O) PROCEDURES

- **Before takeoff**
  - ► MAN CAB ALT knob.....9 O'CLOCK (MAN position)
  - ► CAB PRESS MODE SEL.....MAN
- **During climb**
  - ► MAN CAB ALT knob.....INCREASE
  - ► CAB ALT RATE..... ADJUST 1 000 ft/min UP MAX

ft	14 000	17 000	20 000	22 000	24 000	25 000
TARGET CAB ALT (ft)	2 500	3 300	4 200	5 200	6 200	6 700

- **When CAB ALT = TARGET CAB ALT**
  - ► MAN CAB ALT knob..... AS RQRD TO MAINTAIN TARGET CAB ALT
- **Before descent**
  - ▪ **If CAB ALT above landing elevation**
  - ► MAN CAB ALT knob..... DECREASE
  - ► CAB ALT RATE.....ADJUST 400 ft/min DN
  - ▪ **If CAB ALT below landing elevation**
  - ► MAN CAB ALT knob.....INCREASE
  - ► CAB ALT RATE..... ADJUST 1 000 ft/min UP MAX
- **After landing**
  - ► MAN CAB ALT knob..... MAX INCREASE

21-31-02 DESCENT RATE SELECTOR

RI	NI	NR	PLACARD	O	M
C	1	0	YES	NO	NO

May be inoperative.

21-31-03 ELECTROPNEUMATIC OUTFLOW VALVE

RI	NI	NR	PLACARD	O	M
-	1	0	YES	NO	YES

May be inoperative provided that:
(a) It is secured in closed position, and
(b) Flight is conducted with manual pressurization: **Refer to Item 21-31-01 Automatic Pressurization Control.**

MAINTENANCE (M) PROCEDURES
Refer to MP ATR-A-21-31-20-01ZZZ-560Z-A.

21-31-04-01 CABIN PRESS INDICATION – ALT
Cabin Press Indication – CAB ALT Indication on SD Cabin

RI	NI	NR	PLACARD	O	M
C	1	0	YES	YES	NO

May be inoperative provided that △P indication is operative.

OPERATIONAL (O) PROCEDURES

► Check △P based on table below:

ft	14 000	17 000	20 000	22 000	24 000	25 000
TARGET (PSI)	4.8	5.4	5.8	5.9	6	6.35

21-31-04-02 CABIN PRESS INDICATION – RATE
Cabin Press Indication – CAB RATE Indication on SD Cabin

RI	NI	NR	PLACARD	O	M
C	1	0	YES	YES	NO

May be inoperative.

OPERATIONAL (O) PROCEDURES

► Check cabin altitude based on table below (△P = 6 PSI):

ft	14 000	17 000	20 000	22 000	24 000	25 000
TARGET CAB ALT (ft)	2 500	3 300	4 200	5 200	6 200	6 700

21-31-04-03 CABIN PRESS INDICATION – PRESSURE
Cabin Press Indication – △P Indication on SD Cabin

RI	NI	NR	PLACARD	O	M
C	1	0	YES	YES	NO

May be inoperative.

OPERATIONAL (O) PROCEDURES

► Check cabin altitude based on table below (△P = 6 PSI):

ft	14 000	17 000	20 000	22 000	24 000	25 000
TARGET CAB ALT (ft)	2 500	3 300	4 200	5 200	6 200	6 700

21-31-05 DITCH FUNCTION (IF INSTALLED)

RI	NI	NR	PLACARD	O	M
C	1	0	YES	NO	NO

May be inoperative.

21-35-01 DUMP FUNCTION

RI	NI	NR	PLACARD	O	M
C	1	0	YES	YES	NO

May be inoperative provided that manual pressurization is operative.

OPERATIONAL (O) PROCEDURES

- **In case of rapid depressurization of the aircraft is needed:**
  - ► MAN RATE knob..... MAN
  - ► CAB PRESS MODE SEL..... MAN
  - ► MAN RATE knob.....MAX INCREASE

NOTE: *Residual ⧋P may be cancelled using the cockpit communication door.*

21-50-01 PACK

RI	NI	NR	PLACARD	O	M
C	2	1	YES	YES	NO

One may be inoperative provided:
(a) Maximum operating altitude is limited to 17 000 ft,
(b) Associated pack is selected OFF, and
(c) Opposite pack and bleed valve are operative.
←

OPERATIONAL (O) PROCEDURES
As an in-flight failure could imply total loss of pressurization, fuel consumption at 10 000 ft must be taken into account to compute the trip fuel.

NOTE: *With a pack selected OFF, engine torque parameter of the related side will increase.*

► PL (Affected Engine) ..... ADJUST TO OTHER ENGINE

21-51-01 PACK VALVE

RI	NI	NR	PLACARD	O	M
-	2	1	YES	YES	YES

One may be inoperative provided that:
(a) It is secured closed, and
(b) Associated pack is considered inoperative: **Refer to Item 21-50-01 Pack**.

OPERATIONAL (O) PROCEDURES

As an in-flight failure could imply total loss of pressurization, fuel consumption at 10 000 ft must be taken into account to compute the trip fuel.

NOTE: *With a pack selected OFF, engine torque parameter of the related side will increase.*

► PL (Affected Engine).....ADJUST TO OTHER ENGINE

MAINTENANCE (M) PROCEDURES

Refer to MP ATR-A-21-11-50-00ZZZ-560Z-A.

21-51-02 PACK FAULT ALERT

RI	NI	NR	PLACARD	O	M
-	2	1	YES	YES	NO

One may be inoperative provided that associated pack is considered inoperative.
Refer to Item 21-50-01 Pack.

OPERATIONAL (O) PROCEDURES

As an in-flight failure could imply total loss of pressurization, fuel consumption at 10 000 ft must be taken into account to compute the trip fuel.

NOTE: *With a pack selected OFF, engine torque parameter of the related side will increase.*

► PL (Affected Engine)..... ADJUST TO OTHER ENGINE

21-51-03 COOLING FAN
Turbo Fan

RI	NI	NR	PLACARD	O	M
C	2	0	YES	NO	NO

One or both may be inoperative provided that:
(a) Affected pack is not used on ground, and
(b) Climb, cruise and drift-down planning must ensure that a minimum OEI speed of 150 kt IAS can be maintained.

21-51-05 AUX AFT CARGO VENTILATION VALVE (IF INSTALLED)

RI	NI	NR	PLACARD	O	M
C	1	0	YES	NO	NO

May be inoperative provided that no combustible material is carried in the associated compartment.

21-61-01 PACK AUTO TEMPERATURE CONTROL

21-61-01A (A)

RI	NI	NR	PLACARD	O	M
C	2	1	YES	YES	NO

One may be inoperative provided that associated pack MAN temperature control is operative.

OPERATIONAL (O) PROCEDURES

► TEMP SEL (affected side) ..... MAN

► Adjust manually the associated pack discharge temperature to obtain the same duct indication as the opposite pack.

- OR -

21-61-01B (B)

RI	NI	NR	PLACARD	O	M
-	2	1	YES	NO	NO

One may be inoperative provided that associated pack is considered inoperative.
Refer to Item 21-50-01 Pack.

21-61-02 PACK MAN TEMPERATURE CONTROL

21-61-02A (A)

RI	NI	NR	PLACARD	O	M
C	2	1	YES	NO	NO

One may be inoperative provided that associated pack AUTO temperature control is operative.

- OR -

21-61-02B (B)

RI	NI	NR	PLACARD	O	M
-	2	1	YES	NO	NO

One may be inoperative provided that associated pack is considered inoperative.
Refer to Item 21-50-01 Pack.

21-61-03 OVHT ALERT

RI	NI	NR	PLACARD	O	M
C	2	1	YES	YES	NO

One may be inoperative provided that associated DUCT temperature indication is operative.

OPERATIONAL (O) PROCEDURES

► Monitor and maintain DUCT indication below 92°C / 200°F.

21-63-01-01 COMPT TEMP – FLIGHT DECK INDICATION

RI	NI	NR	PLACARD	O	M
C	1	0	YES	NO	NO

May be inoperative.

21-63-01-02 COMPT TEMP – CABIN INDICATIONS

RI	NI	NR	PLACARD	O	M
C	2	0	YES	NO	NO

One or both may be inoperative.

21-63-02 DUCT INDICATION

21-63-02A (A)

RI	NI	NR	PLACARD	O	M
C	2	0	YES	NO	NO

One or both may be inoperative provided that AUTO temperature control is operative for the associated zones.

- OR -

21-63-02B (B)

RI	NI	NR	PLACARD	O	M
C	2	0	YES	NO	NO

One or both may be inoperative provided that COMPT indication and MAN temperature control are operative for the associated zones.

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22-16-01 PITCH WHEEL

RI	NI	NR	PLACARD	O	M
C	1	0	YES	NO	NO

May be inoperative provided that one TCS is operative.

22-16-02 TOUCH CONTROL STEERING (TCS)

RI	NI	NR	PLACARD	O	M
C	2	0	YES	YES	NO

One or both may be inoperative.

OPERATIONAL (O) PROCEDURES

- **Prior to any manual control movement:**
  - ► AUTOPILOT.....DISENGAGE

22-16-03 COUPLING FUNCTION PUSHBUTTON

RI	NI	NR	PLACARD	O	M
C	1	0	YES	YES	NO

May be inoperative provided that the **pilot flying** is on the selected side.

OPERATIONAL (O) PROCEDURES

In flight, autopilot remains coupled to the side selected before pushbutton failure. For a new dispatch, it will be automatically coupled to the LH side after being energized.

22-16-04 AUTOPILOT QUICK DISCONNECT PUSHBUTTON (QUICK RELEASE CONTROL)

22-16-04A (A)

RI	NI	NR	PLACARD	O	M
C	2	1	YES	NO	YES

One may be inoperative provided that the **pilot flying** side autopilot quick disconnect pushbutton is operative.

MAINTENANCE (M) PROCEDURES

Procedure to follow to check override means on ground:

- ► AVIONICS VENT EXHAUST MODE .....NORM
- ► STICK PUSHER .....OFF
- ► AIR BLEEDS.....OFF
- ► RADAR .....OFF
- ► Switch WOW on Flight position, angle of attack probes positioned in normal angle of attack. (Horizontal position).
- ► Engage the autopilot by pressing AP pb on FGCP.
- ► Check that the AP disconnection is operative by pressing the non-affected Quick Release Control pb.

Check the other AP disconnection means (AP must be engaged before each actions):

- ► Normal pitch trim rocker on captain control wheel (nose up and nose down).
- ► Normal pitch trim rocker on first officer control wheel (nose up and nose down).
- ► Stand by pitch trim rockers (up and down).
- ► Push, then pull on captain control wheel (repeat the same actions on first officer control wheel).
- ► Force on rudder pedals (only on a captain or first officer side): one time left and one time right.
- ► Press on YD pb.

- OR -

22-16-04 AUTOPILOT QUICK DISCONNECT PUSHBUTTON (QUICK RELEASE CONTROL) (Cont'd)

22-16-04B (B)

RI	NI	NR	PLACARD	O	M
-	2	0	YES	NO	NO

One or both may be inoperative provided that autopilot is considered inoperative and not used. **Refer to Item 22-18-01 Autopilot Function.**

22-18-01 AUTOPILOT FUNCTION

←

RI	NI	NR	PLACARD	O	M
B	1	0	YES	YES	NO

May be inoperative.

NOTE: *1) Any mode which is operative may be used, refer to appropriate operative mode.*
2) If AFCS Computer is inoperative Autopilot, Yaw Damper, and Flight Director functions are lost.

Refer to Item 22-18-02 Flight Director Function
Refer to Item 22-18-03 Yaw Damper Function

OPERATIONAL (O) PROCEDURES
CAT 2 operations are not possible.

NOTE: *In the event of smoke or fumes in the cockpit, the SMOKE/FUMES procedure must be performed in manual flight, as autopilot function is not available.*

22-18-02 FLIGHT DIRECTOR FUNCTION

RI	NI	NR	PLACARD	O	M
-	1	0	YES	NO	NO

May be inoperative.
Refer to Item 22-18-03 Yaw Damper Function.

NOTE: *If AFCS Computer is inoperative Autopilot, Yaw Damper, and Flight Director functions are lost.*

Refer to Item 22-18-01 Autopilot Function.

22-18-03 YAW DAMPER FUNCTION

22-18-03A RCU Inoperative (A)

RI	NI	NR	PLACARD	O	M
A	1	0	YES	YES	NO

May be inoperative provided that the operations are limited to two flights.

NOTE: *1) The Autopilot is inoperative. Refer to item 22-18-01 autopilot function.*
2) If AFCS Computer is inoperative, Autopilot, Yaw Damper and Flight Director functions are lost.
Refer to Item 22-18-02 Flight Director Function

OPERATIONAL (O) PROCEDURES

- **If RCU is inoperative**
  - ► PF will keep his feet on the pedals to keep rudder movement under control.

←

- OR -

22-18-03B RCU Operative (B)

RI	NI	NR	PLACARD	O	M
B	1	0	YES	NO	YES

May be inoperative provided that RCU is operative.

NOTE: *1) The Autopilot is inoperative, Refer to item 22-18-01 Autopilot function.*
2) If AFCS Computer is inoperative, Autopilot, Yaw Damper and Flight Director functions are lost.

Refer to Item 22-18-02 Flight Director Function

MAINTENANCE (M) PROCEDURES

To verify that Rudder Releasable Centering Unit (RCU) is operative – Refer to MP ATR-A-27-21-30-00ZZZ-320Z-A.

22-50-01 FGCP – VS PUSHBUTTON

22-50-01A (A)

RI	NI	NR	PLACARD	O	M
C	1	0	NO	YES	NO

May be inoperative provided that IAS is operative.

OPERATIONAL (O) PROCEDURES

► Use the IAS or basic mode for altitude change.
► Disengage AP mode for altitude change.

- OR -

22-50-01B (B)

RI	NI	NR	PLACARD	O	M
-	1	0	NO	NO	NO

May be inoperative provided that FD is considered inoperative.
Refer to Item 22-18-02 Flight Director Function.

22-50-02 FGCP – ALT PUSHBUTTON

22-50-02A (A)

RI	NI	NR	PLACARD	O	M
C	1	0	YES	YES	NO

May be inoperative provided that ALT SEL is operative and used.

OPERATIONAL (O) PROCEDURES
IAS or VS or basic mode may be used for altitude changes in ALT SEL mode.
The Altitude can be held using VS mode with 0 ft/min selected target.

- OR -

22-50-02B (B)

RI	NI	NR	PLACARD	O	M
C	1	0	NO	YES	NO

May be inoperative provided that VS mode with a 0 ft/min selection target is used for ALT HOLD.

OPERATIONAL (O) PROCEDURES
IAS or VS or basic mode may be used for altitude changes in ALT SEL mode.
The Altitude can be held using VS mode with 0 ft/min selected target.

- OR -

22-50-02C (C)

RI	NI	NR	PLACARD	O	M
-	1	0	NO	NO	NO

May be inoperative provided that FD is considered inoperative.
Refer to Item 22-18-02 Flight Director Function

22-50-03 FGCP – ALT SEL KNOB

RI	NI	NR	PLACARD	O	M
B	1	0	YES	YES	NO

May be inoperative provided that ALT pb is operative.

OPERATIONAL (O) PROCEDURES

IAS or VS or basic mode may be used for altitude change with ALT pb selection at requested altitude.

22-50-05 FGCP – STBY PUSHBUTTON

RI	NI	NR	PLACARD	O	M
C	1	0	YES	YES	NO

May be inoperative.

OPERATIONAL (O) PROCEDURES

Flight crew have to deselect the current selected FD mode.

22-50-06 FGCP – HDG SEL PUSHBUTTON

22-50-06A (A)

RI	NI	NR	PLACARD	O	M
C	1	0	YES	YES	NO

May be inoperative provided that AP Basic Modes are operative.

OPERATIONAL (O) PROCEDURES
HDG HOLD mode may be used as required. HDG changes must be done manually or with TCS function.

- OR -

22-50-06B (B)

RI	NI	NR	PLACARD	O	M
-	1	0	YES	NO	NO

May be inoperative provided that FD is considered inoperative.
Refer to Item 22-18-02 Flight Director Function

22-50-07 FGCP – HDG KNOB

RI	NI	NR	PLACARD	O	M
C	1	0	YES	NO	NO

May be inoperative.

22-50-08 FGCP – APP PUSHBUTTON

RI	NI	NR	PLACARD	O	M
C	1	0	YES	YES	NO

May be inoperative.

OPERATIONAL (O) PROCEDURES

ILS and LNAV approaches are possible without FD.
CAT 2 operations are prohibited.

22-50-09 FGCP – NAV PUSHBUTTON

RI	NI	NR	PLACARD	O	M
C	1	0	YES	NO	NO

May be inoperative provided that LNAV or VOR mode is not used.

22-50-10 FGCP – BC PUSHBUTTON

RI	NI	NR	PLACARD	O	M
C	1	0	YES	YES	NO

May be inoperative.

OPERATIONAL (O) PROCEDURES

Manual flight may be used.

22-50-11 FGCP – FD PUSHBUTTON

22-50-11A (A)

RI	NI	NR	PLACARD	O	M
C	2	0	YES	YES	NO

One or both may be inoperative.

OPERATIONAL (O) PROCEDURES

Operations are to be conducted by DAY using VMC flying conditions. For the purpose of repositioning aircraft, coupling positioned to operating AF/D side. No auto-pilot used below 1000 ft.

FD bars can't be removed on affected side.

But AP is fully operational with FD bars displayed on both sides.

- OR -

22-50-11B (B)

RI	NI	NR	PLACARD	O	M
-	2	0	YES	NO	NO

One or both may be inoperative provided that FD function is considered inoperative.

Refer to Item 22-18-02 Flight Director Function

22-50-12 FGCP – VNAV PUSHBUTTON

RI	NI	NR	PLACARD	O	M
C	1	0	YES	YES	NO

May be inoperative provided VNAV mode is not used.

OPERATIONAL (O) PROCEDURES

Flight planning for approach, departure, and planned route must take into account loss of Vertical Guidance in VNAV mode.

22-60-01 ICP - SPD TGT KNOB

RI	NI	NR	PLACARD	O	M
C	2	1	YES	YES	NO

One may be inoperative.

OPERATIONAL (O) PROCEDURES

Manual SPD may be modified by SPD TGT knob of the non-affected side.
Task sharing may be affected. No special procedure if speed AUTO mode is used during all the flight.

22-60-02 ICP – (AUTO – MAN) PUSHBUTTON

22-60-02A (A)

RI	NI	NR	PLACARD	O	M
C	2	1	YES	YES	NO

One may be inoperative.

OPERATIONAL (O) PROCEDURES

PNF on functional side provides input to PF on non-operative side.

Taking into account the following:

- Auto or manual SPD mode may be modified by AUTO/MAN pb of the non affected side.
- Task sharing may be affected.

- OR -

22-60-02B (B)

RI	NI	NR	PLACARD	O	M
C	2	0	YES	NO	NO

One or both may be inoperative provided that AUTO speed is not used.

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23-11-01 HF SYSTEM (IF INSTALLED)

RI	NI	NR	PLACARD	O	M
D	1	0	NO	NO	NO

23-12-01-01 VHF SYSTEM

RI	NI	NR	PLACARD	O	M
C	2	1	YES	YES	NO

May be inoperative when flying VFR over routes navigated by visual reference to landmarks provided that:

(a) VHF 1 is operative, and

(b) The regulations applicable to the aerodrome(s) intended to be used do not require two VHF to be operative.

NOTE: *For all routes, consider the need for TCAS.*

OPERATIONAL (O) PROCEDURES

Crews are to be guided by the remarks above.

NOTE: *The #1 VHF is powered by the Emergency Bus. It may be possible to interchange that unit with a serviceable one i.e. #2 VHF.*

23-22-01 SELCAL

RI	NI	NR	PLACARD	O	M
D	1	0	NO	NO	NO

May be inoperative.

23-24-01 ACARS (IF INSTALLED)

RI	NI	NR	PLACARD	O	M
D	1	0	YES	NO	NO

May be inoperative provided that procedures do not require its use.

23-31-01-01 PASSENGER ADDRESS SYSTEM – PASSENGER CONFIGURATION

RI	NI	NR	PLACARD	O	M
A	1	0	YES	YES	YES

May be inoperative provided that:

(a) Alternate normal, abnormal and emergency procedures and/or operating restrictions are established and used, and

(b) Operations are limited to three calendar days or three flights whichever occurs first.

OPERATIONAL (O) PROCEDURES

Normal Procedures

Before takeoff, cabin crew will make a direct voice communication with all passengers. All preflight announcements will be completed prior to takeoff. All announcements are made by direct voice communications with passengers using megaphone.

Emergency Procedures

Cockpit crew uses interphone system to notify cabin attendant. If necessary, all emergency ground and inflight announcements will be made direct voice communications with passengers or via megaphone.

Check that cabin attendant's interphone system is fully operational.

PROCEDURES

1) Placard Passenger Address System at the pilot's audio control panel in the flight compartment and the PA switch light at the Cabin Attendant handset control unit in the passenger compartment.
2) Make appropriate entry in the Aircraft Technical Log.

MAINTENANCE (M) PROCEDURES

Open, secure and tag C/B COM/PUBLIC ADDRESS.

23-44-01-01 INTERPHONE SYSTEM – GROUND CALL HORN

RI	NI	NR	PLACARD	O	M
C	1	0	NO	YES	NO

May be inoperative.

OPERATIONAL (O) PROCEDURES

Use any available means to establish communicate with Ground Staff.

23-44-01-02 INTERPHONE SYSTEM - FLIGHT DECK TO FLIGHT DECK

RI	NI	NR	PLACARD	O	M
-	1	1	YES	NO	NO

Refer to Audio Control Panel, and Refer to Headset, and Refer to Hand Microphone.

23-44-01-03 INTERPHONE SYSTEM (FLIGHT DECK TO CABIN – CABIN TO FLIGHT DECK)

RI	NI	NR	PLACARD	O	M
B	1	0	YES	YES	NO

May be inoperative provided that:
(a) Passenger Address is operative, and
(b) Alternate normal and abnormal and emergency procedures are established and used, and
(c) Flight deck door automatic locking system (if installed) is verified to operate normal.

OPERATIONAL (O) PROCEDURES
Use PA and Megaphone in an emergency.

23-44-01-04 INTERPHONE SYSTEM – CABIN TO CABIN

RI	NI	NR	PLACARD	O	M
C	1	0	YES	YES	NO

May be inoperative provided alternate procedures are established and used.

NOTE: *Use of PA & Megaphone in an emergency.*

OPERATIONAL (O) PROCEDURES

Use of PA & Megaphone in an emergency.

23-44-01-05 INTERPHONE SYSTEM (FLIGHT CREW TO GROUND – GROUND TO FLIGHT CREW)

RI	NI	NR	PLACARD	O	M
C	1	0	YES	YES	NO

May be inoperative provided that alternate normal and abnormal, and emergency procedures are established and used as per Operations Manual Part A – General, Document No.: BW-OM-0001 Section 12.

OPERATIONAL (O) PROCEDURES

Use of hand signals for communication.

23-44-01-06 INTERPHONE SYSTEM – ALERTING SYSTEM

23-44-01-06A (A)

RI	NI	NR	PLACARD	O	M
C	1	0	YES	NO	NO

Visual signal may be inoperative on the flight deck.

- OR -

23-44-01-06B (B)

RI	NI	NR	PLACARD	O	M
C	1	0	YES	NO	NO

Both visual and aural signals may be inoperative in the cabin provided that Passenger Address is operative from the flight deck.

23-44-01-07 INTERPHONE SYSTEM - HANDSETS

23-44-01-07A (A)

RI	NI	NR	PLACARD	O	M
C	2	1	YES	NO	NO

Handsets at non-required cabin crew seat may be inoperative.

NOTE: *Any handset(s) function(s) that operate normally may be used by the cabin crew.*

- OR -

23-44-01-07B (B)

RI	NI	NR	PLACARD	O	M
C	2	1	YES	YES	NO

May be inoperative provided that:
(a) At least 50 % of cabin handsets operate normally
(b) Alternate procedures are established and used.

NOTE: *1) Any handset(s) function(s) that operate normally may be used by the cabin crew.*
2) An operative handset at an inoperative cabin crew seat shall not be counted to satisfy the 50 % requirement.

OPERATIONAL (O) PROCEDURES

Normal Procedures

All inflight announcements are made by the operational handset.

Check that cabin attendant's interphone system is fully operational for communication to flight deck.

23-44-02 CABIN CHIME SYSTEM (FLIGHT DECK TO CABIN – CABIN SIGNS)

RI	NI	NR	PLACARD	O	M
C	1	0	NO	YES	NO

May be inoperative provided that:

(a) Passenger Address is operative, and

(b) Alternate normal and abnormal and emergency procedures are established and used.

OPERATIONAL (O) PROCEDURES

When necessary, flight crew must directly inform the cabin crew member that SIGNS are ON.

23-50-01 AUDIO CONTROL PANEL

RI	NI	NR	PLACARD	O	M
C	2	2	YES	NO	NO

May be inoperative provided that CAPT panel and F/O panel are operative except VOICE ONLY key which may be inoperative.

NOTE: *There may be components of the audio control panel inoperative for which the crew should refer to the associated item; however, the panel is still adequate for flight.*

23-50-02 HEADSET

RI	NI	NR	PLACARD	O	M
C	-	2	NO	NO	NO

One may be inoperative provided that the associated boom set (headset function) is operative for each crew member on flight deck duty.

23-50-03 HAND MICROPHONE

RI	NI	NR	PLACARD	O	M
C	2	0	NO	NO	NO

One or more may be inoperative provided that the associated boom set is operative for each crew member on flight deck duty.

23-50-04 BOOM SET

RI	NI	NR	PLACARD	O	M
C	2	2	NO	NO	NO

One may be inoperative provided that the associated hand mike and headset are operative for each crew member on flight deck duty.

23-50-05 COCKPIT LOUDSPEAKER

RI	NI	NR	PLACARD	O	M
C	2	1	YES	NO	NO

May be inoperative.

23-51-01 PUSH-TO-TALK (PTT) SELECTOR ON CONTROL WHEEL

RI	NI	NR	PLACARD	O	M
C	2	0	NO	NO	NO

One or both may be inoperative provided that:

(a) INT / RAD selector on ACP are operative,

(b) RAD PTT Pushbutton on Nose Wheel Steering Control Handwheel is operative, and

(c) There is no permanent/constant radio transmission.

23-52-01 RADIO MANAGEMENT SYSTEM

RI	NI	NR	PLACARD	O	M
C	2	1	NO	NO	NO

One may be inoperative provided that:

(a) IESI is operative, and

(b) VHF 1 is operative, and

(c) VOR 1 is operative, and

(d) ILS 1 is operative.

←

23-70-01-01 VIDEO SURVEILLANCE SYSTEM

23-70-01-01A PASSENGER CONFIGURATION (A)

RI	NI	NR	PLACARD	O	M
A	1	0	YES	YES	NO

May be inoperative provided that:
(a) Alternate procedures are established and used,
(b) Cabin/Flight Deck Interphone system is checked operative, and
(c) Repairs are made within three flight days.

OPERATIONAL (O) PROCEDURES
Procedures to exit Flight Deck during Flight
Prior to unlocking the door the flight deck crew shall:
1) Contact cabin crew advising of visit to lavatory.
2) Cabin attendant in charge verifies vacancy of lavatory and locks its.
3) Cabin attendant in charge accesses the key to cabin-to-cargo door and makes way to the front of cabin opens doors to cargo area and locks door behind them.
4) No. 2 cabin attendant contacts flight deck via interphone advises that cabin crew member is positioned outside the cockpit door and gives the password.
5) Flight crew ensure clear zone is secure (by looking through peephole) and exits cabin crew hands over the key to the flight crew member exiting the flight deck, enters flight deck compartment and secures the door.
6) Flight crew uses peep-hole to ensure area in passenger cabin is secure and opens cabin-to cargo door from inside and locks door (keeping the key in their possession).

Procedures to re-enter Flight Deck during Flight
1) No. 2 cabin attendant notifies flight deck that flight crew is ready to re-enter.
2) Flight crew opens cabin to cargo door using the key and locks it behind them.
3) No. 2 cabin attendant contacts flight deck via interphone, notified them that the flight crew is in position to re-enter and gives the password.
4) Cabin crew occupying flight deck ensure clear zone is secure by looking through the peep hole and opens flight deck compartment door, flight crew hand over the key re-enters and secures cockpit door.
5) No. 2 cabin attendant ensures outside cabin to cargo door is secure and exits, returns to rear position and replaces the key.

- **During preliminary cockpit preparation**
  ► CABIN/FLIGHT DECK INTERPHONE SYSTEM.....CHECK
  ► COCKPIT ACCESS PROCEDURES .....DEFINE AND BRIEF CREW

- OR -

23-70-01-01 VIDEO SURVEILLANCE SYSTEM (Cont'd)

23-70-01-01B PASSENGER CONFIGURATION (B)

RI	NI	NR	PLACARD	O	M
C	1	0	YES	YES	NO

May be inoperative provided that:
(a) The flight deck door viewing port operates normally,
(b) Cabin/Flight Deck Interphone system is checked operative, and
(c) Alternate procedures are established and used.

OPERATIONAL (O) PROCEDURES

Procedures to exit Flight Deck during Flight

Prior to unlocking the door the flight deck crew shall:

1) Contact cabin crew advising of visit to lavatory.
2) Cabin attendant in charge verifies vacancy of lavatory and locks its.
3) Cabin attendant in charge accesses the key to cabin-to-cargo door and makes way to the front of cabin opens doors to cargo area and locks door behind them.
4) No. 2 cabin attendant contacts flight deck via interphone advises that cabin crew member is positioned outside the cockpit door and gives the password.
5) Flight crew ensure clear zone is secure (by looking through peephole) and exits cabin crew hands over the key to the flight crew member exiting the flight deck, enters flight deck compartment and secures the door.
6) Flight crew uses peep-hole to ensure area in passenger cabin is secure and opens cabin-to cargo door from inside and locks door (keeping the key in their possession).

Procedures to re-enter Flight Deck during Flight

1) No. 2 cabin attendant notifies flight deck that flight crew is ready to re-enter.
2) Flight crew opens cabin to cargo door using the key and locks it behind them.
3) No. 2 cabin attendant contacts flight deck via interphone, notified them that the flight crew is in position to re-enter and gives the password.
4) Cabin crew occupying flight deck ensure clear zone is secure by looking through the peep hole and opens flight deck compartment door, flight crew hand over the key re-enters and secures cockpit door.
5) No. 2 cabin attendant ensures outside cabin to cargo door is secure and exits, returns to rear position and replaces the key.

- **During preliminary cockpit preparation**
  - ► CABIN/FLIGHT DECK INTERPHONE SYSTEM ..... CHECK
  - ► COCKPIT ACCESS PROCEDURES ..... DEFINE AND BRIEF CREW

- OR -

23-70-01-01 VIDEO SURVEILLANCE SYSTEM (Cont'd)

23-70-01-01C PASSENGER CONFIGURATION (C)

RI	NI	NR	PLACARD	O	M
D	1	0	NO	NO	NO

May be inoperative provided that procedures do not require its use.

23-70-01-02 VIEWING PORTS

23-70-01-02A Passenger Configuration (A)

RI	NI	NR	PLACARD	O	M
A	1	0	YES	YES	NO

May be inoperative provided that:

(a) Alternate procedures are established and used.
(b) Cabin/Flight Deck Interphone system is checked operative, and
(c) Repairs are made within three flight days.

OPERATIONAL (O) PROCEDURES

Procedures to exit Flight Deck during Flight

Prior to unlocking the door the flight deck crew shall:

1) Contact cabin crew advising of visit to lavatory.
2) Cabin attendant in charge verifies vacancy of lavatory and locks its.
3) Cabin attendant in charge accesses the key to cabin-to-cargo door and makes way to the front of cabin opens doors to cargo area and locks door behind them.
4) No. 2 cabin attendant contacts flight deck via interphone advises that cabin crew member is positioned outside the cockpit door and gives the password.
5) Flight crew ensure clear zone is secure (by looking through peephole) and exits cabin crew hands over the key to the flight crew member exiting the flight deck, enters flight deck compartment and secures the door.
6) Flight crew uses peep-hole to ensure area in passenger cabin is secure and opens cabin-to cargo door from inside and locks door (keeping the key in their possession).

Procedures to re-enter Flight Deck during Flight

1) No. 2 cabin attendant notifies flight deck that flight crew is ready to re-enter.
2) Flight crew opens cabin to cargo door using the key and locks it behind them.
3) No. 2 cabin attendant contacts flight deck via interphone, notified them that the flight crew is in position to re-enter and gives the password.
4) Cabin crew occupying flight deck ensure clear zone is secure by looking through the peep hole and opens flight deck compartment door, flight crew hand over the key re-enters and secures cockpit door.
5) No. 2 cabin attendant ensures outside cabin to cargo door is secure and exits, returns to rear position and replaces the key.

- **During preliminary cockpit preparation**
  - ► CABIN/FLIGHT DECK INTERPHONE SYSTEM ..... CHECK
  - ► COCKPIT ACCESS PROCEDURES ..... DEFINE AND BRIEF CREW

- OR -

23-70-01-02B Passenger Configuration (B)

RI	NI	NR	PLACARD	O	M
C	1	0	YES	YES	NO

May be inoperative provided that:

(a) Flight Deck Video Surveillance System is installed and operates normally,
(b) Cabin/Flight Deck Interphone system is checked operative, and
(c) Alternate procedures are established and used.

OPERATIONAL (O) PROCEDURES

Procedures to exit Flight Deck during Flight

Prior to unlocking the door the flight deck crew shall:

1) Contact cabin crew advising of visit to lavatory.
2) Cabin attendant in charge verifies vacancy of lavatory and locks its.
3) Cabin attendant in charge accesses the key to cabin-to-cargo door and makes way to the front of cabin opens doors to cargo area and locks door behind them.
4) No. 2 cabin attendant contacts flight deck via interphone advises that cabin crew member is positioned outside the cockpit door and gives the password.
5) Flight crew ensure clear zone is secure (by looking through peephole) and exits cabin crew hands over the key to the flight crew member exiting the flight deck, enters flight deck compartment and secures the door.
6) Flight crew uses peep-hole to ensure area in passenger cabin is secure and opens cabin-to cargo door from inside and locks door (keeping the key in their possession).

Procedures to re-enter Flight Deck during Flight

1) No. 2 cabin attendant notifies flight deck that flight crew is ready to re-enter.
2) Flight crew opens cabin to cargo door using the key and locks it behind them.
3) No. 2 cabin attendant contacts flight deck via interphone, notified them that the flight crew is in position to re-enter and gives the password.
4) Cabin crew occupying flight deck ensure clear zone is secure by looking through the peep hole and opens flight deck compartment door, flight crew hand over the key re-enters and secures cockpit door.
5) No. 2 cabin attendant ensures outside cabin to cargo door is secure and exits, returns to rear position and replaces the key.

- **During preliminary cockpit preparation**
  - ► CABIN/FLIGHT DECK INTERPHONE SYSTEM.....CHECK
  - ► COCKPIT ACCESS PROCEDURES .....DEFINE AND BRIEF CREW

- OR -

23-70-01-02C Passenger Configuration (C)

RI	NI	NR	PLACARD	O	M
D	1	0	NO	NO	NO

May be inoperative provided that procedures do not require its use.

23-71-01 COCKPIT VOICE RECORDER

RI	NI	NR	PLACARD	O	M
A	1	0	YES	NO	NO

One or more may be inoperative provided that:

(a) Operations are limited to eight further consecutive flights or 72 hrs whichever occurs first, with the cockpit voice recorder unserviceable, and

(b) Any flight data recorder required to be carried is operative.

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24-21-01 AC INVERTER

RI	NI	NR	PLACARD	O	M
A	2	1	YES	YES	NO

Inverter #2 only may be inoperative, provided that:
(a) The flight is conducted in day VMC only, and
(b) AP use is prohibited below 1000 ft AGL, and
(c) Operations are limited to not more than two calendar days, and
(d) TLU manual mode is operative.

OPERATIONAL (O) PROCEDURES
Check of TLU manual mode:

- **During cockpit preparation**
  - ▶ Select manually HI SPD and check the following (after approximately 30 s):
    - Alert is generated: master caution, single chime, FLT CTL TLU on FWS and TLU FAULT light
    - TLU LO SPEED label disappeared
    - Rudder travel is limited.
  - ▶ Select LO SPD and check the following (after approximately 30 s):
    - Alert stops
    - TLU LO SPEED label is displayed
    - Rudder travel is no longer limited.
  - ▶ Select AUTO.

24-21-02 AC INV FAULT LIGHT

RI	NI	NR	PLACARD	O	M
C	2	0	YES	YES	NO

May be inoperative provided that affected inverter is checked operative prior to each departure.

OPERATIONAL (O) PROCEDURES

- **After engines start**
  - ▶ DC BT pb.....ISOL
  - ▶ Check no amber light or flag on instruments come on.
  - ▶ DC BTC pb.....NORM

24-21-03 AC GENERATION BUS TIE RELAY (BTR)

RI	NI	NR	PLACARD	O	M
-	1	0	YES	NO	NO

May be inoperative provided that DC generation Bus Tie Contactor is considered inoperative.

Refer to Item 24-32-02 DC Generation Bus Tie Contactor (BTC)

24-21-04 AC BUS OFF LIGHT

RI	NI	NR	PLACARD	O	M
C	2	0	YES	NO	NO

One or both may be inoperative.

24-22-01 ACW GENERATION (GENERATOR AND (OR) RELATED GCU)

RI	NI	NR	PLACARD	O	M
B	2	1	YES	YES	YES

One may be inoperative provided that:
(a) HYD AUX pump is operative, and
(b) Two engines taxi is performed, and
(c) Aircraft is not operated on narrow runways (width < 30 m (98 ft)), and
(d) Aircraft is not operated into known or forecast icing conditions, and
(e) AFM penalties are applied, and
(f) Ground operations above 8500 ft (when authorized), including taxi, do not exceed 10 minutes.

OPERATIONAL (O) PROCEDURES
Refer to AFM – DEVIATION GUIDE SUPPLEMENTS Dispatch with One ACW Generator Channel Inoperative.

MAINTENANCE (M) PROCEDURES
This procedure has to be performed before the next flight following the ACW generator channels fault triggering to detect a mechanical damage (bearing failure…)

► Perform a visual inspection of the failed alternator:
  ► Look for overheating and/or presence of debris/chips in the alternator-cooling outlet.
  ► Check for unusual noise from the alternator by manually rotating the propeller.

■ **If check satisfactory:**
  ► Repeat this inspection on a daily basis until the ACW generator failure is repaired.

■ **If check not satisfactory:**
  ► Replace the failed ACW generator before flight.

24-22-02 ACW GEN FAULT LIGHT

RI	NI	NR	PLACARD	O	M
C	2	1	YES	NO	NO

One may be inoperative provided that ACW BTC operates normally.

24-22-03 ACW GENERATION BUS TIE CONTACTOR (BTC)

RI	NI	NR	PLACARD	O	M
B	1	0	YES	YES	NO

May be inoperative in the ISOL position provided that:
(a) Both ACW generator channels are operative, and
(b) HYD X FEED pb is operative and is selected ON before takeoff, then OFF during cruise and reselected ON before landing.

OPERATIONAL (O) PROCEDURES

- ► Open hydraulic X FEED before takeoff.
- ► Close it after flaps retraction.
- ► Open it in approach.
- ► Close it after landing.

NOTE: *Two-engine taxi is recommended.*

24-22-04 ACW BUS OFF LIGHT

RI	NI	NR	PLACARD	O	M
C	2	0	YES	NO	NO

One or both may be inoperative.

24-30-01 DC GENERATION (GENERATOR AND (OR) RELATED GCU)

RI	NI	NR	PLACARD	O	M
A	2	1	YES	YES	NO

May be inoperative provided that:
(a) TRU is checked operative prior to each departure, and
(b) Two engines taxi is performed, and
(c) Aircraft does not line up until 6 mins elapsed after operative generator comes on line, and
(d) Operations are limited to two calendar days, and
(e) Aircraft is not dispatched from Piarco.
←

OPERATIONAL (O) PROCEDURES

Check of remaining generator feeder integrity:

6 mins are necessary to establish the temperature difference between the feeders if one line is broken.
So aircraft should not line up until 6 mins elapsed after operative generator comes on line.

Operational test of TRU:

During the test, the ACW electrical network must be available (The AC GPU connected or ACW generators running)

► Checking are performed only on MAIN ELEC PWR panel (29VU)
  ► Switch OFF DC EXTERNAL POWER pb (if selected ON)
  ► Switch OFF DC GEN 1 pb & DC GEN 2 pb (if selected ON).

On MAIN ELEC PWR panel

► Check that the following caution lights are switched ON:
  1) BATTERY ARROWS,
  2) AC BUS 1 OFF and AC BUS 2 OFF,
  3) INVERTER 2 FAULT,
  4) Both DC BUS OFF,
  5) SHED LEGEND OF DC SVCE/UTLY BUS

► Check that the following systems are supplied:
  - VHF 1,
  - FUEL QTY Indicator,
  - FLAPS POSITION indicator

24-30-01 DC GENERATION (GENERATOR AND (OR) RELATED GCU)
(Cont'd)

On MAIN ELEC PWR panel

► Press the TRU pb: then check that:
- The TRU pb is switched ON,
- The TRU ARROW caution light is switched ON,
- The BATTERIES ARROWS caution lights are not illuminated,
- The UNDV legend of OVRD/UNDV pb is not illuminated.

NOTE: *If the test is performed with engine running, IAS FAIL flag is triggered on Capt. PFD as soon as DC Generators are selected OFF and it remains displayed as long as TRU is selected ON.*

► Check that the BATTERY AMMETER shows zero load either if BAT sw is placed in EMER or MAIN position.

► Check that the following systems are still supplied:
- VHF 1,
- FUEL QTY Indicator,
- FLAPS POSITION indicator

On MAIN ELEC PWR panel

► switch off the TRU pb; then check that:
- The TRU pb is not illuminated,
- The TRU ARROW caution light is not illuminated,
- The BATTERY ARROWS caution lights are switched ON.

NOTE: *After the test, the flight crew must set operative DC GEN to ON Position.*

Set Landing Elevation.

24-30-02 DC TRANSFORMER RECTIFIER UNIT (TRU)

RI	NI	NR	PLACARD	O	M
B	1	0	YES	NO	NO

May be inoperative.

24-32-01 DC GEN FAULT LIGHT

RI	NI	NR	PLACARD	O	M
C	2	1	YES	NO	NO

One may be inoperative provided that DC Bus Tie Contactor is operative.

24-32-02 DC GENERATION BUS TIE CONTACTOR (BTC)

RI	NI	NR	PLACARD	O	M
B	1	0	YES	YES	NO

May be inoperative in the ISOL position provided that:
(a) Both DC generator channels are operative, and
(b) DC BUS OFF lights are operative, and
(c) Both inverters are operative.

OPERATIONAL (O) PROCEDURES

NOTE: *Two-engine taxi is recommended.*

CAUTION

► While aircraft is on ground, If LH DC Generator Fails, then automatic pressurization digital controller supplied by DC BUS 1 is Lost. Therefore, both outflow valves will close and aircraft will be pressurized. In this case, use manual pressurization to depressurize the aircraft.

24-32-03 DC SVCE & UTLY BUS CONTROL SYSTEM

RI	NI	NR	PLACARD	O	M
C	1	0	YES	NO	NO

May be inoperative in the OFF position.
Refer to Item 33-10-02-01 Cabin Lights – Passenger Configuration.

24-32-04 DC BUS OFF LIGHT

RI	NI	NR	PLACARD	O	M
C	2	0	YES	NO	NO

One or both may be inoperative.

24-41-01 AC EXTERNAL POWER CONNECTION SYSTEM

RI	NI	NR	PLACARD	O	M
C	1	0	YES	NO	NO

May be inoperative.

24-46-01 DC EXTERNAL POWER CONNECTION SYSTEM

RI	NI	NR	PLACARD	O	M
C	1	0	YES	NO	NO

May be inoperative.

24-77-01 ELECTRICAL POWER MEASUREMENT ASSEMBLY (LH MAINTENANCE PANEL)

RI	NI	NR	PLACARD	O	M
C	1	0	NO	NO	NO

May be inoperative.

24-77-01 ELECTRICAL POWER MEASUREMENT ASSEMBLY (LH MAINTENANCE PANEL)

RI	NI	NR	PLACARD	O	M
C	1	0	NO	NO	NO

May be inoperative.

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25-11-01 FLIGHT CREW SEAT - VERTICAL AND RECLINE ADJUSTMENTS

RI	NI	NR	PLACARD	O	M
B	2	0	YES	NO	NO

Vertical and recline adjustments may be inoperative provided that:

(a) Associated seat is secured or locked in a position acceptable to the flight crew member, and

(b) Horizontal adjustment is operative for each flight crew member.

NOTE: *A seat with an inoperative or missing seat belt or harness is considered inoperative.*

25-11-02 FLIGHT DECK OBSERVER SEAT

RI	NI	NR	PLACARD	O	M
D	1	0	YES	YES	YES

May be inoperative provided that the seat is not occupied and correctly stowed or removed.

NOTE: *A seat with an inoperative or missing seat belt or harness is considered inoperative.*

OPERATIONAL (O) PROCEDURES

When computing the operational CG envelope or load and trim sheet, the CG effect resulting from removal of the flight deck observer seat shall be taken into account in accordance with the applicable approved weight and balance data.

MAINTENANCE (M) PROCEDURES

For removal, Refer to MP ATR-A-25-11-10-01ZZZ-520Z-A.

25-11-03 FLIGHT CREW SEAT - ARMREST

RI	NI	NR	PLACARD	O	M
C	4	-	YES	NO	YES

May be inoperative provided that:

(a) Associated armrest is secured in stowed position or removed, as acceptable to the flight crew member,

(b) Flight crew seat horizontal adjustment is operative for each flight crew member, and

(c) Flight crew seat vertical and recline adjustments are operative for each flight crew member.

MAINTENANCE (M) PROCEDURES

Secure armrest in stowed position or remove armrest per the applicable CMM instructions.

For removal, Refer to MP ATR-A-25-11-10-A0ZZZ-520Z-A.

25-13-11 SUNVISORS

RI	NI	NR	PLACARD	O	M
D	2	0	NO	NO	NO

May be inoperative or missing provided there are no visual restrictions to the flight crew.

25-13-15 SEAT POSITION SIGHT GAUGE (EYE LOCATOR)

RI	NI	NR	PLACARD	O	M
C	1	0	NO	NO	NO

May be missing.

25-13-17 CHART HOLDER

25-13-17A (A)

RI	NI	NR	PLACARD	O	M
C	4	2	YES	NO	NO

May be inoperative or missing provided one is available in each station.

- OR -

25-13-17B (B)

RI	NI	NR	PLACARD	O	M
C	4	2	YES	YES	NO

May be inoperative or missing provided that alternate procedures are established and used.

OPERATIONAL (O) PROCEDURES
Use of knee pad.

25-21-01 REQUIRED CABIN ATTENDANT SEAT

RI	NI	NR	PLACARD	O	M
B	2	1	YES	YES	YES

One seat or seat assembly may be inoperative, provided that:

(a) Inoperative seat or seat assembly is not occupied, and

(b) Cabin attendant displaced by inoperative seat occupies the adjacent cabin attendant seat or the passenger seat most suitable to perform assigned duties, and

(c) Passenger Address is operative and used to communicate with displaced cabin attendant, and

(d) One cabin attendant occupies a seat with a direct access to the communication system, and

(e) Folding type seat is stowed and secured in the retracted position, and

(f) Where a passenger seat is assigned to the displaced cabin attendant it is placarded "FOR CABIN ATTENDANT USE ONLY", and

(g) Alternate normal, abnormal and emergency procedures are established/ approved and used for displaced cabin attendant.

NOTE: *1) A seat with an inoperative or missing seat belt or harness is considered inoperative.*

2) This requirement does not preclude use of passenger seats by cabin attendants carried in excess of the required cabin attendant.

3) #2 1 J/S (AFT) must be operable for dispatch.

OPERATIONAL (O) PROCEDURES

An Unserviceable Jumpseat may be inoperative provided:

- The affected seat is not occupied and
- Is placarded "INOPERATIVE"
- The seat must be stowed in the retracted position
- The passenger seat assigned to the crewmember is placarded "FOR CABIN CREW USE ONLY"
- Each flight is load restricted as a result of crewmember occupying a passenger seat.
- The cabin crewmember assigned to the affected seat must occupy a passenger seat closest to the associated exit.
- The cabin crewmember must be able to reach the assigned emergency exits in essentially the same time as from the normally assigned seat i.e. a 2 or 3 second time difference is considered "essentially the same time".

NOTE: *1) A folding type seat that will not stow automatically is considered inoperative.*

2) A cabin crew seat position with an inoperative lap belt is considered inoperative.

25-21-01 REQUIRED CABIN ATTENDANT SEAT *(Cont'd)*

MAINTENANCE (M) PROCEDURES

Placard Cabin Attendant Seat.

Folding Cabin Attendant's Seat must be secured in the retracted (up) position or be removed as per ATA A-25-21-20.

25-21-02 NON-REQUIRED CABIN ATTENDANT SEAT

RI	NI	NR	PLACARD	O	M
C	2	0	YES	YES	YES

One seat or seat assembly may be inoperative, provided that:
(a) Inoperative seat or seat assembly is not occupied, and
(b) Alternate normal, abnormal and emergency procedures are established/ approved and used for displaced cabin attendant, and
(c) Folding type seat is stowed and secured in the retracted position, and
(d) Where a passenger seat is assigned to the displaced cabin attendant it is placarded: FOR CABIN ATTENDANT USE ONLY.

NOTE: *A seat with an inoperative or missing seat belt or harness is considered inoperative.*

Placarding Procedure
Inoperative Cabin Attendant Seat must be placarded.

OPERATIONAL (O) PROCEDURES
An Unserviceable Jumpseat may be inoperative provided:
- The affected seat is not occupied, and
- Is placarded "INOPERATIVE"
- The seat must be stowed in the retracted position.
- The passenger seat assigned to the crewmember is placarded "FOR CABIN CREW USE ONLY".
- Each flight is load restricted as a result of crewmember occupying a passenger seat.
- The cabin crewmember assigned to the affected seat must occupy a passenger seat closest to the associated exit.
- The cabin crewmember must be able to reach the assigned emergency exits in essentially the same time as from the normally assigned seat i.e. a 2 or 3 second time difference is considered "essentially the same time".

NOTE: *1) A folding type seat that will not stow automatically is considered inoperative.*
2) A cabin crew seat position with an inoperative lap belt is considered inoperative.

MAINTENANCE (M) PROCEDURES
1. Placard Cabin Attendant Seat.
2. Folding Cabin Attendant's Seat must be secured in the retracted (up) position.

NOTE: *Seat pan must be secured in retracted position.*

25-31-01-01 PASSENGER SEATS

RI	NI	NR	PLACARD	O	M
D	-	-	YES	NO	YES

One or more may be inoperative provided that:
(a) Inoperative seat does not block an emergency exit, and
(b) Inoperative seat does not restrict any passenger from access to the main aircraft aisle, and
(c) Affected seat(s) are blocked and placarded "DO NOT OCCUPY", and
(d) Any damage is checked not detrimental to adjacent passenger safety.

NOTE: *A seat with an inoperative or missing seat belt is considered inoperative.*

MAINTENANCE (M) PROCEDURES

1. Placard affected Passenger Seat "DO NOT OCCUPY".
2. Block off inoperative Passenger Seat if it cannot be secured in the upright position, in order to prohibit its use. This may necessitate blocking off aft seat row if passenger access to main aircraft aisle is restricted.
3. Make appropriate entry in the Aircraft Technical Log.

Identify the affected seat(s) and prohibit the use of the affected seat(s) by a placard "DO NOT OCCUPY".
Visual inspection is performed to check for any apparent sharp edge.

25-31-01-02 PASSENGER SEATS – UNDERSEAT BAGGAGE RESTRAINING BARS

RI	NI	NR	PLACARD	O	M
D	-	-	YES	YES	YES

May be inoperative or missing provided that:
(a) Baggage is not stowed under associated seats, and
(b) Associated seats are placarded "DO NOT STOW BAGGAGE UNDER THIS SEAT", and
c) Procedures are established and used to alert cabin crew of inoperative restraining bars, and
d) Any damage is checked not detrimental to adjacent passenger safety.

OPERATIONAL (O) PROCEDURES
Brief cabin crew about affected seats position.

MAINTENANCE (M) PROCEDURES
Visual inspection is performed to check for any apparent sharp edge.

25-31-01-03 PASSENGER SEATS – PASSENGER SEATS ARMRESTS

RI	NI	NR	PLACARD	O	M
D	-	-	NO	NO	YES

May be inoperative or damaged and the affected seat occupied provided that:
(a) The affected armrest does not block an emergency exit, and
(b) The affected armrest is not in such a position that it restricts any passengers from access to the aircraft aisle, and
(c) Any damage is checked not detrimental to adjacent passenger safety, and
(d) Recline function (if installed) is checked operative.

MAINTENANCE (M) PROCEDURES
Recline function is checked operative.
Visual inspection is performed to check for any apparent sharp edge.

25-31-01-04 PASSENGER SEATS – RECLINE FUNCTION

RI	NI	NR	PLACARD	O	M
D	-	-	YES	NO	YES

One or more may be inoperative and the affected seat occupied provided that:

(a) The seat is secured in the takeoff and landing position, and

(b) Any damage to passenger seats and components must not be detrimental to passenger safety.

MAINTENANCE (M) PROCEDURES

Refer to ATR A-25-21-20-00002-560A-A.

Deactivation of Recline Jack on "Light Weight" seat.

25-33-01 GALLEY WASTE COMPARTMENT FLAPPER DOOR

RI	NI	NR	PLACARD	O	M
C	-	0	YES	YES	NO

May be inoperative provided that:
(a) Container is empty, and
(b) Container access is secured to prevent waste introduction into compartment.

OPERATIONAL (O) PROCEDURES
Place Unserviceable Equipment Placard on Door.

25-36-01 NON-ESSENTIAL EQUIPMENT AND FURNISHINGS

RI	NI	NR	PLACARD	O	M
D	21	-	NO	YES	NO

May be inoperative, damaged, or missing provided that the item(s) is deferred in accordance with the NEF Deferral Program.

NOTE: *1) Refer to Section 0: NEF.0.13-1 for NEF Deferral Program.*
2) Refer to Section 0: NEF.0.13-1 for Definition of category.
3) NEF LIST:
1) COCKPIT SUNVISOR
2) COCKPIT ROLLER BLIND
3) READING LIGHT
4) HOT JUG
5) BOARDING MUSIC
6) GALLEY WORKTABLE
7) CURTAIN
8) CABIN WINDOW BLIND
9) CARPET
10) COCKPIT CUP HOLDER
11) PASSENGER MEAL TRAY AND LATCH
12) CABIN MANAGEMENT SYSTEM (CMS)
13) PASSENGER SEAT ARMREST HEIGHT ADJUSTMENT
14) PASSENGER SEAT COVER AND CUSHION
15) INTERIOR COSMETIC ITEM (PAINTING, DECORATION, etc)
16) TOILET SEAT COVER / BUMPER
17) TOILET MIRROR
18) CABIN OVHD STOWAGE DOOR LATCH / ACTUATOR
19) PASSENGER SEAT FAIRING / LINING
20) EFB
21) COCKPIT LINING / FAIRING / COVER

25-40-01 CABIN FLASHLIGHT

RI	NI	NR	PLACARD	O	M
C	4	3	NO	YES	NO

May be inoperative or removed provided:

(a) Crewmember assigned to the affected position has an equivalent operative flashlight readily available.

(b) Inoperative flashlight remains in a certified location or is removed from the aircraft.

OPERATIONAL (O) PROCEDURES

NOTE: *2 required for dispatch in the cabin.*
1 required for dispatch in the cockpit as per proviso (a).

25-41-01 EXTERIOR LAVATORY DOOR ASHTRAY

RI	NI	NR	PLACARD	O	M
D	-	0	YES	NO	NO

May be inoperative or missing provided that flight is non-smoking.

25-41-02 INTERIOR LAVATORY DOOR ASHTRAY

25-41-02A (A)

RI	NI	NR	PLACARD	O	M
C	-	0	YES	NO	NO

May be inoperative or missing provided that:

(a) Lavatory smoke detection system is operative,

(b) Toilet waste receptacle fire extinguishing system is operative, and

(c) Flight is non-smoking.

25-45-01 PORTABLE PROTECTIVE BREATHING EQUIPMENT (IF APPLICABLE)

RI	NI	NR	PLACARD	O	M
D	2	2	YES	YES	YES

Any in excess of that required may be inoperative or missing provided that:

(a) Required distribution of operative PBE units is maintained throughout the aircraft, and

(b) The inoperative PBE unit and its installed location is placarded inoperative, and

(c) The inoperative PBE unit is removed from the aircraft, or remains in an approved stowage, and

(d) Procedures are established and used to alert crew members of inoperative or missing equipment.

NOTE: *2 required for dispatch.*

NOTE: *PBE unit which cannot be stowage may be subject to the requirements of Dangerous Goods regulations.*

OPERATIONAL (O) PROCEDURES

1) Maintain required distribution of operative units throughout the aircraft TTCAR 7.(47).
2) Prior to flight, the pilot in command shall brief cabin crew on alternate emergency procedures that will be used.

MAINTENANCE (M) PROCEDURES

1) One serviceable unit must be provided for each cabin attendant.
2) Ensure inoperative PBE units are removed from the cabin prior to dispatch.
3) Placard removed/missing PBE unit at the storage location.
4) Make appropriate entry in Aircraft Technical Log.

25-45-21 LAVATORY WASTE COMPARTMENT FLAPPER DOOR

RI	NI	NR	PLACARD	O	M
C	-	0	YES	NO	NO

May be inoperative provided that:
(a) Toilet waste receptacle is empty, and
(b) Toilet door is closed locked and placarded, "INOPERATIVE DO NOT ENTER".

NOTE: *Toilet may be used and/or inspected by crew members.*

25-54-23 CARGO RESTRAINT SYSTEM

25-54-23A (A)

RI	NI	NR	PLACARD	O	M
C	4	-	YES	NO	YES

May be inoperative or missing provided acceptable cargo loading limits from Weight and Balance Manual are observed.

MAINTENANCE (M) PROCEDURES
Ensure that Cargo Restraint System is secured and not protruding as to obstruct with Flight Crew or Passengers.

- OR -

25-54-23B (B)

RI	NI	NR	PLACARD	O	M
D	4	-	YES	NO	NO

May be inoperative or missing provided relevant section of the cargo compartment remains empty.

25-60-01-01 AUTOMATIC EMERGENCY LOCATOR TRANSMITTER

25-60-01-01C (C)

RI	NI	NR	PLACARD	O	M
A	1	0	YES	YES	YES

May be inoperative for 6 flights or 25 flight hours whichever occurs first provided alternate procedures are established or used.

OPERATIONAL (O) PROCEDURES
Any in excess of those required regulations.

MAINTENANCE (M) PROCEDURES
Portable ELT must be installed and serviceable.

25-61-01 FIRST AID KITS

RI	NI	NR	PLACARD	O	M
D	2	1	YES	NO	NO

Any in excess of those required may be incomplete or missing.

25-62-01 MEGAPHONES

RI	NI	NR	PLACARD	O	M
A	1	0	YES	YES	YES

May be inoperative for one flight day provided:

(a) Inoperative megaphone(s) is/are removed from the passenger cabin and its location placarded "INOPERATIVE", or it is removed from installed location, secured out of sight and the megaphone and its installed position is placarded "INOPERATIVE",

(b) Required distribution is maintained, and

(c) Procedures are established to alert crewmembers of inoperative or missing megaphones.

OPERATIONAL (O) PROCEDURES

To alert crew members of inoperative or missing megaphones, an oral briefing will be done inflight by cabin crew. Refer: Safety and Emergency Procedures Manual (SEP) Doc. No.: BW-OM-0008, Chapter 7.

MAINTENANCE (M) PROCEDURES

1. Placard inoperative megaphone.
2. Make appropriate entry in the Aircraft Technical Log.

25-64-01 CRASH AXES OR CROWBARS

RI	NI	NR	PLACARD	O	M
D	1	1	NO	NO	NO

Any in excess of those required by TTCAR 7(43).

25-64-02 LIFEJACKETS

RI	NI	NR	PLACARD	O	M
-	-	-	NO	NO	NO

As required by regulations.

NR – One for every passenger onboard

NI – One for every seat onboard and spares for infants and adults as per regulations.

25-65-10 FLIGHT CREW FIREPROOF GLOVES

RI	NI	NR	PLACARD	O	M
D	3	1	NO	NO	NO

Any in excess of one pair may be inoperative or missing, provided that:

(a) At least one serviceable pair of fireproof gloves is available in the cockpit and verified operative prior to flight, and

(b) No operating procedures require the use of the inoperative or missing pair.

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26-12-01 FIRE DETECTION LOOP

RI	NI	NR	PLACARD	O	M
C	4	3	YES	NO	NO

One may be inoperative provided that the affected loop is selected OFF.

26-12-02 FUEL SO LIGHT (INTEGRATED IN CL)

RI	NI	NR	PLACARD	O	M
C	2	0	YES	NO	NO

One or both may be inoperative

26-12-03 ENG 2 NAC OVHT

RI	NI	NR	PLACARD	O	M
C	1	0	YES	NO	YES

May be inoperative provided that hotel mode operation is limited to a maximum tailwind component of 10 kt.

MAINTENANCE (M) PROCEDURES

Open, secure and tag C/B ENG IND/NAC 2 OVHT DET.

26-12-04 ENG 1 NAC OVHT *(IF INSTALLED)*

RI	NI	NR	PLACARD	O	M
C	1	0	YES	NO	NO

May be inoperative.

26-15-01-01 CARGO SMOKE ALERT – PASSENGER CONFIGURATION

RI	NI	NR	PLACARD	O	M
C	-	0	YES	NO	NO

May be inoperative provided that no combustible material is carried in the **associated** compartment.

26-17-01-01 TOILET SMOKE DETECTION SYSTEM (IF INSTALLED) – PASSENGER CONFIGURATION

26-17-01-01A (A)

RI	NI	NR	PLACARD	O	M
C	1	0	YES	YES	YES

May be inoperative provided that:
(a) Toilet waste receptacle is empty,
(b) Toilet door is closed locked and "INOPERATIVE DO NOT ENTER", placard is installed, and
(c) Toilet is not used for storage of any inflammable or combustible materials.

NOTE: *Toilet may be used and/or inspected only by crew members.*

OPERATIONAL (O) PROCEDURES

- **In case of Toilet Smoke Detection System (if installed) – Passenger Configuration**
  Ensure lavatory is visited regularly by the cabin crew during the entire flight duration and the occasional usage of toilet by passenger.

MAINTENANCE (M) PROCEDURES

To provide instructions to verify/test the agent bottle of the lavatory waste receptacle fire-extinguishing system.
Refer to MP ATR-A-26-25-20-00ZZZ-281Z-A.

- OR -

26-17-01-01B (B)

RI	NI	NR	PLACARD	O	M
B	1	0	NO	YES	YES

May be inoperative provided that:
(a) Toilet waste receptacle fire-extinguishing system is verified operative,
(b) Lavatory Waste Compartment Flapper Door is verified operative, and
(c) AFT SMOKE DETECTION FANS are verified operative.
(d) Procedures are established and used to check periodically absence of smoke in affected lavatory.

NOTE: *Toilet may be used and/or inspected only by crew members.*

OPERATIONAL (O) PROCEDURES

- **In case of Toilet Smoke Detection System (if installed) – Passenger Configuration**
  Ensure lavatory is visited regularly by the cabin crew during the entire flight duration and the occasional usage of toilet by passenger.

MAINTENANCE (M) PROCEDURES

To provide instructions to verify/test the agent bottle of the lavatory waste receptacle fire-extinguishing system.
Refer to MP ATR-A-26-25-20-00ZZZ-281Z-A.

26-18-01 AFT CARGO (CURTAIN – DOOR) *(IF INSTALLED)*

RI	NI	NR	PLACARD	O	M
C	1	0	YES	NO	NO

May be inoperative provided that no combustible material is carried in the associated compartment.

26-19-01 AFT SMOKE DETECTION FANS

26-19-01A (A)

RI	NI	NR	PLACARD	O	M
C	2	1	NO	NO	NO

One may be inoperative provided that the remaining fan is checked operative before each departure.

- OR -

26-19-01B (B)

RI	NI	NR	PLACARD	O	M
C	2	0	YES	YES	YES

One or both may be inoperative provided that:
(a) Toilet waste receptacle is empty, and
(b) Toilet door is closed locked and "INOPERATIVE DO NOT ENTER" placard is installed, and
(c) No combustible material is carried in the AFT cargo compartment.

NOTE: *Toilet may be used and/or inspected only by crew members.*

OPERATIONAL (O) PROCEDURES
The Captain must:
1) Ensure toilet waste receptacle is empty.
2) Ensure toilet door is closed locked with "INOPERATIVE DO NOT ENTER" placard.
3) Ensure no combustible material is carried in the AFT cargo compartment.
4) Ensure that all crew members are briefed regarding the use of the toilet.

MAINTENANCE (M) PROCEDURES
To provide instructions close, lock and placard the inoperative lavatory door.

26-21-01 AGENT DISCH LIGHT (ENGINE)

RI	NI	NR	PLACARD	O	M
A	4	2	YES	NO	YES

One or two may be inoperative provided that:

(a) Affected light(s) are not on the same engine fire panel,

(b) Respective extinguisher bottle pressure is checked on the direct reading gauge prior to first flight of each day, and

(c) Operations are limited to two calendar days.

NOTE: *AGENT 1 DISCH on left engine and AGENT 2 DISCH on right engine are associated with Fire Extinguisher Bottle 1.*
AGENT 2 DISCH on left engine and AGENT 1 DISCH on right engine are associated with Fire Extinguisher Bottle 2.

MAINTENANCE (M) PROCEDURES *(CAL)*

Refer to MP ATR-A-26-21-XX-00ZZZ-362Z-A.

26-25-00 LAVATORY FIRE EXTINGUISHER

RI	NI	NR	PLACARD	O	M
C	-	0	YES	YES	YES

May be inoperative provided:
(a) Lavatory waste receptacle is empty,
(b) Associated lavatory door is locked, closed and placarded to prohibit passengers from entering,
(c) Toilet is not used for storage of any inflammable or combustible materials,
(d) Affected lavatory is used only by crew members, and
(e) Lavatory Waste Compartment Flapper Door is verified operative.

OPERATIONS (O) PROCEDURES
Toilet should not be used for stowage of any inflammable or combustible materials. Ensure the associated lavatory door remains locked except for use or inspection by crew members.

MAINTENANCE (M) PROCEDURES
Associated lavatory door is locked, closed and placarded inoperative lavatory.

26-30-01 PORTABLE HAND FIRE EXTINGUISHER

RI	NI	NR	PLACARD	O	M
D	4	3	YES	NO	YES

Any in excess of that required by TTCAR 7.40 may be inoperative or missing provided that:

(a) The inoperative hand fire extinguisher is removed from the aircraft and its installed location is placarded inoperative; or it is removed from the installed location, secured out of sight, and the hand fire extinguisher and its installed location are placarded inoperative,

(b) Required distribution of operative units is maintained throughout the aircraft, and

(c) Procedures are established and used to alert crew members of inoperative or missing equipment.

NOTE: *When determining the location for storage of the inoperative units, compliance with the dangerous goods requirements must be considered.*

MAINTENANCE (M) PROCEDURES *(CAL)*

1) A minimum of three (3) units must be serviceable for dispatch.
   - 1 in cockpit
   - 2 in cabin

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27-20-01 RUDDER DAMPER

RI	NI	NR	PLACARD	O	M
C	1	0	YES	YES	NO

May be inoperative provided that for taxi, takeoff and landing:
(a) Maximum crosswind is limited to 15 kt, and
(b) Maximum total wind is limited to 30 kt.

OPERATIONAL (O) PROCEDURES

Pilot Flying and **Pilot Not Flying** must keep their feet on the pedals during all the taxi phases in order to avoid rudder excessive movements.

Pilot Flying will avoid large pedal movements during takeoff run and landing and will prefer nose wheel steering use.

27-20-02 RUDDER RELEASABLE CENTERING UNIT (RCU)

27-20-02A Yaw Damper Function Inoperative (A)

RI	NI	NR	PLACARD	O	M
-	1	-	YES	YES	NO

Refer to Item 22-18-03 Yaw Damper Function.

OPERATIONAL (O) PROCEDURES

- **If the yaw damper is inoperative or not engaged**
  Pilot Flying will keep his feet on the pedals to be ready to control rudder.

- OR -

27-20-02B Yaw Damper Function Operative (B)

RI	NI	NR	PLACARD	O	M
C	1	0	YES	YES	NO

May be inoperative provided that yaw damper function is operative and used.

OPERATIONAL (O) PROCEDURES

- **If the yaw damper is inoperative or not engaged**
  Pilot Flying will keep his feet on the pedals to be ready to control rudder.

27-23-01 TRAVEL LIMITER UNIT (TLU)

27-23-01A (A)

RI	NI	NR	PLACARD	O	M
C	1	0	YES	NO	NO

May be inoperative in the low speed position provided that speed is limited to **185 kt**.

- OR -

27-23-01B Automatic Mode (B)

RI	NI	NR	PLACARD	O	M
C	1	0	YES	YES	NO

May be inoperative provided that manual mode is operative.

OPERATIONAL (O) PROCEDURES

- **During Cockpit Preparation**
  - ► Select manually HI SPD and check alert is generated after 25 s: master caution, single chime, FLT CTL TLU on FWS and TLU FAULT light. Check rudder travel limited.
  - ► Select LO SPD and check alert stops after approx.30 s and check rudder travel no longer limited.
- **In Flight**
  - ► During an acceleration, select manually HI SPD when reaching 185 kt.
  - ► During a deceleration, select manually LO SPD when reaching 180 kt.

Check of TLU manual mode:

- **During Cockpit Preparation**
  - ► Select manually HI SPD and check the following (after approximately 30 s):
    - Alert is generated: master caution, single chime, FLT CTL TLU on FWS and TLU FAULT light
    - TLU LO SPEED label has disappeared
    - Rudder travel is limited.
  - ► Select LO SPD and check the following (after approximately 30 s):
    - Alert stops
    - TLU LO SPEED label is displayed
    - Rudder travel is no longer limited.

  Do not use AUTO mode.

27-23-01 TRAVEL LIMITER UNIT (TLU) *(Cont'd)*

27-23-01B Automatic Mode (B) *(Cont'd)*

OPERATIONAL (O) PROCEDURES *(Cont'd)*

- **Before TAKEOFF**
  - ► Select manually LO SPD.

- **In Flight**
  - ► During an acceleration, select manually HI SPD when reaching 185 kt and check TLU LO SPEED label disappeared.
  - ► During a deceleration, select manually LO SPD when reaching 180 kt and check TLU LO SPEED label is displayed (after approximately 30 s).

NOTE: *A TLU FAULT alert will trigger transitorily after approximately 30 s.*

27-23-02 TLU FAULT LIGHT

RI	NI	NR	PLACARD	O	M
D	1	0	YES	YES	NO

May be inoperative provided that the aural alert and Master Caution and associated FLT CTL TLU alert on FWS are checked operative prior to each flight.

OPERATIONAL (O) PROCEDURES

- **During Cockpit Preparation**
  - ► Select manually HI SPD and check alert is generated after 25 s: master caution, single chime, FLT CTL TLU on FWS. Check rudder travel limited.
  - ► Select LO SPD and check alert stops after approx. 30 s and check rudder travel no longer limited.
- **In Flight**
  - ► During an acceleration, select manually HI SPD when reaching 185 kt.
  - ► During a deceleration, select manually LO SPD when reaching 180 kt.

27-25-01 YAW TRIM POSITION INDICATOR

RI	NI	NR	PLACARD	O	M
A	1	0	YES	NO	NO

May be inoperative provided that:

(a) Yaw trim system operates normally,

(b) Rudder trim tab is visually verified in neutral position before each departure,

(c) Yaw damper and/or autopilot are not used until the aircraft has been fully trimmed, and

(d) Operations are limited to three calendar days.

27-32-01 PITCH TRIM ROCKER SWITCH

RI	NI	NR	PLACARD	O	M
B	2	1	YES	NO	NO

One may be inoperative provided that standby pitch trim is operative.

27-32-02 STANDBY PITCH TRIM

RI	NI	NR	PLACARD	O	M
C	1	0	YES	NO	NO

May be inoperative provided that both pitch trim rocker switch are operative.

27-32-03 FIRST OFFICER PITCH TRIM ROCKER SWITCH COVER

RI	NI	NR	PLACARD	O	M
-	1	0	YES	NO	YES

May be inoperative or missing provided that associated trim is considered inoperative.
Refer to Item 27-32-01 Pitch Trim Rocker Switch.

MAINTENANCE (M) PROCEDURES

► Open, secure and tag C/B R NORM PITCH TRIM.

27-35-01 PITCH TRIM POSITION INDICATOR

RI	NI	NR	PLACARD	O	M
B	1	0	YES	YES	NO

May be inoperative provided that elevator tabs are:
(a) Visually checked operative prior to start up, and
(b) Positioned neutral prior to takeoff.

OPERATIONAL (O) PROCEDURES

- **DURING Taxi Check List**
  - ► Run the trim from nose up to nose down: from the trim position where the whooler stops, run the trim from nose down to nose up for 13 s.

  NOTE: *This procedure sets the pitch trim to a conservative position for any CG inside the operational CG envelope (neutral position).*
  A TOCG in the center of the operational CG envelope is recommended in order to reduce the efforts of the pilot during the rotation.

  NOTE: *When performing TO CONFIG test, verify that CONFIG FLT CTL alert is only due to Pitch Trim Position Indicator by checking that TLU is in LOW SPEED, GUST LOCK is OFF and FLAPS are in Takeoff configuration.*
- **Before Takeoff**
  - ► Cancel MW alarm by pressing MASTER WARN pb.

  NOTE: *CONFIG FLT CTL message remains displayed on the EWD.*

27-36-01 STICK SHAKER ACTUATOR

27-36-01A Ferry Flight (A)

RI	NI	NR	PLACARD	O	M
A	2	1	YES	YES	NO

One may be inoperative for one Ferry Flight provided that MFC modules 1B and 2B are operative.

OPERATIONAL (O) PROCEDURES

- **During Preliminary Cockpit Preparation**
  - ► TCAS ..... TA ONLY

Limitations

- **During Approach**
  - ► IAS .....NOT LESS THAN V_{min} OPS + 10 kt
  - ► SPD TGT .....MAN SEL
  - ► LDG DIST ..... MULTIPLY BY 1.15
- **Below 1 000 ft AGL**
  - ► AP USE: PROHIBITED

27-36-02 STICK PUSHER ACTUATOR

27-36-02A Ferry Flight (A)

RI	NI	NR	PLACARD	O	M
A	1	0	YES	YES	YES

May be inoperative for one Ferry Flight only, provided that MFC modules 1B and 2B are operative.

OPERATIONAL (O) PROCEDURES

- **During Preliminary Cockpit Preparation**
  - ► TCAS.....TA ONLY
  - ► STICK PUSHER/SHAKER pb..... ON
  - ► CONTROL COLUMN & WHEEL.....CHECK FREE MOVEMENT
  - ► STICK PUSHER/SHAKER FAULT LIGHT ..... CHECK OFF

NOTE: *If control column is stuck, refer to Maintenance Procedure (disassembling of stick pusher).*

Refer to Maintenance Procedures 27-36-02 Stick Pusher Actuator (M).

Perform stick shakers test:

► WARN selector..... STICK PUSHER/SHAKER YES

Limitations

- **During Approach**
  - ► IAS ..... NOT LESS THAN V_{min} OPS + 10 kt
  - ► SPD TGT..... MAN SEL
  - ► LDG DIST.....MULTIPLY BY 1.15
- **Below 1 000 ft AGL**
  - ► AP USE: PROHIBITED

MAINTENANCE (M) PROCEDURES

■ **If control column is stuck**

Stick Pusher Deactivation

► Open, secure and tag C/B FLT CTL / STICK PUSHER / PWR SPLY

Stick Pusher Actuator Removal

Refer to MP ATR-A-27-36-30-00ZZZ-520Z-A.

27-36-03 INDICATIONS AND CONTROLS ON MAINTENANCE PANELS

RI	NI	NR	PLACARD	O	M
C	-	0	NO	NO	NO

May be inoperative.

27-37-01 PITCH ELEVATORS COUPLING SYSTEM

RI	NI	NR	PLACARD	O	M
A	1	0	YES	YES	YES

May be inoperative provided that:
(a) Operations are limited to one Ferry Flight, and
(b) Operations are conducted in compliance with applicable AFM Deviation Guide.

OPERATIONAL (O) PROCEDURES
Refer to AFM – DEVIATION GUIDE SUPPLEMENTS Flight with Pitch Elevators Disconnected.

MAINTENANCE (M) PROCEDURES

- **If "PITCH DISC" warning message appear on EWD and pitch coupling system is disconnected in flight**
  - ► Perform visual inspection as per (Refer to MP ATR-A-55-1X-XX-01ZZZ-310Z-A) before next flight.
- **If "PITCH DISC" warning message appear on EWD and pitch coupling system is disconnected on ground.**
  - ► No maintenance action is required before the Ferry Flight.
- **If "PITCH RECONNECT" warning message is flashing on EWD.**
  - ► Deactivate the elevator reconditioning device as per (Refer to MP ATR-A-27-31-40-00ZZZ-560Z-A).

27-37-02 ELEVATORS CLUTCH RECONDITIONING DEVICE

27-37-02A Ferry Flight (A)

RI	NI	NR	PLACARD	O	M
A	1	0	YES	YES	NO

May be inoperative provided that:
(a) PITCH DISC alert is displayed on FWS, and
(b) Operations are limited to one Ferry Flight, and
(c) Operations are conducted in compliance with applicable AFM Deviation Guide.

OPERATIONAL (O) PROCEDURES
Refer to AFM - DEVIATION GUIDE SUPPLEMENTS Flight with Pitch Elevators Disconnected.

- OR -

27-37-02B (B)

RI	NI	NR	PLACARD	O	M
D	1	0	YES	NO	YES

May be inoperative provided that:
(a) The elevator clutch reconditioning device is deactivated, and
(b) Steep Slope Approach is prohibited (> 4.5°).

MAINTENANCE (M) PROCEDURES
Refer to MP ATR-A-27-31-40-00ZZZ-560Z-A.

27-51-01 FLAPS CONTROL SYSTEM

RI	NI	NR	PLACARD	O	M
A	1	0	YES	YES	NO

May be inoperative provided that:
(a) Flaps are checked in retracted position, and
(b) Operations are conducted in compliance with applicable AFM Deviation Guide, and
(c) Operations are limited to two flights.

OPERATIONAL (O) PROCEDURES
Refer to AFM - DEVIATION GUIDE SUPPLEMENTS Dispatch with Flaps Retracted.

27-55-01 FLAPS POSITION INDICATOR

RI	NI	NR	PLACARD	O	M
A	1	0	YES	YES	NO

May be inoperative provided that:
(a) Flaps position visual check is made prior to takeoff and in flight, and
(b) Operations are limited to two calendar days.

OPERATIONAL (O) PROCEDURES

► Check visually flaps position at each configuration change.

27-55-02 FLAPS UNLOCK WARNING

RI	NI	NR	PLACARD	O	M
A	1	0	YES	NO	NO

May be inoperative provided that:

(a) Flaps control system is considered inoperative: **Refer to Item 27-51-01 Flaps Control System,** and

(b) Operations are limited to two flights.

27-61-01 SPOILER CONTROL SYSTEM

RI	NI	NR	PLACARD	O	M
B	1	0	YES	YES	YES

May be inoperative provided that:
(a) Spoilers are secured in retracted position, and
(b) Operations are conducted in compliance with applicable AFM Deviation Guide.

OPERATIONAL (O) PROCEDURES
Refer to AFM - DEVIATION GUIDE SUPPLEMENTS Dispatch with Spoiler Control System Inoperative.

MAINTNENANCE (M) PROCEDURES
Refer to MP ATR-A-27-61-XX-00ZZZ-560Z-A.

27-65-01 SPOILER POSITION INDICATING SYSTEM

27-65-01A (A)

RI	NI	NR	PLACARD	O	M
C	1	0	YES	YES	NO

May be inoperative provided that proper functioning of the spoilers is checked prior to each departure and monitored throughout the flight.

OPERATIONAL (O) PROCEDURES

- **In case of Spoiler Position Indicating System (A)**
  Prior to each departure in appropriate condition for visual spoiler verification.
  ► Proper functioning of the spoilers is checked

- OR -

27-65-01B (B)

RI	NI	NR	PLACARD	O	M
-	1	0	NO	YES	YES

May be inoperative provided that the Spoiler Control System is considered inoperative. **Refer to Item 27-61-01 Spoiler Control System.**

OPERATIONAL (O) PROCEDURES

- **In case of Spoiler Position Indicating System (B)**
  Refer to AFM - DEVIATION GUIDE SUPPLEMENTS Dispatch with Spoiler Control System Inoperative.

MAINTNENANCE (M) PROCEDURES

Refer to MP ATR-A-27-61-XX-00ZZZ-560Z-A.

27-70-01-01 GUST LOCK SYSTEM - AILERON

RI	NI	NR	PLACARD	O	M
C	1	0	YES	YES	YES

May be inoperative provided that the tailwind during parking does not exceed 40 kt.

NOTE: *Propeller brake is not available.*

OPERATIONAL (O) PROCEDURES

Take special care for parking and taxi (wind effects).
To perform the TO CONFIG Test, set the GUST LOCK lever to OFF position.

MAINTENANCE (M) PROCEDURES

Deactivation of Aileron Gust Lock System

NOTE: *This task is to be performed on the both sides.*

- ► Place GUST LOCK lever on "OFF" position.
- ► Open, secure and tag C/B FLT CTL/GUST LOCK AIL.
- ► Display warning notice in the flight compartment prohibiting operation of gust lock control.
- ► Open access panel 533AB (633AB) then 544AB (644AB).
- ► Retract the actuator, if necessary, by means of an ALLEN KEY.
- ► Check the clearance (8 mm +/- 3 mm) between the cam and the rocking lever.
- ► Close C/B FLT CTL/GUST LOCK AIL and make sure that gust lock actuators do not engage inadvertently.
- ► Open, secure and tag C/B FLT CTL/GUST LOCK AIL.
- ► Make certain that working area is clear and clean of tools and miscellaneous items.
- ► Close access doors.
- ► Remove warning notice in the flight compartment.
- ► Place GUST LOCK lever on "ON" position.

NOTE:

► *After 8 s, AIL LOCK LIT Alert will be triggered.*

- ► Rotate PILOT HANDWHEEL to the right to reach aileron stop.
- ► Install tool P/N S0003024900000 or 98S27105002000 between the PILOT HANDWHEEL and the STEERING HANDWHEEL.

NOTE: ► *The tool shall be removed and stored before leaving the parking; the ailerons shall be kept on the stops manually during the taxiing operations.*

Perform a daily inspection of Aileron hinge fitting stops (on RH and LH)

- ► Install notice prohibiting operation of flight controls in flight compartment.
- ► Install access platform.
- ► Remove fairing 555AB or 655AB according to aileron to be inspected.

27-70-01-01 GUST LOCK SYSTEM – AILERON *(Cont'd)*

MAINTENANCE (M) PROCEDURES (Cont'd)

NOTE:

► *LH and RH wings aileron hinge fitting stops must be inspected.*

► Manually move the aileron until reaching the upper stop.

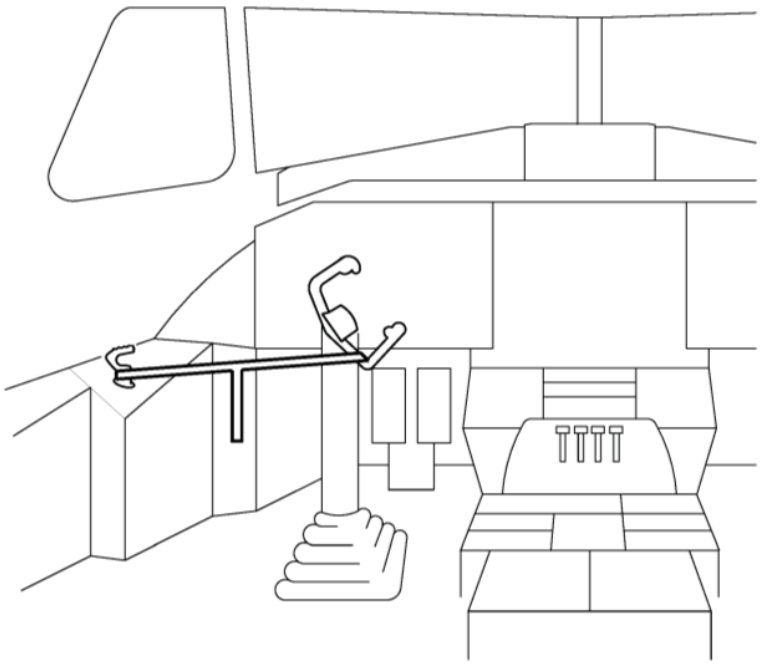
► Perform a Detailed Visual Inspection of aileron center fitting, aileron control fitting and lower stop as per *Refer to NDTM 51-90-00*. In case of findings, contact ATR prior any action.

► Manually move the aileron until reaching the lower stop.

► Perform a Detailed Visual Inspection of aileron center fitting, aileron control fitting and upper stop as per *Refer to NDTM 51-90-00*. In case of findings, contact ATR prior any action.

► Install fairing 555AB or 655AB.

► Remove warning notices and access platform.



27-70-01-02 GUST LOCK SYSTEM - ELEVATOR

RI	NI	NR	PLACARD	O	M
C	1	0	YES	YES	YES

May be inoperative.

NOTE: *Propeller brake is not available.*

OPERATIONAL (O) PROCEDURES

Take special care for parking and taxi (wind effects).

MAINTENANCE (M) PROCEDURES

► Perform a daily inspection of elevator hinge fitting stops (on RH and LH).
- ► Install notice prohibiting operation of flight controls in flight compartment.
- ► Install access platform.
- ► Remove access panels 333CB and 343CB.
- ► Maintain elevators in full nose up position (trailing edge upward).
- ► Perform a Detailed Visual Inspection of elevators inboard hinge fittings and stops as per *Refer NDTM 51-90-00*. In case of findings, contact ATR prior any action.
- ► Install access panels 333CB and 343CB.
- ► Remove warning notices and access platform.

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28-21-01 FEED LO PR LIGHT

RI	NI	NR	PLACARD	O	M
B	2	0	YES	YES	NO

One or both may be inoperative provided that:
(a) FEED LO PR labels are available on SD Engine page, and
(b) FUEL FEED LO PR alert is processed by the FWS, and
(c) Associated pump is operative.

OPERATIONAL (O) PROCEDURES

- **During Cockpit Preparation**
  - ▶ RH and LH PUMPS .....CHECK ON
  - ▶ ON SD ENGINE PAGE, FEED LO PR LABELS…CHECK NOT DISPLAYED
  - ▶ AFFECTED SIDE PUMP ....SELECT OFF
  - ▶ C/B FUEL FEED/AFFECTED SIDE ENGINE / CTL & CAUT.....RESET
  - ▶ ON SD ENGINE PAGE, FEED LO PR LABEL .....CHECK DISPLAYED
  - ▶ ON EWD, FUEL FEED LO PR ALERT.....CHECK DISPLAYED
  - ▶ AFFECTED SIDE PUMP ..... SELECT ON
  - ▶ ASSOCIATED ALERTS.....CHECK NOT DISPLAYED

28-21-02 ELECTRICAL FUEL PUMP

RI	NI	NR	PLACARD	O	M
C	2	1	YES	YES	YES

One may be inoperative provided that:
(a) Fuel X FEED valve is check operative, and
(b) Associated jet pump (and motive flow valve) is operative.
←

OPERATIONAL (O) PROCEDURES

Limitations

Flight planning must take into account engine failure in the inoperative pump side as X FEED would not be possible. Flight time from suitable alternate must not exceed 45 min with one engine operating (elapse time before reaching maxi fuel unbalance).

Maximum certified fuel unbalance must never be exceeded 730 kg

- **During Preliminary Cockpit Preparation**
  - ► FUEL X FEED ..... CHECK OFF
  - ► FUEL PUMP (on affected side).....OFF
  - ► FUEL PUMP (on operative side)..... ON
  - ► ENG FEED LO PR (on operative side) .....CHECK TURN OFF
  - ► ENG FEED LO PR (on affected side) ..... CHECK REMAINS ON
  - ► FUEL X FEED ..... IN LINE
  - ► ENG FEED LO PR (on affected side) .....CHECK TURN OFF
  - ► FUEL PUMP (on operative side).....OFF
  - ► FUEL PUMP (on affected side)..... ON
    (There is no cockpit indication as Circuit breaker of the pump is pulled)
  - ► ENG FEED LO PR 1+2 ..... CHECK TURN ON
    (It is necessary to wait several minutes)
  - ► FUEL X FEED .....X LINE
  - ► FUEL PUMP (on affected side).....OFF
  - ► FUEL PUMP (on operative side).....AS RQRD
- **Before affected engine start**
  - ► FUEL PUMP (on operative side) ..... ON
  - ► FUEL X FEED ..... IN LINE
- **After affected engine start**
  - ► FUEL PUMP (on affected side) ..... ON
    NOTE: *This enables motive flow valve and then jet pump to be operative.*
  - ► FUEL X FEED .....X LINE
  - ► ENG FEED LO PR ..... CHECK REMAINS OFF

28-21-02 ELECTRICAL FUEL PUMP *(Cont'd)*

MAINTENANCE (M) PROCEDURES

Open, secure and tag C/B FUEL / FUEL FEED / ENG 1 (2) ELEC PUMP.

28-21-03 PUMP RUN LIGHT

RI	NI	NR	PLACARD	O	M
C	2	0	YES	NO	NO

One or both may be inoperative provided that associated electrical pumps are operative.

28-21-04 JET PUMP (AND ASSOCIATED MOTIVE FLOW VALVE)

RI	NI	NR	PLACARD	O	M
C	2	1	YES	NO	NO

One may be inoperative provided that:
(a) Fuel X FEED valve is operative, and
(b) Associated electrical pump is operative.

28-21-05 FEEDER TANK JET PUMP

RI	NI	NR	PLACARD	O	M
C	2	1	NO	NO	NO

One may be inoperative provided that:

(a) Fuel X FEED valve is operative,

(b) Both electrical pumps are operative,

(c) Both fuel quantity indication systems are operative, and

(d) Unusable fuel quantity in each tank is raised to 130 kg. Refueling must be done accordingly.

←

NOTE: *The "FUEL LO LVL" may be triggered in case of feeder tank jet pump malfunction on aircraft.*

28-23-01 X FEED VALVE

RI	NI	NR	PLACARD	O	M
C	1	0	YES	YES	NO

May be inoperative provided that:
(a) X FEED valve is checked in closed position, and
(b) Both fuel quantity indications are operative.
←

OPERATIONAL (O) PROCEDURES

Limitations

Flight planning must take into account engine failure as X FEED would not be possible. Flight time from suitable alternate must not exceed 45 min with one engine operating (elapse time before reaching maxi fuel unbalance).

► Dispatch without any asymmetry.

Maximum certified fuel unbalance must never be exceeded 730 kg.

► When selecting RH PUMP ON, check ENG 2 FEED LO PR light turned off and ENG 1 FEED LO PR light remains illuminated.

28-23-02 X FEED FLOW BAR

RI	NI	NR	PLACARD	O	M
-	1	0	YES	YES	NO

May be inoperative provided X FEED Valve is considered inoperative:
Refer to Item 28-23-01 X FEED Valve.

OPERATIONAL (O) PROCEDURES

Limitations

Flight planning must take into account engine failure as X FEED would not be possible. Flight time from suitable alternate must not exceed 45 min with one engine operating (elapse time before reaching maxi fuel unbalance).

► Dispatch without any asymmetry.

Maximum certified fuel unbalance must never be exceeded 730 kg.

► When selecting RH PUMP ON, check ENG 2 FEED LO PR light turned off and ENG 1 FEED LO PR light remains illuminated.

28-25-01-01 REFUELING CONTROL PANEL

RI	NI	NR	PLACARD	O	M
C	1	0	YES	NO	YES

May be inoperative provided that gravity or manual refueling using the handle ring control is performed.

NOTE: *Manual refueling procedure using the handle ring control cannot be used if drainage of the refuel pipe is required (As Refueling Control Panel is inoperative and surge valve cannot be controlled).*

MAINTENANCE (M) PROCEDURES

Gravity Refueling

Refer to MP ATR-A-12-11-28-00ZZZ-211Z-A.

Refueling without the Refuel Control Panel (RCP)

Refer to MP ATR-A-12-11-28-00ZZZ-211Z-A.

NOTE: *Manual refueling procedure using the handle ring control cannot be used if drainage of the refuel pipe is required (As Refueling Control Panel is inoperative and surge valve cannot be controlled).*

28-25-01-02 REFUELING CONTROL PANEL – FUEL PRESELECTOR

RI	NI	NR	PLACARD	O	M
C	1	0	YES	NO	YES

May be inoperative provided that manual refueling is performed.

MAINTENANCE (M) PROCEDURES

Manual Refueling

Refer to MP ATR-A-12-11-28-00ZZZ-211Z-A.

28-25-01-03 REFUELING CONTROL PANEL – PRESELECTED QTY

RI	NI	NR	PLACARD	O	M
C	1	0	YES	NO	YES

May be inoperative provided that manual refueling is performed.

MAINTENANCE (M) PROCEDURES

Manual Refueling

Refer to MP ATR-A-12-11-28-00ZZZ-211-A.

28-25-01-04 REFUELING CONTROL PANEL – REFUELING CONTROL PANEL DOOR INDICATION (ON SD CABIN PAGE)

RI	NI	NR	PLACARD	O	M
C	1	0	NO	YES	YES

May be inoperative provided that:
(a) Refueling Control Panel Door is check closed before each flight, and
(b) Refueling Control Panel is deactivated after each refueling operations.

OPERATIONAL (O) PROCEDURES

Exterior Inspection

► REFUELING CONTROL PANEL DOOR ..... CHECK CLOSED

MAINTENANCE (M) PROCEDURES

Open, secure and tag C/B FUEL / RCP.

28-25-02 PRESSURE REFUELING CAP

RI	NI	NR	PLACARD	O	M
A	1	0	NO	NO	YES

May be inoperative provided that:
(a) Refuel valves are confirmed closed, and
(b) There is no leakage from the refuel coupling, and
(c) Operations are limited to three calendar days.

MAINTENANCE (M) PROCEDURES
Make a note in the Technical Log book to have Refuel/Defuel Valve Cap replaced as per ATR-A-28-50-A0001-720A-A.

28-26-01 LP VALVE POSITION INDICATOR

RI	NI	NR	PLACARD	O	M
C	2	0	YES	NO	NO

One or both may be inoperative.

28-41-01 TANK TEMPERATURE INDICATION

RI	NI	NR	PLACARD	O	M
C	1	0	YES	YES	NO

May be inoperative.

NOTE: *OAT has to be used.*

OPERATIONAL (O) PROCEDURES

NOTE: *Refer to AFM - LIM-5-70-4-2 Fuel Temperature, for temperature limitation.*

28-42-01 FUEL QTY INDICATION

RI	NI	NR	PLACARD	O	M
	2	1	YES	YES	YES

One may be inoperative provided that:
(a) The associated FU indication is operative,
(b) The opposite side Fuel LO LVL alert is checked operative,
(c) The use of the preselected quantity is prohibited during refueling, and
(d) The associated tank fuel quantity is checked using manual (magnetic) indicators after each refueling.
←

NOTE: *1) In addition to the cockpit FUEL QTY INDICATION, the preselected quantity and the repeated fuel quantity indications on the refueling panel must be placarded inoperative.*
2) If the FUEL LO LVL alert is triggered on affected side due to Fuel Quantity Indication inoperative: **Refer to 28.42-03 Fuel LO LVL Alert.**

OPERATIONAL (O) PROCEDURES
► Observe use of magnetic sticks to ascertain fuel quantity: **Refer to FCOM PRO.NOP.NSU.28.3.4 Use of Manual (Magnetic) Indicators.**
► In case of suspected fuel leak or fuel unbalance (an excessive aileron trim indication may help for detection), Maintain X FEED valve CLOSED.
► Monitor fuel consumption using Fuel Flow/Fuel Used Indicator.

NOTE: *FUEL MISMATCH and/or FUEL UNBALANCED FWS cautions may be triggered.*

MAINTENANCE (M) PROCEDURES
► Fuel Aircraft to maximum fuel for Schedule to be flown: **Refer to MP ATR-A-12-11-28-00ZZZ-211Z-A for manual refueling.**
► Check fuel level using manual (magnetic) indicators: **Refer to MP ATR-A-12-11-28-00ZZZ-310Z-A for checking fuel level using manual (magnetic) indicators.**

NOTES:
► *Ensure a flight crew member observes use of magnetic sticks to ascertain fuel quantity. Use roll attitude on the "Alpha Call Ups"* ***ATR-A-31-31-XX-01001-345A-A*** *as the primary source for establishing the roll attitude of the aircraft. The inclinometer reading can be used as a secondary or back up reading.*
► *Alpha Call Ups will give accuracy and only indicates a positive (+) up or a negative (-) down value. It does not indicate or determine which Wing (Left or Right) is up or down. The Inclinometer will determine and verify which Wing is (Left or Right) up or down.*

28-42-01 FUEL QTY INDICATION (Cont'd)

MAINTENANCE (M) PROCEDURES (Cont'd)

► Perform functional test of feeder tank low level sensor to check the opposite side Fuel LO LVL alert: **Refer to MP ATR-A-28-42-70-04ZZZ-340Z-A (FUNCTIONAL TEST OF FEEDER TANK LOW LEVEL SENSOR) to check the opposite side Fuel LO LVL alert.**

► Make an entry in the rectification section of the Technical Log book of the calculated magnetic stick quantities for each tank and the inclinometer reading.

NOTE: *Calculation of fuel quantities using the Magnetic Fuel Level Indicator Sticks (Magnetic Sticks) requires two independent measurements to be corroborated. The second independent measurement can be the Technical Services Engineer in MCC, having received photos of the drip sticks and Alpha Call Ups. An entry would be made in the Technical Log book of the name and authorization number of the Engineer.*

28-42-03 FUEL LO LVL ALERT

RI	NI	NR	PLACARD	O	M
B	2	1	NO	YES	YES

One may be inoperative provided that:
(a) The associated FU indication is operative,
(b) The opposite side Fuel LO LVL alert is checked operative,
(c) The opposite side Fuel Quantity Indication is operative,
(d) X FEED Valve is checked closed, and
(e) The associated tank fuel quantity is checked using manual (magnetic) indicators after each refueling.
←

NOTE: *If Fuel Quantity Indication on affected side is inoperative****: Refer to Item 28-42-01 Fuel QTY Indication.***

OPERATIONAL (O) PROCEDURES
► X FEED valve must be maintained closed in all cases.
Limitations
Flight planning must take into account engine failure as X FEED would not be possible.
Flight time from suitable alternate must not exceed 45 min with one engine operating (elapse time before reaching maxi fuel unbalance).
► Dispatch without any asymmetry.
Maximum certified fuel unbalance must never be exceeded 730 kg.
► In case of Fuel LO LVL on opposite side, LAND ASAP.
► Monitor fuel consumption using Fuel Flow/Fuel Used Indicator.

MAINTENANCE (M) PROCEDURES
Refer to MP ATR-A-12-11-28-00ZZZ-310Z-A for checking fuel level using manual (magnetic) indicators.

Refer to MP ATR-A-28-42-70-04ZZZ-340Z-A (FUNCTIONAL TEST OF FEEDER TANK LOW LEVEL SENSOR) to check Fuel LO LVL alert on opposite side.

28-43-01 MANUAL (MAGNETIC) INDICATORS

RI	NI	NR	PLACARD	O	M
C	4	0	YES	NO	NO

All may be inoperative.

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29-11-01 MAIN HYD PUMP

29-11-01A GREEN HYD Pump (A)

RI	NI	NR	PLACARD	O	M
C	1	0	YES	YES	YES

One may be inoperative provided that:
(a) X FEED valve is open, and
(b) Both ACW GEN are operative, and
(c) ACW BTC is operative, and
(d) DC AUX pump operates normally, and
(e) Hydraulic fluid QTY is checked prior to each departure, and
(f) Aircraft is not operated on narrow runways (width < 30 m (98ft))
(g) TLU is set to LO SPD position for entire flight.
(h) Maximum IAS is limited to 180 kt.
(i) Maximum crosswind component is limited to 15 kt at landing.
(j) Operations are conducted in compliance with applicable AFM Deviation Guide.

OPERATIONAL (O) PROCEDURES
Refer to AFM - Dispatch with Green Hydraulic Pump Inoperative.

MAINTENANCE (M) PROCEDURES
Refer to MP ATR-A-29-XX-XX-00ZZZ-281Z-A to check the level of hydraulic fluid.

- OR-

29-11-01B BLUE HYD Pump (B)

RI	NI	NR	PLACARD	O	M
C	1	0	YES	YES	YES

One may be inoperative provided that:
(a) X FEED valve is open, and
(b) Both ACW GEN are operative, and
(c) ACW BTC is operative, and
(d) DC AUX pump operates normally, and
(e) Hydraulic fluid QTY is checked prior to each departure, and
(f) Aircraft is not operated on narrow runways (width < 30 m (98 ft)).

OPERATIONAL (O) PROCEDURES
Two-engine taxi is recommended.

MAINTENANCE (M) PROCEDURES
Refer to MP ATR-A-29-XX-XX-00ZZZ-281Z-A to check the level of hydraulic fluid.

29-11-02 DC AUX PUMP

RI	NI	NR	PLACARD	O	M
C	1	0	YES	YES	NO

May be inoperative provided that:
(a) Aircraft is not operated on narrow runways (width < 30 m (98 ft)),
(b) HYD BLUE Pump and HYD GREEN Pump operate normally,
(c) Hydraulic X-FEED valve operates normally,
(d) Both ACW generators systems operate normally,
(e) Wheels are chocked until ACW BUS is powered, and
(f) Both EECs operate normally.

OPERATIONAL (O) PROCEDURES
Ensure wheels are chocked until ACW BUS is powered.

- **After both propellers are released:**
  ► BRAKE HANDLE.....PARKING
  ► HYD BLUE PUMP ..... OFF
  Monitor Blue HYD SYS PRESS decreasing on HYD SYS PRESS indicator.
  ► HYD X FEED..... ON
  Monitor Blue HYD SYS PRESS increasing on HYD SYS PRESS indicator.
  ► HYD X FEED..... OFF
  Monitor Blue HYD SYS PRESS decreasing on HYD SYS PRESS indicator.
  ► HYD BLUE PUMP ..... ON
  Monitor Blue HYD SYS PRESS increasing on HYD SYS PRESS indicator.

29-21-01 X FEED VALVE

RI	NI	NR	PLACARD	O	M
C	1	0	YES	NO	YES

May be inoperative provided that:
(a) X FEED valve is in the closed position, and
(b) Hydraulic fluid QTY is checked prior to each departure.

MAINTENANCE (M) PROCEDURES
Refer to MP ATR-A-29-XX-XX-00ZZZ-281Z-A to check the level of hydraulic fluid.

29-31-01 HYD PWR - LO LVL LIGHT

RI	NI	NR	PLACARD	O	M
C	2	1	YES	YES	YES

One may be inoperative provided that:
(a) Hydraulic fluid QTY is checked prior to each departure, and
(b) X FEED valve is not used in flight.

OPERATIONAL (O) PROCEDURES

► Monitor associated HYD SYS pressure indication throughout the flight.

MAINTENANCE (M) PROCEDURES

Refer to MP ATR-A-29-XX-XX-00ZZZ-281Z-A to check the level of hydraulic fluid.

29-32-01-01 LO PR LIGHT – MAIN HYD PUMP

RI	NI	NR	PLACARD	O	M
C	2	1	YES	YES	NO

One may be inoperative provided that the associated HYD SYS pressure indication is operative.

OPERATIONAL (O) PROCEDURES

► Monitor associated HYD SYS pressure indication throughout the flight.

29-32-01-02 LO PR LIGHT – AUX HYD PUMP

RI	NI	NR	PLACARD	O	M
C	1	0	YES	NO	NO

May be inoperative provided that the BLUE HYD SYS pressure indication is operative.

29-32-02-01 PRESSURE INDICATION – HYD SYS Indication
Pressure Indication – HYD SYS Indication (on SD Hydraulic)

RI	NI	NR	PLACARD	O	M
C	2	0	YES	NO	NO

One or both may be inoperative provided that the associated LO PR alert is operative.

29-32-02-02 PRESSURE INDICATION – BRAKE ACCU INDICATION
Pressure Indication – BRAKE ACCU Indication (on SD Hydraulic)

RI	NI	NR	PLACARD	O	M
C	1	0	YES	NO	NO

May be inoperative provided that the brake accumulator pressure is checked on the hydraulic bay Indication prior to each flight.

29-32-03 ACCUMULATORS PRESSURE INDICATOR (HYDRAULIC BAY)

RI	NI	NR	PLACARD	O	M
C	3	0	NO	NO	NO

All may be inoperative.

29-33-01 HYD PWR – OVHT ALERT (BLUE PUMP OR GREEN PUMP)

RI	NI	NR	PLACARD	O	M
-	2	1	YES	YES	NO

One may be inoperative provided that:

(a) Associated main hydraulic pump is considered inoperative and is not used, **Refer to Item 29-11-01 Main HYD Pump**, and

(b) The other main hydraulic pump and associated OVHT light is operative.

←

OPERATIONAL (O) PROCEDURES

During the cockpit preparation, the associated main hydraulic pump is set to OFF.

29-33-02 HYD PWR – AUX PUMP – OVHT ALERT

RI	NI	NR	PLACARD	O	M
-	1	0	YES	YES	NO

May be inoperative provided that:

(a) Both main hydraulic pumps operate normally,

(b) X-FEED valve is operative,

(c) Both ACW generator systems operate normally,

(d) Associated AUX hydraulic pump is considered inoperative and not used,
Refer to Item 29-11-02 DC AUX Pump.

(e) Both EEC's operate normally, and

(f) Wheels are chocked until ACW BUS is powered.

OPERATIONAL (O) PROCEDURES

The DC AUX pump is set to OFF.

Ensure wheels are chocked until ACW BUS is powered.

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30-11-01 DE ICE VALVE (PRSOV)

RI	NI	NR	PLACARD	O	M
C	2	1	YES	NO	NO

One may be inoperative in closed position provided that aircraft is not operated into known or forecast icing conditions.

30-11-02 AIRFRAME DE ICING BOOTS SYSTEM

RI	NI	NR	PLACARD	O	M
B	1	0	YES	NO	NO

May be inoperative provided that aircraft is not operated into known or forecast icing conditions.

30-11-03-01 DE ICING – SAT BELOW -20°C (ENGINE AIRFRAME FAST - SLOW MODE SELECTOR)

30-11-03-01A (A)

RI	NI	NR	PLACARD	O	M
C	1	0	YES	NO	NO

May be inoperative provided that MODE SEL AUTO is operative.

- OR -

30-11-03-01B (B)

RI	NI	NR	PLACARD	O	M
C	1	0	YES	NO	NO

May be inoperative provided that aircraft is not operated into known or forecast icing conditions when mode FAST is inoperative.

30-11-03-02 DE ICING OVRD SYSTEM

RI	NI	NR	PLACARD	O	M
C	1	0	YES	NO	NO

May be inoperative.

30-11-04 DE ICING – AFR AIR BLEED FAULT LIGHT

RI	NI	NR	PLACARD	O	M
B	1	0	YES	YES	NO

May be inoperative provided that AIRFRAME FAULT light is checked operative.

OPERATIONAL (O) PROCEDURES

With Bleed Supply:

► AFR AIR BLEED.....CHECK ON
► DE-ICING AIRFRAME.....ON
► AIRFRAME FAULT LIGHT ..... CHECK TURNED OFF
► AFR AIR BLEED.....OFF
► AIRFRAME FAULT LIGHT ..... CHECK TURNED ON
► AFR AIR BLEED.....ON

30-11-06 DE ICING AND ANTI ICING – MODE SEL AUTO

RI	NI	NR	PLACARD	O	M
C	1	0	YES	NO	NO

May be inoperative provided that MAN mode is operative.

30-21-01 ENGINE DE ICING BOOTS SYSTEM

RI	NI	NR	PLACARD	O	M
C	2	1	YES	NO	NO

One may be inoperative provided that aircraft is not operated into known or forecast icing conditions.

30-21-02 DE ICING – ENG AIRFRAME FAULT ALERT

RI	NI	NR	PLACARD	O	M
B	3	0	YES	NO	NO

All may be inoperative provided that:
(a) Associated device is considered inoperative, and
(b) Aircraft is not operated into known or forecast icing conditions.
←

30-30-01 PROBE HEATING SYSTEM - PITOT

RI	NI	NR	PLACARD	O	M
B	3	2	YES	NO	NO

One may be inoperative provided that aircraft is not operated into known or forecast icing conditions.

30-30-02 PROBE HEATING SYSTEM – STATIC

RI	NI	NR	PLACARD	O	M
B	6	5	YES	NO	NO

One may be inoperative provided that both STBY are operative.

30-31-01 PROBE HEATING SYSTEM - TAT

RI	NI	NR	PLACARD	O	M
B	2	1	YES	YES	NO

One may be inoperative provided that valid TAT is selected.

OPERATIONAL (O) PROCEDURES

► Select valid TAT in VCP ENG OAT page.

30-31-02 PROBE HEATING SYSTEM - ALPHA

RI	NI	NR	PLACARD	O	M
B	2	1	YES	NO	NO

One may be inoperative provided that aircraft is not operated into known or forecast icing conditions.

30-31-04 PROBES HTG FAULT LIGHTS (ALPHA, TAT, STAT)

RI	NI	NR	PLACARD	O	M
B	7	4	YES	YES	NO

May be inoperative provided that:
(a) Two static alerts are operative,
(b) One TAT alert is operative,
(c) One alpha alert is operative, and
(d) Heating system associated to failed alerts are checked operative prior to each departure.

OPERATIONAL (O) PROCEDURES

Operational procedure for checking TAT heating:

NOTE: *During this check, all probes on the affected side – including PITOT, ALPHA, TAT, and SAT – will be energized (heating ON).*

CAUTION

► Ensure that all probe covers on the affected side are removed before starting the test.

► Ensure that ground personnel are informed to avoid inadvertent contact with heated probes.

► Verify the TAT value on the TAT indicator

► On the Right-Hand Maintenance Panel, locate the WEIGHT ON WHEELS section (panel 702V)

► GUST LOCK..... OFF

► CABIN TEMPERATURE..... <30°C

► AVIONICS VENT EXHAUST MODE..... NORM

► STICK PUSHER / SHAKER pb..... OFF

► Open the safety guard and place the MODE switch to FLT position

► Set the affected TAT Probe heating to the ON position

► Observe the TAT indicator for increase in temperature

NOTE: *A duration of approximately one minute is sufficient to observe significant increase in TAT values.*

► Return the MODE switch to the NORM position and close the safety guard

► Set the affected TAT Probe heating back to the OFF position

► Confirm that the TAT value returns to its initial reading

30-31-05 PITOT PROBES HTG FAULT LIGHTS

30-31-05A (A)

RI	NI	NR	PLACARD	O	M
B	3	2	YES	NO	YES

May be inoperative provided that:
(a) Two pitot alerts are operative, and
(b) Pitot heating system associated to failed alert is checked operative prior to each departure.

MAINTENANCE (M) PROCEDURES
Refer to ATR-A-34-11-10-00ZZZ-366Z-A to check pitot probes heating resistance.

- OR -

30-31-05B (B)

RI	NI	NR	PLACARD	O	M
-	3	2	YES	NO	NO

May be inoperative provided that:
(a) Probe heating system – Pitot is considered inoperative, Refer to Item 30-30-01 Probe Heating System – Pitot, and
(b) Aircraft is not operated into known or forecast icing conditions.
←

30-53-01 ANTI ICING HORNS SYSTEM

RI	NI	NR	PLACARD	O	M
A	2	1	YES	NO	NO

May be inoperative provided that:
(a) Aircraft is not operated into known or forecast icing conditions, and
(b) Operations are limited to three calendar days.
←

30-53-02 ANTI ICING - HORNS FAULT LIGHT

RI	NI	NR	PLACARD	O	M
A	2	1	YES	NO	NO

One may be inoperative provided that:

(a) Associated system is considered inoperative, Refer to Item 30-53-01 Anti Icing Horns System, and

(b) Operations are limited to three calendar days.

30-60-01 WINDSHIELD HEATING SYSTEM

30-60-01A (A)

RI	NI	NR	PLACARD	O	M
C	2	1	YES	NO	NO

One may be inoperative provided that aircraft is not operated into known or forecast icing conditions.

- OR -

30-60-01B (B)

RI	NI	NR	PLACARD	O	M
C	2	1	YES	NO	NO

One may be inoperative provided that both side windows heating are operative.

- OR -

30-60-01C (C)

RI	NI	NR	PLACARD	O	M
C	2	0	YES	NO	NO

May be inoperative provided that:
(a) Both packs are operative,
(b) Aircraft is not operated into known or forecast icing conditions, and
(c) Both side windows heating are operative.

30-60-02 SIDE WINDOWS HEATING SYSTEM

30-60-02A (A)

RI	NI	NR	PLACARD	O	M
C	2	1	YES	NO	NO

One side window heating may be inoperative provided that flight is not conducted into known or forecast icing conditions.

- OR -

30-60-02B (B)

RI	NI	NR	PLACARD	O	M
C	2	0	YES	NO	NO

Both may be inoperative provided that:
(a) Flight is not conducted into known or forecast icing conditions.
(b) Both windshield heating are operative, and
(c) Both ACW GEN are operative.

30-60-03 SIDE WNDOWS FAULT LIGHT

30-60-03A (A)

RI	NI	NR	PLACARD	O	M
C	1	0	YES	YES	NO

May be inoperative provided that associated alert is checked operative.

OPERATIONAL (O) PROCEDURES

► SIDE WINDOWS pb ..... ON
► C/B ANTI/DE ICING/WSHLD HTG R IND L/SIDE WDO/L CTL ..... PULL
► MC light + SC ..... CHECK
► A-ICING WINDOWS amber on EWD ..... CHECK
► SIDE WINDOWS amber on MFD ..... CHECK
► C/B ANTI/DE ICING/WSHLD HTG R IND L/SIDE WDO/L CTL ..... PRESS
► SIDE WINDOWS pb ..... OFF

- OR -

30-60-03B (B)

RI	NI	NR	PLACARD	O	M
-	1	0	YES	NO	NO

May be inoperative provided that both side window heating are considered inoperative. Refer to Item 30-60-02 Side Windows Heating System.

30-60-04 WINDSHIELD HTG FAULT LIGHT

30-60-04A (A)

RI	NI	NR	PLACARD	O	M
C	2	1	YES	YES	NO

One may be inoperative provided that respective heating system is checked operative prior to each departure.

OPERATIONAL (O) PROCEDURES

► Check by hand the correct functioning of the respective heating system.

- OR -

30-60-04B (B)

RI	NI	NR	PLACARD	O	M
-	2	1	YES	NO	NO

One may be inoperative provided that associated windshield heating is considered inoperative. Refer to Item 30-60-01 Windshield Heating System.

30-61-01 PROPELLER ANTI ICING SYSTEM

RI	NI	NR	PLACARD	O	M
C	2	0	YES	NO	NO

One or both may be inoperative provided that aircraft is not operated into known or forecast icing conditions.

30-61-02 ANTI ICING – PROP ON LIGHT

RI	NI	NR	PLACARD	O	M
C	2	0	YES	YES	NO

May be inoperative provided that FAULT alert is operative.

OPERATIONAL (O) PROCEDURES

- **While engines are running with CL at AUTO**
  - ► BRAKE HANDLE.....PARKING
  - ► ACW generation.....OFF
  - ► PROP ANTI-ICING affected ..... ON
  - ► FAULT alert.....CHECK GENERATED
  - ► PROP ANTI-ICING affected .....AS RQRD
  - ► ACW generation..... ON

30-61-03 MODE SEL SAT BELOW -10°C (PROPELLER ANTI ICING MODE SELECTOR)

RI	NI	NR	PLACARD	O	M
C	1	0	YES	NO	NO

May be inoperative provided that aircraft is not operated into known or forecast icing conditions.

30-61-04 ANTI ICING - PROP - FAULT LIGHT

RI	NI	NR	PLACARD	O	M
C	2	0	YES	NO	NO

One or both may be inoperative provided that aircraft is not operated into known or forecast icing conditions.

30-65-01-01 WINDSHIELD WIPER SYSTEM

RI	NI	NR	PLACARD	O	M
C	2	0	YES	NO	NO

One or both may be inoperative provided that:

(a) No precipitation is forecasted during a period from one hour before until one hour after the estimated time of departure and arrival at the takeoff and destination aerodromes, and

(b) Affected wipers are not part of the equipment required for the intended operation.

NOTE: *1) Takeoff and destination aerodromes include any takeoff and destination alternate aerodromes required by the operational rules.*

2) Check Flight Manual for minimum required equipment for CAT II approaches and low visibility takeoff.

30-65-01-02 WINDSHIELD WIPERS – HIGH SPEED FUNCTION (FAST)

RI	NI	NR	PLACARD	O	M
C	2	0	YES	NO	NO

One or both may be inoperative provided that the associated low speed function is operative.

30-65-01-03 WINDSHIELD WIPERS – LOW SPEED FUNCTION (SLOW)

RI	NI	NR	PLACARD	O	M
C	2	0	YES	NO	NO

One or both may be inoperative provided that the associated high speed function is operative.

30-70-01 ICE & RAIN PROTECTION SYSTEMS (RH MAINTENANCE PANEL)

RI	NI	NR	PLACARD	O	M
-	1	0	NO	NO	NO

May be inoperative provided:

(a) Airplane is not operated in known or forecast icing conditions, and

(b) Associated system is considered inoperative.

Refer to Item 30-60-01 Windshield Heating System

Refer to Item 30-61-01 Propeller Anti Icing System

30-80-01 ICE DETECTION SYSTEM

RI	NI	NR	PLACARD	O	M
B	1	0	YES	NO	YES

May be inoperative provided that:
(a) The aircraft is not operated into known or forecast icing conditions, and
(b) The ice detection system is deactivated.

MAINTENANCE (M) PROCEDURES

► Open, secure and tag C/B ANTI/DE ICING/ ICE DET.

30-80-03 ICE EVIDENCE PROBE IEP

RI	NI	NR	PLACARD	O	M
-	1	1	NO	NO	NO

NO GO.

NOTE: *Refer to Item 33-49-02 IEP Light.*

30-80-04 ICING AOA LIGHT

RI	NI	NR	PLACARD	O	M
C	1	0	YES	YES	NO

May be inoperative provided that the ICING AOA green label on FMA is checked operative.

OPERATIONAL (O) PROCEDURES

- **While engines are running with CL at AUTO:**
  - ► HORNS.....ON
  - ► ICING AOA label on FMA..... CHECK CAME ON
  - ► HORNS.....OFF
  - ► ICING AOA pb .....PRESS
  - ► ICING AOA label on FMA..... CHECK TURNED OFF

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31-21-01 CLOCK

CLOCK FUNCTION

RI	NI	NR	PLACARD	O	M
C	2	0	YES	NO	NO

May be inoperative provided that an accurate timepiece is operative on the flight deck indicating the time in hours, minutes and seconds.

NOTE: *On the basis that the required timepiece(s) does not need to be approved, an accurate pilot's wristwatch that indicates hours, minutes and seconds would be acceptable. On the EWD a FMS time indication with hours, minutes and seconds would also be acceptable.*

31-31-03-01 MPC – FDAU FUNCTION

RI	NI	NR	PLACARD	O	M
-	1	0	YES	YES	NO

May be inoperative provided that:

(a) The DFDR is considered inoperative (Refer to Item 31-32-01 Digital Flight Data Recorder – DFDR), and

NOTE: *No parameter acquisition.*

(b) The NL indication is inoperative (Refer to Item 77-11-02 NL Indicator), and

(c) The APM and EWD Flight Time functions are inoperative (Refer to Item 31-31-03-02 MPC with FDAU Function – APM and Refer to Item 31-31-03-03 MPC with FDAU Function – EWD Flight Time).

OPERATIONAL (O) PROCEDURES

The ELEC DC ESS alert is inoperative on EWD (Flight Warning) and on MFD (ELEC s-page). During the flight, if the following systems are simultaneously lost:

- GREEN PUMP,
- FUEL QUANTITY,
- MCP F/O,
- EFCP F/O,
- IOM -22,
- IOM -23

The following failure procedure must be applied: **DC ESS BUS OFF**.

The ELEC DC EMER alert is inoperative on EWD (Flight Warning) and on MFD (ELEC s-page). During the flight, if the following systems are simultaneously lost:

- BLUE PUMP,
- VHF 1,
- MCP CAPT,
- EFCP CAPT,
- CLOCK CAPT.

The following failure procedure must be applied: **DC EMER BUS OFF**

The ELEC AC STBY alert is inoperative on EWD (Flight Warning) and on MFD (ELEC s-page). During the flight, if the following indications are simultaneously lost on the Flight Control window (on EWD):

- PITCH Trim,
- ROLL Trim,
- YAW Trim,
- FLAP POSITION,

The following failure procedure must be applied: **AC STBY BUS FAULT**.

31-31-03-02 APM

RI	NI	NR	PLACARD	O	M
C	1	0	YES	YES	NO

May be inoperative provided that APM is selected OFF.

OPERATIONAL (O) PROCEDURES

► APM ..... OFF

► Monitor the ICING CONDITION and IAS & V/S (for NORMAL, ABNORMAL and EMERGENCY procedures for flight in ICING and SEVERE ICING CONDITIONS):
Refer to AFM - SEVERE ICING DETECTION Abnormal Procedure for severe icing indications information
Refer to AFM - ADDITIONAL NORMAL PROCEDURES Procedure for Icing Conditions

► Review Recommended Maximum Icing Flight Level.

31-31-03-03 MPC – EWD FLIGHT TIME

RI	NI	NR	PLACARD	O	M
D	1	0	NO	NO	NO

May be inoperative.

31-31-03-04 MPC - DMU ACMS (MAINTENANCE)

RI	NI	NR	PLACARD	O	M
C	1	0	YES	NO	NO

May be inoperative.

31-31-04 PRINTER

RI	NI	NR	PLACARD	O	M
D	1	0	NO	NO	NO

May be inoperative provided that operational procedures do not require its use.

31-32-01 DIGITAL FLIGHT DATA RECORDER - DFDR

RI	NI	NR	PLACARD	O	M
A	1	0	YES	NO	NO

May be inoperative provided that:

(a) Aircraft does not exceed 8 further consecutive flights with the Flight Data Recorder unserviceable, and

(b) Not more than 72 hrs have elapsed since the Flight Data Recorder was found to be unserviceable, and

(c) Any Cockpit Voice Recorder required to be carried is operative.

NOTE: *DFDR is considered to be inoperative when any of the following conditions exists:*

- *Loss of the flight recording function is evident to the flight crew during pre-flight check e.g., by means of a system status monitor, or*
- *The need for maintenance has been identified by the system monitors, where available, with the setting of an indicator and the cause of that setting has not been determined, or*
- *Analyses of recorded data or maintenance actions have shown that more than 5% of the total number of individual parameters (variable and discrete (Non-Mandatory)) required to be recorded for the particular aircraft are not being recorded properly.*

Flight recorder systems shall be considered unserviceable if there is a significant period of poor-quality data, unintelligible signals, or if one or more of the mandatory parameters are not recorded correctly.

←

31-32-03 QAR

May be inoperative provided procedures do not require its use.

Perform DFDR DATA download (Refer 31-31-31-OPT-10010-001).
Ensure the data card is correctly labelled with the aircraft registration and date.
Place labelled data card in aircraft work package.

31-32-04 DFDR REQUIRED PARAMETERS – VARIABLE AND DISCRETE

RI	NI	NR	PLACARD	O	M
A	19	18	YES	NO	YES

Up to 5% of the total number of required parameters may be inoperative for a maximum of 90 calendar days or until the next maintenance inspection, whichever occurs first.

NOTE: *DFDR is considered inoperative if more than 5% of the total number of required parameters are not recorded properly. Refer to Item 31-32-01 Digital Flight Data Recorder – DFDR.*

Refer to Table 1 for TTCAA Mandatory & Non-Mandatory parameters.

MAINTENANCE (M) PROCEDURES

All required DFDR parameters are reviewed as per **TTCAR No. 7 49:03 (c)** and work instructions issued to rectify all parameters found defective/not recorded.

TABLE 1 – TTCAA MANDATORY & NON-MANDATORY PARAMETERS

Mandatory	Non-Mandatory		
Time (UTC when available, otherwise relative time count or GPS time sync)	Trailing edge flap or/and cockpit control selection	Selected speed (all pilot selectable modes of operation)	Operational stall protection,stick shaker and pusher activation
Pressure-altitude	Leading edge flap or/and cockpit control selection	Selected Mach (all pilot selectable modes of operation)	Primary navigation system reference (GNSS, INS, VOR/DME, MLS, Loran C, localiser glideslope)
Indicated airspeed or calibrated airspeed	Thrust reverser position	Selected vertical speed (all pilot selectable modes of operation)	Ice detection
Heading (primary flight crew reference)	Ground spoiler/speed brake selection (Selection and position)	Selected heading (all pilot selectable modes of operation)	Engine warning each engine vibration
Normal acceleration	Autopilot/auto throttle/AFCS mode and engagement status	Selected flight path (all pilot selectable modes of operation) [course/DSTRK, path angle, final approach path (IRNAV/IAN)]	Engine warning each engine vibration engine over temperature
Pitch attitude	Radio altitude	Selected Decision Height	Engine warning each engine oil pressure low
Roll attitude	Vertical beam deviation (ILS/GPS/GLS glide path, MLS elevation, IRNAV/IAN vertical deviation)	EFIS display format (pilot, co-pilot)	Engine warning each engine oil pressure low engine over speed
Radio transmission Keying	Horizontal beam deviation (ILS/GPS/GLS localiser, MLS azimuth,IRNAV/IAN lateral deviation)	Multi-function/engine/ alerts display format	Yaw Trim Surface Position
Power on each engine	GPWS (ground proximity warning system) /TAWS/GCAS status (selection of terrain display mode including pop-up display status) and (terrain alerts, both cautions and warnings, and advisories)	AC electrical bus status	Roll Trim Surface Position
Outside air temperature	Angle of attack	DC electrical bus status	Yaw or sideslip angle
Longitudinal acceleration	Hydraulics, each system (low pressure)	Engine bleed valve position	De-icing and/or anti-icing systems selection
Lateral acceleration	Navigation data (latitude/longitude, ground speed and drift angle)	APU bleed valve position	Hydraulic pressure (each system)
Pilot input and/or control surface position-primary controls (pitch, roll, yaw)	Landing gear and gear selector position	Computer failure	Loss of cabin pressure
Pitch trim position	Groundspeed	Engine thrust command	Cockpit trim control input Position Pitch
Marker beacon passage	Additional engine parameters (EPR, N1, indicated vibration level, N2, EGT, fuel flow, fuel cut-off lever position, N3)	Engine thrust target	Cockpit trim control input Position Roll
Master warning	TCAS/ACAS (traffic alert and collision avoidance system)	Computed centre of gravity	Cockpit trim control input Position Yaw
Each NAV frequency selection	Windshear warning	Fuel quantity in CG trim tank	All cockpit flight control input forces (control wheel, control column, rudder pedal)
Air/ground status	Selected barometric setting (pilot, co-pilot)	Head up display in use	Event marker
Brakes (left and right brake pressure, left and right brake pedal position)	Selected altitude (all pilot selectable modes of	Para visual display on/off	Date
			ANP or EPE or EPU

31-48-01-01 MULTI FUNCTION COMPUTER (MFC) MODULE 1A

NO GO.

31-48-01-02 MULTI FUNCTION COMPUTER (MFC) MODULE 2A

NO GO.

31-48-01-03 MULTI FUNCTION COMPUTER (MFC) MODULE 1B

NO GO.

31-48-01-04 MULTI FUNCTION COMPUTER (MFC) MODULE 2B

NO GO.

31-48-06 MFC MODULE FAULT LIGHT

NO GO.

31-53-02 MASTER WARNING (MW) LIGHT AND ASSOCIATED CANCEL FUNCTION

NO GO.

31-53-03 MASTER CAUTION (MC) LIGHT AND ASSOCIATED CANCEL FUNCTION

NO GO.

31-53-05 EMER AUDIO CANCEL

RI	NI	NR	PLACARD	O	M
C	1	0	YES	NO	NO

May be inoperative.

31-53-06 TO CONFIG TEST

RI	NI	NR	PLACARD	O	M
B	1	0	YES	NO	NO

May be inoperative.

31-60-01 DISPLAY UNIT (DU)

RI	NI	NR	PLACARD	O	M
C	5	4	YES	YES	NO

One Display unit may be inoperative among DU #1 and DU #3 and DU #5 provided that:
(a) CAPT // PFD /MFD/EWD pb-sw reversion is checked operative, and
(b) F/O // PFD/MFD/EWD pb-sw reversion is checked operative.

OPERATIONAL (O) PROCEDURES
During Preliminary Cockpit Preparation

- ► Check on the DU #2 that the CAPT sw allows to swap the format displays: PFD, EWD and MFD.
- ► Check on the DU #4 that the F/O sw allows to swap the format displays: PFD, EWD and MFD.

31-60-02-01 MCP

RI	NI	NR	PLACARD	O	M
B	2	1	NO	YES	NO

One may be inoperative provided that:
(a) The MCP on the **Pilot Flying** side is operative, and
(b) Both radio tuning through MCDU are checked operative.

OPERATIONAL (O) PROCEDURES
The flight crew have to check that the radio tuning is operative through at least one RMA on both MCDU.

31-60-02-02 MCP – COM pb

RI	NI	NR	PLACARD	O	M
C	2	1	NO	NO	NO

One may be inoperative provided that both MCDU are operative.

31-60-02-03 MCP – NAV pb

RI	NI	NR	PLACARD	O	M
C	2	1	NO	NO	NO

One may be inoperative provided that both MCDU are operative.

31-60-02-04 MCP – SURV pb

RI	NI	NR	PLACARD	O	M
B	2	1	NO	NO	NO

One may be inoperative provided that the SURV pb of the **Pilot Flying** side is operative.

31-60-02-05 MCP – ESC pb

RI	NI	NR	PLACARD	O	M
C	2	0	NO	NO	NO

One or both may be inoperative.

31-60-02-06 MCP – ENTER pb and MULTI DIRECTIONAL PAD

RI	NI	NR	PLACARD	O	M
B	2	1	NO	NO	NO

One may be inoperative provided that both MCDU are operative.

31-60-02-07 MCP – NUMERIC KEYBOARD

RI	NI	NR	PLACARD	O	M
C	2	1	NO	NO	NO

One may be inoperative provided that both MCDU are operative.

31-60-03-01 EFCP

RI	NI	NR	PLACARD	O	M
A	2	1	NO	NO	NO

The right side EFCP may be inoperative provided that the operations are limited to two calendar days.

31-60-03-02 EFCP – MFD CONTROL – SYS PUSHBUTTON

RI	NI	NR	PLACARD	O	M
A	2	1	NO	NO	NO

One may be inoperative provided that:

(a) The SYS pb of the **Pilot Monitoring** side is operative, and

(b) Operations are limited to two calendar days.

31-60-03-03 EFCP – MFD CONTROL – ND PUSBUTTON

RI	NI	NR	PLACARD	O	M
B	2	1	NO	NO	NO

One may be inoperative provided that the ND pb of the **Pilot Flying** side is operative.

31-60-03-04 EFCP – MFD CONTROL – PERF PUSHBUTTON

RI	NI	NR	PLACARD	O	M
B	2	1	NO	NO	NO

One may be inoperative provided that:
(a) PERF pb of the **Pilot Flying** side is operative, and
(b) MCP of the **Pilot Flying** is operative.

31-60-03-05 EFCP - MFD CONTROL – MISC PUSHBUTTON

RI	NI	NR	PLACARD	O	M
D	2	0	NO	NO	NO

One or both may be inoperative.

31-60-03-06 EFCP – MFD CONTROL – MAP PUSHBUTTON

RI	NI	NR	PLACARD	O	M
D	2	0	NO	NO	NO

One or both may be inoperative.

31-60-03-07 EFCP - MFD CONTROL – VID PUSHBUTTON

RI	NI	NR	PLACARD	O	M
D	2	0	NO	NO	NO

One or both may be inoperative.

31-60-03-08 EFCP – NAV CONTROL – BRG PUSHBUTTON

RI	NI	NR	PLACARD	O	M
B	4	3	NO	NO	NO

One may be inoperative provided that BRG #1 is available on the **Pilot Flying** side.

31-60-03-09 EFCP – NAV CONTROL – RANGE PUSHBUTTON

RI	NI	NR	PLACARD	O	M
B	2	1	NO	NO	NO

One may be inoperative.

31-60-03-10 EFCP – NAV CONTROL – FORMAT ROTARY SELECTOR

RI	NI	NR	PLACARD	O	M
B	2	1	NO	NO	NO

One may be inoperative.

31-60-03-11 EFCP – PROC CONTROL – PROC UP PUSHBUTTON

RI	NI	NR	PLACARD	O	M
C	2	1	NO	NO	NO

One may be inoperative.

31-60-03-12 EFCP – PROC CONTROL – PROC DOWN PUSHBUTTON

RI	NI	NR	PLACARD	O	M
C	2	1	NO	NO	NO

One may be inoperative.

31-60-03-13 EFCP – PROC CONTROL – ENTER PROC PUSHBUTTON

RI	NI	NR	PLACARD	O	M
C	2	1	NO	NO	NO

One may be inoperative.

31-60-03-14 EFCP – PROC CONTROL – PROC MENU PUSHBUTTON

RI	NI	NR	PLACARD	O	M
C	1	0	NO	NO	NO

May be inoperative.

31-60-03-15 EFCP – PROC CONTROL – MAN DEL PUSHBUTTON

RI	NI	NR	PLACARD	O	M
C	1	0	NO	NO	NO

May be inoperative.

31-60-03-16 EFCP – PROC CONTROL – RCL PUSHBUTTON

NO GO.

31-60-03-17 EFCP – PROC CONTROL – CLR PUSHBUTTON

RI	NI	NR	PLACARD	O	M
B	1	0	NO	NO	NO

May be inoperative.

31-60-04-01 ICP

RI	NI	NR	PLACARD	O	M
-	2	1	NO	NO	NO

One may be inoperative provided that associated ADC is considered inoperative.
Refer to Item 34-11-01 Air Data Computer (ADC)

31-60-04-02 ICP – BARO SET ROTARY SELECTOR

RI	NI	NR	PLACARD	O	M
-	2	1	NO	NO	NO

One may be inoperative provided that associated ADC is considered inoperative.
Refer to Item 34-11-01 Air Data Computer (ADC).

31-60-04-03 ICP – PUSH STD pb

RI	NI	NR	PLACARD	O	M
C	2	0	NO	NO	NO

One or both may be inoperative.

31-60-04-04 ICP – DH MDA SELECTION

RI	NI	NR	PLACARD	O	M
C	2	0	NO	NO	NO

One or both may be inoperative provided that approach procedures do not require DH use.

31-60-04-05 ICP – DH MDA ROTARY SELECTOR

RI	NI	NR	PLACARD	O	M
C	2	0	NO	NO	NO

One or both may be inoperative provided that approach procedures do not require DH use.

31-60-04-06 ICP – SPD TGT KNOB

Refer to SPD TGT Knob

31-60-04-07 ICP – AUTO MAN pb

Refer to AUTO – MAN Pushbutton.

31-60-05-01 MCDU

RI	NI	NR	PLACARD	O	M
C	2	1	NO	NO	NO

One may be inoperative.

31-60-06 DU SWITCH REVERSION

←

RI	NI	NR	PLACARD	O	M
C	2	0	NO	YES	NO

One or both may be inoperative provided that all Display Units are operative.

OPERATIONAL (O) PROCEDURES

During Preliminary Cockpit Preparation

For the proper use of the manual DU reconfiguration and considering task sharing:
The Pilot Flying must be on the side with the DU not failed.

31-70-01 MFC – INDICATIONS AND CONTROLS (RH MAINTENANCE PANEL)

RI	NI	NR	PLACARD	O	M
C	1	0	NO	NO	NO

May be inoperative.

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INTENTIONALLY
LEFT BLANK

32-31-01 ANTI RETRACTION SYSTEM (LEVER LATCH)

RI	NI	NR	PLACARD	O	M
C	1	0	YES	NO	NO

May be inoperative.

32-31-02 LANDING GEAR LEVER INTEGRATED RED LIGHT

RI	NI	NR	PLACARD	O	M
C	1	0	YES	NO	NO

May be inoperative.

32-31-03 LANDING GEAR RETRACTION OR EXTENSION AND UPLOCKING SYSTEM

RI	NI	NR	PLACARD	O	M
A	1	0	YES	YES	NO

May be inoperative provided that:
(a) Operations are limited to two calendar days,
(b) Operations are performed with landing gear down, and
(c) Operations are conducted in compliance with applicable AFM Deviation Guide.
←

OPERATIONAL (O) PROCEDURES
Refer to AFM - DEVIATION GUIDE SUPPLEMENTS Dispatch with Landing Gear Down.

32-32-04 BRAKE ACCU PRESSURE INDICATOR (ON MAINTENANCE PANEL)

RI	NI	NR	PLACARD	O	M
D	1	0	NO	NO	NO

May be inoperative.

32-42-01 WHEEL BRAKE

RI	NI	NR	PLACARD	O	M
C	4	3	YES	YES	YES

One may be inoperative provided that:
(a) Anti-skid system is operative,
(b) Affected brake is deactivated,
(c) Landing gear is left down for 1 min after takeoff except in case of emergency, and
(d) Operations are conducted in compliance with applicable AFM Deviation Guide.

OPERATIONAL (O) PROCEDURES
Refer to AFM - DEVIATION GUIDE SUPPLEMENTS Dispatch with One Wheel Brake Deactivated.

MAINTENANCE (M) PROCEDURES
For deactivation, Refer to MP ATR-A-32-42-50-01ZZZ-560Z-A.

NOTE: *Dispatch is not possible with wheel brake removed.*

32-42-02 ANTI-SKID SYSTEM

RI	NI	NR	PLACARD	O	M
C	1	0	YES	YES	NO

May be inoperative provided that operations are conducted in compliance with applicable AFM Deviation Guide.

OPERATIONAL (O) PROCEDURES

Refer to AFM - DEVIATION GUIDE SUPPLEMENTS Dispatch with Antiskid System Inoperative.

32-42-03 BRK TEMP ALERT

RI	NI	NR	PLACARD	O	M
C	1	0	YES	NO	NO

May be inoperative provided that:
(a) Landing gear is left down for 1 min after takeoff except in case of emergency, and
(b) A minimum turnaround of 20 min is observed.

32-45-01 PRKG BRK

PRKG BRK ON Indication

RI	NI	NR	PLACARD	O	M
C	1	0	YES	NO	NO

May be inoperative provided that the parking brake is operative.

32-51-01 NOSE WHEEL STEERING

RI	NI	NR	PLACARD	O	M
C	1	0	YES	YES	YES

May be inoperative provided that:
(a) All brakes are operative,
(b) Maximum crosswind component is limited to 15 kt, and
(c) Aircraft is not operated on narrow runways (width < 30 m (98 ft)).

OPERATIONAL (O) PROCEDURES
Maximum demonstrated crosswind component for takeoff and landing is 15 kt.
► Apply asymmetrical thrust and differential braking for taxi and line up as required.
► Avoid sharp turns while using brakes.
Powerback operations are prohibited.

MAINTENANCE (M) PROCEDURES
Open, secure and tag C/B L/G-BRK/NOSE WHEEL STRG.

32-61-01 LANDING GEAR POSITION INDICATING SYSTEM

RI	NI	NR	PLACARD	O	M
B	2	1	YES	YES	NO

May be inoperative.

OPERATIONAL (O) PROCEDURES

←

The illumination of the Green System "LO PR" amber light on the HYD PWR overhead panel will be inhibited when the landing gear control lever is in DOWN position.

As soon as three green lights are illuminated on the operative panel, landing gear is down.

- **When the landing gear is down:**
  - ► Monitor that the hydraulic green pressure remains stable around 3 000 psi.

NOTE: *If the hydraulic green pressure drops significantly, Refer to AFM – HYD BLUE (GREEN) LO PR Abnormal Procedure.*

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33-10-01 COCKPIT LIGHTS

33-10-01A (A)

RI	NI	NR	PLACARD	O	M
C	-	-	NO	NO	NO

May be inoperative provided that:
(a) Sufficient lighting to illuminate all instruments, controls and placards is assured, and
(b) RH dome and STBY instruments integrated lights are operative.

- OR –

33-10-01B (B)

RI	NI	NR	PLACARD	O	M
C	-	0	NO	NO	NO

May be inoperative for daylight operations.

33-10-02-01 CABIN LIGHTS – PASSENGER CONFIGURATION

RI	NI	NR	PLACARD	O	M
C	-	14	NO	NO	NO

One or more may be inoperative provided that:
(a) **14** operating ceiling lights are regularly distributed within the cabin, and
(b) Lighting must be sufficient for cabin attendant to perform its duties.

33-16-01-01 ANN LIGHT – TEST FUNCTION

RI	NI	NR	PLACARD	O	M
A	1	0	YES	YES	NO

May be inoperative provided that operations are limited to two calendar days.

OPERATIONAL (O) PROCEDURES

Following warnings have to be tested before the first flight of the day, to check red lights illumination.

PITCH DISCONNECT

- ► The SELECTOR on the LH maintenance panel has to be used.
- ► Set the SELECTOR to the function PITCH DISCONNECT.
- ► Depress TEST PTT pb.
- ► Check the warnings: CRC, MW light flashing red and PITCH DISCONNECT red message on EWD.
- ► After the check set SELECTOR to NORM FLT.

LANDING GEAR NOT DOWN

- ► On LH MAINTENANCE panel, set the SELECTOR to LDG GEAR NOT DOWN.
- ► Set PL to FI or FLAPS 30.
- ► Depress TEST PTT pb.
- ► Check the warnings: CRC, MW light flashing red and LDG GEAR NOT DN on FWS.
- ► After the check set SELECTOR to NORM FLT.

TO CONFIG

- ► Depress TO CONFIG TEST pb while at least one out of the following four conditions is not met:
  - ► Pitch trim set in the TO range.
  - ► FLAPS set to 15°.
  - ► PWR MGT set to TO.
  - ► RUD TLU green light. Come on
- ► Check the warnings: CRC, MW light flashing red and CONFIG on FWS.

FLAPS UNLOCK

- ► Set FLAPS 15.
- ► Depress and maintain FLAPS UNLK PTT pb on RH maintenance panel.
- ► Check warning is activated when the flaps position passes over the selected position minus 3° on the flaps position indicator.

33-16-01-02 ANN LIGHT – DIM FUNCTION

33-16-01-02A (A)

RI	NI	NR	PLACARD	O	M
B	1	0	YES	NO	NO

May be inoperative.

- OR -

33-16-01-02B Daylight Operations (B)

RI	NI	NR	PLACARD	O	M
C	1	0	YES	NO	NO

May be inoperative for Daylight Operations.

33-26-01 CABIN SIGNS

RI	NI	NR	PLACARD	O	M
C	-	0	YES	NO	NO

May be inoperative provided that:

(a) Passenger address system is operative and is used to advise passengers and cabin attendant when seat belts should be fastened and use of personal electronic devices is prohibited, and

(b) Cabin attendant call and interphone are operative.

NOTE: *For Cargo Operations, no MEL Item entry is applicable as the system is not required.*

33-31-01 NOSE WHEEL WELL LIGHT

RI	NI	NR	PLACARD	O	M
C	1	0	NO	NO	NO

May be inoperative.

33-34-01 CARGO BAY LIGHT

RI	NI	NR	PLACARD	O	M
C	-	0	NO	NO	NO

May be inoperative.

33-37-01 MAIN WHEEL WELL LIGHT

RI	NI	NR	PLACARD	O	M
C	2	0	NO	NO	NO

One or both may be inoperative.

33-41-01 NAV LIGHTS SYSTEM

RI	NI	NR	PLACARD	O	M
C	1	0	YES	NO	NO

May be inoperative for daylight operation.

NOTE: *NAV Lights System includes 3 lights (1 green, 1 red and 1 white) and the associated system.*

33-42-01 LAND LIGHTS

33-42-01A (A)

RI	NI	NR	PLACARD	O	M
C	2	1	YES	NO	NO

One may be inoperative.

- OR -

33-42-01B Daylight Operations (B)

RI	NI	NR	PLACARD	O	M
C	2	0	YES	NO	NO

May be inoperative for Daylight Operations.

33-43-01 STROBE LIGHT *(IF INSTALLED)*

RI	NI	NR	PLACARD	O	M
D	3	0	YES	NO	NO

One or more STROBE lights may be inoperative provided that all BEACON lights are operative.

33-46-01 TAXI AND TO LIGHTS

33-46-01A (A)

RI	NI	NR	PLACARD	O	M
A	2	1	YES	NO	NO

One or both may be inoperative provided that one LAND light (L or R) is operative.

- OR -

33-46-01B Daylight Operations (B)

RI	NI	NR	PLACARD	O	M
C	2	0	YES	NO	NO

One or both may be inoperative for Daylight Operations.

33-47-01 LOGO LIGHTS *(IF INSTALLED)*

RI	NI	NR	PLACARD	O	M
D	2	0	YES	NO	NO

One or both may be inoperative.

33-48-01 BEACON LIGHT

33-48-01A (A)

RI	NI	NR	PLACARD	O	M
C	2	1	YES	YES	NO

One may be inoperative provided that all STROBE lights are operative

NOTE: *If the center fuselage BEACON light is inoperative, alternative procedures must be developed and used when aircraft is on ground with engine(s) running.*

OPERATIONAL (O) PROCEDURES

Ensure the area around the aircraft is clear of personnel prior to engine start and during ground engine running. Operate the anti-collision light switch as if the lights were operating normally.

- OR -

33-48-01B Daylight Operations (B)

RI	NI	NR	PLACARD	O	M
C	2	0	YES	YES	NO

One or both may be inoperative for Daylight Operations provided that all STROBE lights are operative.

NOTE: *If the center fuselage BEACON light is inoperative, alternate procedures must be developed and used when aircraft is on ground with engine(s) running.*

OPERATIONAL (O) PROCEDURES

Ensure the area around the aircraft is clear of personnel prior to engine start and during ground engine running.

Operate the anti-collision light switch as if the lights were operating normally.

33-49-01 WING INSPECTION LIGHTS

33-49-01A (A)

RI	NI	NR	PLACARD	O	M
B	2	1	YES	NO	NO

One may be inoperative provided that flight is not conducted into known or forecast icing conditions.

NOTE: *If right Wing Inspection Light is inoperative, HOTEL mode is prohibited.*

- OR -

33-49-01B Daylight Operations (B)

RI	NI	NR	PLACARD	O	M
C	2	0	YES	NO	NO

One or both may be inoperative for Daylight Operations.

33-49-02 IEP LIGHT

33-49-02A (A)

RI	NI	NR	PLACARD	O	M
B	1	0	NO	YES	NO

May be inoperative provided that the flight is not conducted into known or forecast icing conditions.

OPERATIONAL (O) PROCEDURES

► In case of inadvertent icing conditions encountered, keep the anti-icing systems activated until the end of the Flight (Icing AOA illuminated).

- OR -

33-49-02B Daylight Operations (B)

RI	NI	NR	PLACARD	O	M
C	1	0	NO	NO	NO

May be inoperative for Daylight Operations.

33-50-01 EMER EXIT (DISARM) LIGHT

RI	NI	NR	PLACARD	O	M
C	1	0	YES	NO	NO

May be inoperative provided that EMER EXIT light system is operative and armed prior to each departure.

33-50-02-01 PHOTOLUMINESCENT FLOOR PROXIMITY EMERGENCY ESCAPE PATH MARKING TAPE

RI	NI	NR	PLACARD	O	M
C	-	-	NO	NO	NO

One or more photoluminescent tape portion may be inoperative provided that:
(a) No more than one meter of photoluminescent is inoperative on each strip, and
(b) This inoperative part is neither facing nor continuous to an emergency exit.

NOTE: *For Cargo Operations, no MEL item entry is applicable as the system is not required.*

33-50-03 CEILING EMERGENCY LIGHTING

RI	NI	NR	PLACARD	O	M
B	-	-	NO	NO	NO

May be inoperative provided that a maximum of one in four consecutive overhead emergency lights may be inoperative.

NOTE: *For Cargo Operations, no MEL item entry is applicable as the system is not required.*

33-50-04 EMERGENCY EXIT SIGNS

33-50-04A (A)

RI	NI	NR	PLACARD	O	M
-	-	-	YES	NO	NO

One may be inoperative provided that:
(a) It does not indicate more than one emergency exit, and
(b) Associated exit is considered inoperative. Refer to Item 52-21-01-01 Emergency Exit (Including Passenger-Crew Doors, Service Door, but Excluding Flight Deck Emergency Exits) – Passenger or Combi Configuration.

←

- OR -

33-50-04B (B)

RI	NI	NR	PLACARD	O	M
C	-	-	YES	NO	NO

Up to 50% of the bulbs/LEDS may be inoperative in one or more signs provided that the sign remains legible.

←

33-50-05 EMEGENCY EXIT HANDLE LIGHTING

RI	NI	NR	PLACARD	O	M
-	2	1	YES	NO	NO

One may be inoperative provided that associated emergency exit is considered inoperative.

Refer to Item 52-21-01-01 Emergency Exit (Including Passenger-Crew Doors, Service Door, but Excluding Flight Deck Emergency Exits) – Passenger or Combi Configuration.

33-51-01 EXTERIOR EMER LIGHTS

33-51-01A (A)

RI	NI	NR	PLACARD	O	M
-	-	-	NO	NO	NO

One may be inoperative provided that the associated emergency exit is considered inoperative.
Refer to Item 52-21-01-01 Emergency Exit (Including Passenger – Crew Doors, but Excluding Flight Deck Emergency Exits) – Passenger or Combi Configuration

- OR -

33-51-01B Daylight Operations (B)

RI	NI	NR	PLACARD	O	M
B	-	-	NO	NO	NO

One light may be inoperative for Daylight Operations only.

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←

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34-11-01 AIR DATA COMPUTER (ADC)

←

RI	NI	NR	PLACARD	O	M
A	2	1	YES	YES	NO

One ADC may be inoperative provided that:
(a) The flight is conducted in day VMC only, and
(b) The TLU manual mode is operative, and
(c) The IESI is operative, and
(d) All the IOM DC are operative, and
(e) The operations are limited to two calendar days, and
(f) The operations are conducted in compliance with AFM AFCS limitations, Refer to AFM – Automatic Flight Control System.

←

OPERATIONAL (O) PROCEDURES

► ADC switching must be set to valid ADC
TLU manual mode check:
- ► Select HI SPD and check alert is generated after 25 s: MC, SC, FLT CTL TLU on FWS and TLU FAULT light.
- ► Check rudder travel is limited.
- ► Select AUTO and check alert disappears after 15 s.
- ► Check rudder travel is not limited.

■ **If ADC 1 is inoperative**
► LANDING ELEVATION.....SET PRESSURE ALTITUDE

34-16-01 ALTITUDE ALERTING SYSTEM

RI	NI	NR	PLACARD	O	M
B	1	0	YES	YES	NO

May be inoperative provided that:
(a) Autopilot altitude hold mode is operative, and
(b) Both main altimeters are operative, and
(c) The altitude alerting system is not part of the required equipment for intended operations.

OPERATIONAL (O) PROCEDURES

- **In flight**
  Flight crew should check on altimeters that there is no altitude excursion.
- **During climb, descent and altitude capture phases**
  Flight crew must ensure strict adherence to standardized callouts.

34-20-01 AHRS

←

RI	NI	NR	PLACARD	O	M
A	2	1	YES	YES	YES

One may be inoperative, provided that:
(a) The flight is conducted in day VMC only, and
(b) The operations are conducted in compliance with AFM AFCS limitations, (Refer to AFM – Automatic Flight Control System), and
(c) The IESI is operative, and
(d) Operations are limited to two calendar days.

OPERATIONAL (O) PROCEDURES

► ATT/HDG (affected side) ..... ALTERNATE SYS

MAINTENANCE (M) PROCEDURES

- **On ground if AHRS NOT ALIGN alert is triggered for more than 3 minutes**
  ► C/B ATT/HDG/AHRS NORM SPLY (affected side).....RESET
- **If AHRS NOT ALIGN alert remains**
  ► Open, secure and tag C/B ATT/HDG/AHRS NORM SPLY (affected side).

34-22-01-01 IESI

←

RI	NI	NR	PLACARD	O	M
A	1	0	YES	NO	NO

IESI may be inoperative provided that:
(a) The flight is conducted in day VMC only, and
(b) Both AHRS are operative, and
(c) Both ADC are operative, and
(d) Operations are limited to two calendar days.
←

34-22-01-02 IESI – STD BARO BUTTON

RI	NI	NR	PLACARD	O	M
C	1	0	YES	NO	NO

May be inoperative provided that the SET rotary sw and the SEL pb operates normally.

34-22-01-03 IESI – BRIGHTNESS BUTTON

RI	NI	NR	PLACARD	O	M
C	2	0	NO	NO	NO

One or both may be inoperative provided that the brightness level is acceptable for the flight crew on the affected side.

34-33-01 MARKER BEACON SYSTEM

34-33-01A (A)

RI	NI	NR	PLACARD	O	M
C	2	0	YES	NO	NO

One or both may be inoperative under IFR operations provided approach procedures do not require marker fixes.

- OR -

34-33-01B (B)

RI	NI	NR	PLACARD	O	M
D	2	0	YES	NO	NO

One or both may be inoperative for flight under VFR operations.

34-36-01 ILS

34-36-01 A (A)

RI	NI	NR	PLACARD	O	M
C	2	1	NO	YES	NO

One may be inoperative under IFR operations provided that:
(a) The remaining ILS is operative, and
(b) Landing capability is limited to CAT I.

NOTE: *If ILS 2 glideslope signal is inoperative, GPWS Reactive modes – Glideslope Deviation (mode 5) is inhibited. Refer to Item 34-48-03-03 GPWS – GPWS Reactive modes – Glideslope Deviation (Mode 5).*

OPERATIONAL (O) PROCEDURES
Flight planning must ensure that, when the destination requires an ILS approach, the designated alternate aerodrome(s) provides at least one suitable non-ILS approach.

- OR -

34-36-01B (B)

RI	NI	NR	PLACARD	O	M
C	2	0	NO	NO	NO

May be inoperative under IFR operations provided approaches and missed approaches where navigation is based on ILS are not included in the flight plan.

NOTE: *If ILS 2 glideslope signal is inoperative, GPWS Reactive modes – Glideslope Deviation (mode 5) is inhibited. Refer to Item 34-48-01-05 TAWS – GPWS Reactive modes – Glideslope Deviation (Mode 5).*

- OR -

34-36-01C (C)

RI	NI	NR	PLACARD	O	M
D	2	0	NO	NO	NO

One or both may be inoperative for flight in accordance with VFR operations.

34-38-01-01 AIRPORT NAVIGATION FUNCTION *(IF INSTALLED)* - AIRPORT NAVIGATION FUNCTION

RI	NI	NR	PLACARD	O	M
D	2	0	YES	NO	NO

One or both may be inoperative.

NOTE: *Database(s) which is/are out of date is/are considered to be inoperative.*

34-38-01-02 AIRPORT NAVIGATION FUNCTION *(IF INSTALLED)* - AIRPORT DATABASES

34-38-01-02A (A)

RI	NI	NR	PLACARD	O	M
C	-	0	NO	NO	NO

One or more may be inoperative provided that airport maps required for the flight have not been amended in current database.

NOTE: *Database(s) which is/are out of date is/are considered to be inoperative.*

- OR -

34-38-01-02B (B)

RI	NI	NR	PLACARD	O	M
D	-	0	NO	NO	NO

One or more may be inoperative provided that the airport navigation function is not used.

NOTE: *Database(s) which is/are out of date is/are considered to be inoperative.*

34-41-01 WEATHER RADAR

RI	NI	NR	PLACARD	O	M
C	1	0	YES	NO	NO

May be inoperative provided that:

(a) For daylight operations ONLY as required by TTCAR 7.27.

(b) The weather reports or forecast available to the captain indicate that cumulonimbus clouds or other potentially hazardous weather conditions, which could be detected by the system in working order are unlikely to be encountered on the intended flight route.

34-42-01 RADIO ALTIMETER SYSTEM

RI	NI	NR	PLACARD	O	M
-	1	0	YES	YES	NO

One or more may be inoperative provided that:

(a) GPWS is considered inoperative, Refer to Item 34-48-01-01 TAWS - GPWS Reactive Modes and TERRAIN Predictive Modes, and

(b) TCAS is considered inoperative, Refer to Item 34-57-01-01 Traffic Alert and Collision Avoidance System – TCAS System.

OPERATIONAL (O) PROCEDURES

- **During Final Cockpit Preparation**
  - ► GPWS .....OFF

  TCAS is failed when RA is failed.
- **During Descent Preparation**
  - ► ARRIVAL BRIEFING .....ANNOUNCE STICK PUSHER INHIBITION BELOW 500 FT AGL INOPERATIVE
- **During Approach**
  - ► SPD TGT..... MAN SEL

34-48-01-01 TAWS – GPWS REACTIVE MODES AND TERRAIN PREDICTIVE MODES

RI	NI	NR	PLACARD	O	M
A	-	0	YES	YES	NO

One or more may be inoperative provided that:

(a) Operations are not conducted into TNCM, TDPD, TLPC, TAPA or TVSA unless in day VMC, and

(b) Repairs or replacements are carried out within six further flights, or twenty-five flying hours or two calendar days, whichever occurs first.

NOTE: *GPWS and TERRAIN Callouts are lost.*

OPERATIONAL (O) PROCEDURES

► GPWS ..... OFF

► TERR ..... OFF

1) Flight crew must maintain increased awareness of airplane altitude and flight path.
2) Flight crew should make use of all available systems (autopilot, ILS, FMS).
3) The Pilot Monitoring should use standard calls to alert the Pilot Flying of any parameter approaching limits.
4) The Pilot Monitoring should monitor the flight path during takeoff, approach or landing, and alert the Pilot Flying should any of the following conditions exist:
   (a) Excessive descent rate.
   (b) Excessive terrain closure.
   (c) Excessive bank angle
   (d) Altitude loss after takeoff or go-around.
   (e) Unsafe terrain clearance when not in landing configuration.
   (f) Excessive deviation below glideslope on approach.
   (g) Abnormal radio altitude indications.
5) Conduct a thorough review of all Minimum Safe Altitudes (MSA's) during the Arrival and Departure Briefings.

34-48-01-02 TAWS – GPWS REACTIVE MODES

RI	NI	NR	PLACARD	O	M
A	-	0	YES	YES	NO

One or more may be inoperative provided that:

(a) TERRAIN Predictive modes are operative,

(b) Operations are not conducted into TNCM, TDPD, TLPC, TAPA or TVSA unless in day VMC, and

(c) Repairs or replacements are carried out within six further flights, or twenty-five hours or two calendar days, whichever occurs first.

NOTE: *GPWS Callouts are lost.*

OPERATIONAL (O) PROCEDURES

► GPWS.....OFF

1) Flight crew must maintain increased awareness of airplane altitude and flight path.
2) Flight crew should make use of all available systems (autopilot, ILS, FMS).
3) Monitor for abnormal radio altitude indications.
4) Conduct a thorough review of all Minimum Safe Altitudes (MSA's) during the Arrival and Departure Briefings.

34-48-01-03 TAWS – TERRAIN PREDICTIVE MODES

RI	NI	NR	PLACARD	O	M
A	-	0	YES	YES	NO

One or more may be inoperative provided that:

(a) TERRAIN Predictive modes are operative,

(b) Operations are not conducted into TNCM, TDPD, TLPC, TAPA or TVSA unless in day VMC, and

(c) Repairs or replacements are carried out within six further flights, or twenty-five hours or two calendar days, whichever occurs first.

NOTE: *GPWS Callouts are lost.*

OPERATIONAL (O) PROCEDURES

► TERR ..... OFF

1) Flight crew must maintain increased awareness of airplane altitude and flight path.
2) Flight crew should make use of all available systems (autopilot, ILS, FMS).
3) The Pilot Monitoring should use standard calls to alert the Pilot Flying of any parameter approaching limits.
4) The Pilot Monitoring should monitor the flight path during takeoff, approach or landing, and alert the Pilot Flying should any of the following conditions exist:
   (a) Excessive descent rate.
   (b) Excessive terrain closure.
   (c) Excessive bank angle
   (d) Altitude loss after takeoff or go-around.
   (e) Unsafe terrain clearance when not in landing configuration.
   (f) Excessive deviation below glideslope on approach.
   (g) Abnormal radio altitude indications.
5) Conduct a thorough review of all Minimum Safe Altitudes (MSA's) during the Arrival and Departure Briefings.

34-48-01-04 TAWS – TEST MODE

RI	NI	NR	PLACARD	O	M
A	-	0	YES	YES	NO

(a) TAWS is considered inoperative, Refer to Item 34-48-01-01 TAWS – GPWS Reactive modes and TERRAIN Predictive modes,
(b) Operations are not conducted into TNCM, TDPD, TLPC, TAPA or TVSA unless in day VMC, and
(c) Repairs or replacements are carried out within six further flights, or twenty-five flying hours or two calendar days, whichever occurs first.

NOTE: *GPWS and TERRAIN Callouts are lost.*

OPERATIONAL (O) PROCEDURES

► GPWS .....OFF
► TERR .....OFF

1) Flight crew must maintain increased awareness of airplane altitude and flight path.
2) Flight crew should make use of all available systems (autopilot, ILS, FMS).
3) The Pilot Monitoring should use standard calls to alert the Pilot Flying of any parameter approaching limits.
4) The Pilot Monitoring should monitor the flight path during takeoff, approach or landing, and alert the Pilot Flying should any of the following conditions exist:
   (a) Excessive descent rate.
   (b) Excessive terrain closure.
   (c) Excessive bank angle
   (d) Altitude loss after takeoff or go-around.
   (e) Unsafe terrain clearance when not in landing configuration.
   (f) Excessive deviation below glideslope on approach.
   (g) Abnormal radio altitude indications.
5) Conduct a thorough review of all Minimum Safe Altitudes (MSA's) during the Arrival and Departure Briefings.

34-48-01-05 TAWS – GPWS REACTIVE MODES – GLIDESLOPE DEVIATION (MODE 5)

34-48-01-05A (A)

RI	NI	NR	PLACARD	O	M
B	-	0	NO	YES	NO

One or more may be inoperative.

OPERATIONAL (O) PROCEDURES
The Pilot Monitoring will call out any excessive glideslope deviation.

- OR

34-48-01-05B DAY VMC FLIGHT ONLY (B)

RI	NI	NR	PLACARD	O	M
C	-	0	NO	YES	NO

One or more may be inoperative provided that the flight is conducted in day VMC only.

OPERATIONAL (O) PROCEDURES
The Pilot Monitoring will call out any excessive glideslope deviation.

34-50-01 VOR

RI	NI	NR	PLACARD	O	M
C	-	-	YES	YES	NO

One or more may be inoperative provided that:

(a) The navigation systems required for each segment of the intended flight route are operative, and

(b) Alternate procedures are established and used, where applicable.

OPERATIONAL (O) PROCEDURES

Use of alternate available NAV Systems i.e., GPS other VOR NDB.

34-50-02 DME

RI	NI	NR	PLACARD	O	M
C	-	-	YES	YES	NO

One or more may be inoperative provided that:

(a) The navigation systems required for each segment of the intended flight route are operative, and

(b) Alternate procedures are established and used, where applicable.

OPERATIONAL (O) PROCEDURES

Use of alternate available NAV Systems i.e., GPS other VOR NDB.

34-50-03 ADF

RI	NI	NR	PLACARD	O	M
C	-	-	YES	YES	NO

One or more may be inoperative provided that:

(a) The navigation systems required for each segment of the intended flight route are operative, and

(b) Alternate procedures are established and used, where applicable.

OPERATIONAL (O) PROCEDURES

Use of alternate available NAV Systems i.e., GPS other VOR NDB.

34-52-02 SSR TRANSPONDER SYSTEM – MODE S FUNCTION *(IF INSTALLED)*

34-52-02A (A)

RI	NI	NR	PLACARD	O	M
D	-	-	NO	NO	NO

Any in excess of those required for the intended route may be inoperative.

NOTE: *1) Altitude reporting, provided by an ATC Mode S Transponder, is required for TCAS II operation. Refer to item 34-57-01-01 Traffic Alert and Collision Avoidance System – TCAS System for flight with TCAS inoperative.*

2) Elementary Surveillance (ELS) function is required in European Mode S designated airspace.

- OR -

34-52-02B (B)

RI	NI	NR	PLACARD	O	M
C	-	0	NO	NO	NO

One or more may be inoperative provided permission is obtained from the Air Navigation Service Provider(s) when required for the intended flight route.

NOTE: *1) Altitude reporting, provided by an ATC Mode S Transponder, is required for TCAS II operation. Refer to item 34-57-01-01 Traffic Alert and Collision Avoidance System – TCAS System for flight with TCAS inoperative.*

2) Elementary Surveillance (ELS) function is required in European Mode S designated airspace.

34-52-03 SSR TRANSPONDER SYSTEM – ENHANCED SURVEILLANCE FUNCTION *(IF INSTALLED)*

34-52-03A Single European Sky Operations (A)

RI	NI	NR	PLACARD	O	M
A	-	0	NO	NO	NO

One or more Downlinked Aircraft Parameters which provide Enhanced Surveillance, may be inoperative when required for the intended route provided operations are limited to three consecutive days

- OR -

34-52-03B (B)

RI	NI	NR	PLACARD	O	M
C	-	0	NO	NO	NO

One or more Downlinked Aircraft Parameters which provide Enhanced Surveillance, may be inoperative when required for the intended route.

- OR -

34-52-03C (C)

RI	NI	NR	PLACARD	O	M
D	-	0	NO	NO	NO

One or more Downlinked Aircraft Parameters which provide Enhanced Surveillance, may be inoperative when not required for the intended route.

34-52-04 SSR TRANSPONDER SYSTEM – EXTENDED SQUITTER TRANSMISSION FUNCTION (ADS-B OUT) *(IF INSTALLED)*

34-52-04A Single European Sky Operations (A)

RI	NI	NR	PLACARD	O	M
A	-	0	NO	NO	NO

One or more extended squitter transmissions may be inoperative when required for the intended flight route provided operations are limited to three consecutive days.

- OR -

34-52-04B (B)

RI	NI	NR	PLACARD	O	M
C	-	0	NO	NO	NO

One or more extended squitter transmissions may be inoperative when required for the intended route.

- OR -

34-52-04C (C)

RI	NI	NR	PLACARD	O	M
D	-	0	NO	NO	NO

One or more extended squitter transmissions may be inoperative when not required for the intended route.

34-57-01-01 TRAFFIC ALERT AND COLLISION AVOIDANCE SYSTEM – TCAS SYSTEM

RI	NI	NR	PLACARD	O	M
A	1	0	YES	YES	NO

May be inoperative for ten calendar days provided that the system is set to STBY.

OPERATIONAL (O) PROCEDURES

1) Operate within RADAR environment only.
2) Dispatch to CCS not permitted.
3) The associated TCAS C/B is pulled for TCAS deactivation.

Limitation

► TCAS: ..... STBY

NOTE: *TCAS on the VCP should automatically be switched to STBY mode. If not, manually switch the TCAS to STBY mode using the VCP and MCP controls.*

34-58-01 GPS – GNSS

34-58-01A No GPS Operative (A)

RI	NI	NR	PLACARD	O	M
C	1	0	YES	YES	NO

May be inoperative provided that:
(a) The navigation systems required for each segment of the intended flight route are operative, and
(b) Alternate procedures are established and used, where applicable.
←

NOTE: *1) TERRAIN is inoperative, Refer to Item 34-48-01-03 TAWS – TERRAIN Predictive modes*
2) GNSS parameters are no longer available for Transponder System, Refer to Item 34-52-03 SSR Transponder System - Enhanced Surveillance Function (if installed) and, Refer to SSR Transponder System – Extended Squitter Transmission Function (ADS-B OUT) (if installed).

OPERATIONAL (O) PROCEDURES
Dispatch with inoperative or expired Navigational Database:
1) Enter flight plan into FMCS (waypoint, etc) using standard FMCS procedures.
2) Cross check FMCS waypoints using current flight crew navigation charts.
3. Manually tune and identify approach navigation radios.

- OR -

34-58-01B One GPS Operative (B)

RI	NI	NR	PLACARD	O	M
C	2	1	YES	YES	NO

One or more may be inoperative provided that:
(a) The navigation systems required for each segment of the intended flight route are operative, and
(b) Alternate procedures are established and used, where applicable.
←

OPERATIONAL (O) PROCEDURES
Verify all Resolution Advisory (RA) displays and audio functions are operative.

34-61-01 FLIGHT MANAGEMENT SYSTEM

RI	NI	NR	PLACARD	O	M
C	2	0	YES	YES	NO

One or both may be inoperative provided that operations/procedures do not require its use.

OPERATIONAL (O) PROCEDURES

Dispatch with inoperative or expired Navigational Database:

1) Enter flight plan into FMCS (waypoint, etc) using standard FMCS procedures.
2) Cross check FMCS waypoints using current flight crew navigation charts.
3. Manually tune and identify approach navigation radios.

Refer to AFM - LIMITATIONS Chapter.

34-61-02 NAVIGATION DATABASES

34-61-02A (A)

RI	NI	NR	PLACARD	O	M
C	-	0	NO	NO	NO

One or more may be inoperative for the intended route where conventional (non-RNAV) navigation is sufficient, provided that:

(a) Current aeronautical information (e.g. charts) is available for the entire route and for the airports intended to be used, and

(b) Navigation database information is disregarded.

NOTE: *Database(s) which is/are out of date is/are considered to be inoperative.*

- OR -

34-61-02B (B)

RI	NI	NR	PLACARD	O	M
C	-	1	NO	NO	NO

Any in excess of one may be inoperative provided that:

(a) The operative database is up to date for routes, departures, arrival and approach procedures that require the use of navigational Database for RNAV, and

(b) This up to date Database is readily available to the flight crew member(s) responsible for navigation.

NOTE: *Database(s) which is/are out of date is/are considered to be inoperative.*

- OR -

34-61-02 NAVIGATION

34-61-02C (C)

RI	NI	NR	PLACARD	O	M
A	-	0	NO	NO	NO

One or more may be inoperative for a maximum of 10 calendar days provided that:

(a) Area Navigation (RNAV) departure, arrival and approach procedures do not depend on the data amended in the current database cycle or Conventional (Non-RNAV) departure, arrival and approach procedures, when available, or ANSP assistance are used as an alternative to RNAV procedures which have been amended in the current database cycle,

(b) Before each flight, current aeronautical information is used to verify the database Navigation Fixes, the coordinates, frequencies, status (as applicable) and suitability of Navigation Facilities required for the intended route, and

(c) Radio navigation aids, which are required to be flown for departure, arrival and approach procedures and which have been amended in the current database cycle, are manually tuned and identified.

NOTE: *Database(s) which is/are out of date is/are considered to be inoperative.*

34-61-02 NAVIGATION DATABASES

34-61-02A (A)

RI	NI	NR	PLACARD	O	M
C	-	0	NO	NO	NO

One or more may be inoperative for the intended route where conventional (non-RNAV) navigation is sufficient, provided that:

(a) Current aeronautical information (e.g. charts) is available for the entire route and for the airports intended to be used, and

(b) Navigational database information is disregarded.

NOTE: *Database(s) which is/are out of date is/are considered to be inoperative.*

- OR -

34-61-02B (B)

RI	NI	NR	PLACARD	O	M
C	-	1	NO	NO	NO

Any in excess of one may be inoperative provided that:

(a) The operative database is up to date for routes, departures, arrival and approach procedures that require the use of navigational Database for RNAV, and

(b) This up to date Database is readily available to the flight crew member(s) responsible for navigation.

NOTE: *Database(s) which is/are out of date is/are considered to be inoperative.*

- OR -

34-61-02 NAVIGATION

34-61-02C (C)

RI	NI	NR	PLACARD	O	M
A	-	0	NO	NO	NO

One or more may be inoperative for a maximum of 10 calendar days provided that:

(a) Area Navigation (RNAV) departure, arrival and approach procedures do not depend on the data amended in the current database cycle or Conventional (Non-RNAV) departure, arrival and approach procedures, when available, or ANSP assistance are used as an alternative to RNAV procedures which have been amended in the current database cycle,

(b) Before each flight, current aeronautical information is used to verify the database Navigation Fixes, the coordinates, frequencies, status (as applicable) and suitability of Navigation Facilities required for the intended route, and

(c) Radio navigation aids, which are required to be flown for departure, arrival and approach procedures and which have been amended in the current database cycle, are manually tuned and identified.

NOTE: *Database(s) which is/are out of date is/are considered to be inoperative.*

←

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35-11-01 MAIN SUPPLY – LO PR LIGHT

RI	NI	NR	PLACARD	O	M
B	1	0	YES	NO	NO

May be inoperative provided that maximum altitude is limited to 10 000 ft pressure altitude.

35-12-10 OXYGEN OVERBOARD DISCHARGE INDICATOR (GREEN DISC)

RI	NI	NR	PLACARD	O	M
C	1	0	NO	YES	YES

May be damaged or missing provided that the oxygen pressure is checked by direct reading before each flight.

OPERATIONAL (O) PROCEDURES

Refer to FCOM LIM-5-35-2 Minimum Bottle Pressure to ensure that the remaining oxygen quantity is sufficient for the intended flight.

MAINTENANCE (M) PROCEDURES

Refer to MP ATR-A-35-12-XX-00ZZZ-362Z-A.

35-13-01 FLIGHT DECK PRESSURE INDICATION

RI	NI	NR	PLACARD	O	M
B	1	0	YES	NO	YES

May be inoperative provided that oxygen pressure is checked on the bottle direct reading gauge prior to each departure.

MAINTENANCE (M) PROCEDURES

1) Check crew oxygen cylinder pressure indicator prior to each takeoff.
2) Ensure supply is above minimum required for flight.

35-13-02 OBSERVER SEAT OXYGEN MASK

35-13-02A (A)

RI	NI	NR	PLACARD	O	M
C	1	0	YES	YES	NO

May be inoperative provided that:
(a) Observer seat is not occupied,
(b) PED storage and use is prohibited in the cockpit, and
(c) Conventional documents in paper format are available and used.

OPERATIONAL (O) PROCEDURES

- ► Required set of documentation must be made available before flight departure. Conventional paper procedures will be used to ensure the conduct of the flight.
- ► Alternative means of performances and weight and balance calculations must be used.
- ► Use of paper takeoff and landing card is compulsory.

- OR -

35-13-02B (B)

RI	NI	NR	PLACARD	O	M
C	1	0	YES	YES	NO

May be inoperative provided that:
(a) Maximum altitude is limited to 10 000 ft,
(b) PED storage and use is prohibited in the cockpit, and
(c) Conventional documents in paper format are available and used.

OPERATIONAL (O) PROCEDURES

- ► Required set of documentation must be made available before flight departure. Conventional paper procedures will be used to ensure the conduct of the flight.
- ► Alternative means of performances and weight and balance calculations must be used.
- ► Use of paper takeoff and landing card is compulsory.

35-15-01 BOTTLE GAUGE

RI	NI	NR	PLACARD	O	M
B	1	0	YES	NO	NO

May be inoperative provided that flight deck pressure indication is operative.

35-20-01 PASSENGER OXYGEN SYSTEM – PASSENGER CONFIGURATION

RI	NI	NR	PLACARD	O	M
B	1	0	YES	YES	NO

May be inoperative provided that maximum altitude is limited to 10 000 ft pressure altitude.

OPERATIONAL (O) PROCEDURES

1) Prior to flight, the pilot in command shall brief cabin crew on the alternate emergency procedures that will be used.

←

35-20-03 PASSENGER SERVICE UNIT

RI	NI	NR	PLACARD	O	M
B	-	-	YES	YES	YES

One or more passenger service units may be inoperative provided that:

(a) Affected seats are blocked and placarded to prevent occupancy, and

(b) Units are operative for all operative passenger seats, toilet compartment (if applicable) and cabin attendant location (if applicable).

OPERATIONAL (O) PROCEDURES

1) Prior to flight, the pilot in command shall brief cabin crew on the alternate emergency procedures that will be used.

MAINTENANCE (M) PROCEDURES

Ensure that there are portable oxygen units for at least 10 percent of the passengers.

35-35-01 PORTABLE OXYGEN BOTTLE

RI	NI	NR	PLACARD	O	M
D	2	1	YES	YES	YES

Any in excess of those required may be inoperative or missing provided:
(a) Required distribution of operative units is maintained throughout the aircraft,
(b) The inoperative unit is:
- removed from the aircraft and its installed location is placarded inoperative; or
- removed from the installed location, secured out of sight, and the unit and its installed location are placarded inoperative, and

(c) Procedures are established and used to alert crew members of inoperative or missing equipment.

Note: *When determining the location for storage of the inoperative units, compliance with the dangerous goods requirements must be considered.*

MAINTENANCE (M) PROCEDURES

1) Until it can be service or removed from the airplane at the next available maintenance base, stow inoperative unit(s) in a location where they will not be used.
2) If units at a specific storage location are inoperative or missing, remove or obscure the associated location placarding for the affected units.
3) Ensure required distribution of serviceable units is maintained throughout the airplane.

OPERATION NOTE

During flight when crew portable oxygen is administered to a passenger, use of the gaseous bottle should be discontinued when the gauge pressure reads approximately 1,100 p.s.i.
This will ensure that each serviceable bottle has reserve oxygen available in case of a depressurization requirement.
Smoking is prohibited when oxygen is being administered. Relevant speech to be made by Purser.

35-65-01 OXYGEN MANUAL RELEASE TOOL

35-65-01A (A)

RI	NI	NR	PLACARD	O	M
C	4	3	YES	NO	NO

One may be inoperative or missing provided that the remaining tool(s) is (are) checked operative before each departure.

- OR –

35-65-01B (B)

RI	NI	NR	PLACARD	O	M
C	4	3	YES	NO	NO

May be inoperative or missing provided that Passenger Oxygen System is considered inoperative.
Refer to Item 35-20-01 Passenger Oxygen System – Passenger Configuration.

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36-11-01 HP VALVE

RI	NI	NR	PLACARD	O	M
C	2	1	YES	NO	YES

One may be inoperative provided that it is deactivated.

MAINTENANCE (M) PROCEDURES

Refer to MP ATR-A-36-11-60-01ZZZ-560Z-A.

36-11-02-01 BLEED VALVE – PASSENGER CONFIGURATION

RI	NI	NR	PLACARD	O	M
C	2	1	YES	YES	YES

One may be inoperative provided that:
(a) Associated bleed valve is secured CLOSED,
(b) Maximum operating altitude is limited to 17 000 ft,
(c) Associated bleed valve is selected OFF, and
(d) Opposite bleed valve and pack are operative.
←

OPERATIONAL (O) PROCEDURES
As an in-flight failure could imply total loss of pressurization, fuel consumption at 10 000 ft must be taken into account to compute the trip fuel.

NOTE: *With one bleed INOP, whilst on ground, with both engines running, the crossfeed valve remains open with the associated X FEED indicator illuminated.*

LIMITATIONS
► LARGE RAPID POWER CHANGE: AVOID

MAINTENANCE (M) PROCEDURES
Refer to MP ATR-A-36-11-60-00ZZZ-560Z-A.

36-11-03 BLEED FAULT ALERT

36-11-03A (A)

RI	NI	NR	PLACARD	O	M
C	2	1	YES	NO	NO

One may be inoperative provided that associated pack FAULT alert is operative.

- OR -

36-11-03B (B)

RI	NI	NR	PLACARD	O	M
-	2	1	YES	NO	NO

One may be inoperative provided that associated bleed valve is considered inoperative and selected OFF:
Refer to Item 36-11-02-01 BLEED Valve – Passenger Configuration.

36-11-04 GROUND X FEED VALVE

RI	NI	NR	PLACARD	O	M
C	1	0	YES	NO	YES

May be inoperative provided that it is secured closed.

MAINTENANCE (M) PROCEDURES

Refer to MP ATR-A-36-11-20-00ZZZ-560Z-A.

36-11-05 GRD X FEED – X VALVE OPEN LIGHT

RI	NI	NR	PLACARD	O	M
C	1	0	YES	NO	YES

May be inoperative provided that X FEED valve is checked closed.

MAINTENANCE (M) PROCEDURES

Check of Crossfeed Valve:

Location: Under passenger compartment, access doors 231DF and 231EF.

► Open, secure and tag C/B AIR BLEED/X FEED.
► Open access doors 231DF and 231EF.
► Check that mechanical indicator shows closed.
► Close access doors.
► Install placard.

36-11-06 OVHT ALERT

RI	NI	NR	PLACARD	O	M
-	2	1	YES	NO	NO

One may be inoperative provided that associated bleed valve is considered inoperative and selected OFF.

Refer to Item 36-11-02-01 BLEED Valve – Passenger Configuration.

36-22-01 LEAK ALERT

RI	NI	NR	PLACARD	O	M
-	2	1	YES	NO	NO

One may be inoperative provided that:

(a) Associated bleed valve is considered inoperative and selected OFF,
Refer to Item 36-11-02-01 BLEED Valve - Passenger Configuration, and

(b) Ground X Feed valve is secured CLOSED,
Refer to Item 36-11-04 Ground X Feed Valve.

36-70-01 AIR BL LEAK PTT BUTTON (RH MAINTENANCE PANEL)

RI	NI	NR	PLACARD	O	M
C	1	0	NO	NO	NO

May be inoperative provided pneumatic system is operative.

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42-11-01 IOM DC

NO GO

42-11-02 IOM AP

RI	NI	NR	PLACARD	O	M
-	2	0	NO	NO	NO

One or both may be inoperative.
Refer to Item 22-18-01 Autopilot Function
Refer to Item 22-18-03 Yaw Damper Function

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46-25-01-01 ELECTRONIC FLIGHT BAG – SYSTEM DEVICE

46-25-01-01A (A)

RI	NI	NR	PLACARD	O	M
C	2	0	YES	YES	YES

OPERATIONAL (O) PROCEDURES

One EFB inoperative

On inoperative side revert to using paper charts and documents.

Both EFB inoperative

Revert to using paper charts and documents.

MAINTENANCE (M) PROCEDURES

Secure or remove associated EFB and hardware.

46-25-01-02 ELECTRONIC FLIGHT BAG – MOUNTING DEVICE

46-25-01-02A (A)

RI	NI	NR	PLACARD	O	M
C	2	0	YES	YES	YES

OPERATIONAL (O) PROCEDURES – 1 Mounting Device Inoperable

1) When not in use, the affected PED should be stored in a secure location so as not to obstruct any aircraft components such as panels, switches, lights, instruments, displays or flight controls.
2) Ensure that all taxi routes, SIDs, STARs and approach procedures in use are reviewed thoroughly prior to any critical phases of flight.
   NOTE: *Critical phases of flight include, but are not limited to, any time the aircraft is moving under its own power below 10,000 ft.*
3) Verify that the relevant information pertaining to all SIDs, STARs and IAPs in use are cross-checked with the data displayed on the FMS and the Navigation Displays.
4) During taxi, the Pilot Monitoring will reference the appropriate taxiway charts and provide detailed guidance to the Pilot Flying. Exercise extreme caution at all times. Should any doubt exist about the location of the aircraft along the taxiway route, STOP IMMEDIATELY, so that the PF may review the appropriate charts.

OPERATIONAL (O) PROCEDURES – Both Mounting Devices Inoperable

Revert to using paper charts and documents.

NOTE: *Paper charts and documents may not be available at stations other than TTPP.*

MAINTENANCE (M) PROCEDURES

Secure or remove associated EFB and hardware.

46-25-01-03 ELECTRONIC FLIGHT BAG – POWER CONNECTION

46-25-01-03A (A)

RI	NI	NR	PLACARD	O	M
C	2	0	YES	YES	NO

May be inoperative provided:

(a) Each PED onboard has a minimum internal battery charge of 80% prior to dispatch.

(b) Paper charts and documents may be substituted for each PED with an internal battery charge of less than 80%.

NOTE: *Paper charts and documents may not be available at stations other than TTPP.*

OPERATIONAL (O) PROCEDURES

1) If practicable, ensure that all PEDs onboard are fully charged prior to dispatch.
2) Consider switching off affected PEDs when not in use to preserve internal battery charge.
3) Consider dispatching with paper enroute navigation charts onboard to preserve PED battery charge during the cruise portion of the flight.
4) After dispatch, use the spare PED as necessary to substitute any PED with a low battery charge.
5) Use any remaining power connections to charge PEDs on a rotational basis. Use only approved power connections.
6) If possible, ensure a minimum battery charge of 20% for all PEDs in use prior to top of descent.

46-25-01-04 ELECTRONIC FLIGHT BAG – DATA CONNECTIVITY

46-25-01-04A (A)

RI	NI	NR	PLACARD	O	M
C	2	0	YES	YES	NO

May be inoperative provided:

(a) EFB data is current and will remain current for the duration of each flight sector.

(b) Paper charts and documents are substituted for any EFB that will not remain current for the duration of each flight sector.

NOTE: *Paper charts and documents may not be available at stations other than TTPP.*

OPERATIONAL (O) PROCEDURES

Check the EFB database to ensure that data is current and will remain current for the duration of each flight sector or revert to using paper charts and documents.

NOTE: *EFB data can be manually updated or PEDs replaced at TTPP. Dispatch should only be allowed if return to this Station can be reasonably assured.*

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52-11-01 DOOR ALERT SYSTEM

RI	NI	NR	PLACARD	O	M
B	1	0	YES	YES	YES

May be inoperative provided that affected doors are checked locked closed prior to each flight.

NOTE: *Maintenance Procedure is applicable for Flight Compartment Overhead Hatch only.*

OPERATIONAL (O) PROCEDURES

► Check visually that the doors are latched, locked and closed prior to each flight.

MAINTENANCE (M) PROCEDURES

► Check that flight compartment overhead hatch is closed:
- ► Manually operate handle to ensure the shoot bolt operates normally.
- ► Close and latch the hatch.
- ► Verify the handle is flush in the closed position.

NOTE: *Applicable only for Flight Compartment Overhead Hatch.*

52-11-02 SECONDARY HANDRAIL *(IF INSTALLED)*

RI	NI	NR	PLACARD	O	M
D	1	0	YES	NO	YES

May be inoperative provided that the rail is secured in retracted position or removed.

MAINTENANCE (M) PROCEDURES

For removal, Refer to MP ATR-A-52-11-20-A0ZZZ-520Z-A.

52-12-01 PASSENGER DOOR TORQUE TUBE

52-12-01A Ferry Flight (A)

RI	NI	NR	PLACARD	O	M
A	1	0	YES	NO	NO

May be inoperative for one Ferry Flight to a base where repair is possible provided that:
(a) Ground personnel assistance for opening and closing operation is available, and
(b) Door is not operated from inside the aircraft (except in case of emergency), and
(c) It is checked that nobody is right below the door during the operation, and
(d) All possibly concerned personnel are informed about the unbalanced condition.

52-21-01-01 EMERGENCY EXIT (INCLUDING PASSENGER – CREW DOORS, SERVICE DOOR, BUT EXCLUDING FLIGHT DECK EMERGENCY EXITS) – PASSENGER OR COMBI CONFIGURATION

RI	NI	NR	PLACARD	O	M
A	4	3	YES	YES	YES

One may be inoperative for a maximum of five flights provided that:

(a) The passenger number reduction and distribution policy, and cabin safety procedures are established and used, and

(b) The affected emergency exit is closed and locked, and

(c) A conspicuous barrier, strap or rope and a placard stating "DO NOT USE" are placed across the affected emergency exit prior to passenger boarding, and

(d) The affected emergency exit is not used for passenger boarding, nor for any purpose whilst passengers are on board, and

(e) Visual indications (illuminated and non-illuminated) directing passengers to the affected emergency exit are obscured, and

(f) All crew members are briefed on the location and condition of the affected emergency exit, passenger distribution and modified cabin safety procedures, and

(g) The affected emergency exit and blocked seating layout are checked before each flight by the relevant cabin crew member, and

(h) The escape path to the affected emergency exit is check by the relevant cabin crew member to be unobstructed before each takeoff and landing, and

(i) Extended overwater flight is prohibited.

NOTE: *If the affected emergency exit is operative mechanically, it may still be used for evacuation in the case of emergency.*

OPERATIONAL (O) PROCEDURES

Flight crew members and passengers must be brief prior to takeoff and landing regarding exit that cannot be used for emergency evacuation.

52-21-01-01 EMERGENCY EXIT (INCLUDING PASSENGER – CREW DOORS, SERVICE DOOR, BUT EXCLUDING FLIGHT DECK EMERGENCY EXITS) – PASSENGER OR COMBI CONFIGURATION *(Cont'd)*

MAINTENANCE (M) PROCEDURES

Configure the door, exit signs and seats as appropriate.

1. Cover the emergency exit sign of the affected door so it is not visible.
2. Cover or otherwise obscure the floor proximity escape path marking system lights unique to the inoperative door.
3. Block the inoperative door with a conspicuous barrier strap or rope and placard INOPERATIVE.
4. Block unusable passenger seats with conspicuous barrier tape or ropes and placard DO NOT OCCUPY as follows:

(9Y-TTA, 9Y-TTB, 9Y-TTC, 9Y-TTD, 9Y-TTE, 9Y-TTG, 9Y-TTH, 9Y-TTJ) ONLY

Inoperative Door	Block out Seat Rows
L or R Emergency Exit	1 and 2
Service Door	10 to 17

(9Y-TTI, 9Y-TTF) ONLY

Inoperative Door	Block out Seat Rows
L or R Emergency Exit	1, 2 & 3
Service Door	11 to 19

52-31-01 CARGO DOOR MOTOR ACTUATOR *(IF INSTALLED)*

52-31-01A (A)

RI	NI	NR	PLACARD	O	M
C	1	0	NO	YES	YES

May be inoperative provided that:
(a) An approved mechanical device (crank) installed in the aircraft is used for cargo door operation, and
(b) The cargo door is checked closed, latched and locked prior to each flight.

OPERATIONAL (O) PROCEDURES
► Check visually that cargo door is closed, latched and locked prior to each flight.

MAINTENANCE (M) PROCEDURES
Refer to MP ATR-A-52-31-XX-01ZZZ-540Z-A.

- OR -

52-31-01B (B)

RI	NI	NR	PLACARD	O	M
A	1	0	NO	YES	NO

May be inoperative provided that:
(a) Cargo door is not used, and an alternate way is defined to load/unload the forward cargo.
(b) The cargo door is checked closed, latched, and locked prior to each flight, and
(c) Operations are limited to a maximum of 90 calendar days.

OPERATIONAL (O) PROCEDURES
► Check visually that cargo door is closed, latched and locked prior to each flight.

MAINTENANCE (M) PROCEDURES
Refer to MP ATR-A-52-31-30-01ZZZ-520Z-A.

52-31-01 CARGO DOOR MOTOR ACTUATOR (IF INSTALLED) *(Cont'd)*

- OR -

52-31-01C (C)

RI	NI	NR	PLACARD	O	M
A	1	0	NO	YES	NO

May be inoperative provided that:
(a) Cargo door is not used, and an alternate way is defined to load/unload the forward cargo,
(b) The cargo door is checked closed, latched, and locked prior to each flight, and
(c) Operations are limited to a maximum of 90 calendar days.

OPERATIONAL (O) PROCEDURES

► Check visually that cargo door is closed, latched and locked prior to each flight.

52-31-02 CARGO DOOR CONTROL SYSTEM *(IF INSTALLED)*

RI	NI	NR	PLACARD	O	M
C	1	0	YES	NO	NO

May be inoperative provided that cargo door is visually checked locked closed.

52-31-04 CARGO DOOR HOLD OPEN STRUT

RI	NI	NR	PLACARD	O	M
B	1	0	NO	YES	NO

May be inoperative provided that:
(a) Actuation of cargo door is not allowed,
(b) Cargo door is not used, and an alternate way is defined to load/unload the forward cargo, and
(c) Cargo door is checked closed, latched and locked prior to each flight.

OPERATIONAL (O) PROCEDURES

► Check visually that cargo door is closed, latched and locked prior to each flight.
Make a note in the Technical Log book to have Cargo Door Actuator replaced as per MP ATR-A-52-31-30-01001-720A-A.

52-42-01 SERVICE DOOR – HOLD OPEN DEVICE

52-42-01A (A)

RI	NI	NR	PLACARD	O	M
-	1	0	NO	NO	NO

May be inoperative provided that service door is maintained in closed position.
Refer to Item 52-21-01-01 Emergency Exit (Including Passenger – Crew Doors, Service Door, but Excluding Flight Deck Emergency Exits) – Passenger or Combi Configuration

←

- OR -

52-42-01B (B)

RI	NI	NR	PLACARD	O	M
C	1	-	YES	NO	YES

May be inoperative provided that the approved mechanical device to maintain the hook in open position is installed.

MAINTENANCE (M) PROCEDURES
Refer to MP ATR-A-53-67-XX-00ZZZ-560Z-A.

52-50-01-01 COCKPIT DOOR LOCKING SYSTEM

RI	NI	NR	PLACARD	O	M
B	1	0	YES	YES	NO

May be inoperative provided that the complete system is deactivated.

OPERATIONAL (O) PROCEDURES

► Move the manual lock bolt(s) to CLOSE position.

NOTE: *When the door is locked with the manual bolt(s), the emergency access to the cockpit is unavailable.*

Two crew members are mandatory to stay in the cockpit permanently.

NOTE: *Any part of the CDLS may be inoperative provided that the complete system is deactivated in the unlocked position and is not required by operational regulations for the intended flight.*

52-50-01-02 COCKPIT DOOR LOCKING SYSTEM – SYSTEM (CONTROL UNIT, TOGGLE SWITCH, LIGHTS ON CONTROL PANEL, BUZZER)

RI	NI	NR	PLACARD	O	M
A	1	0	YES	YES	NO

May be inoperative provided that operations are limited to two calendar days.

OPERATIONAL (O) PROCEDURES

► Move the manual lock bolts(s) to CLOSE position.

NOTE: *When the door is locked with the manual bolt(s), the emergency access to the cockpit is unavailable.*

Two crew members are mandatory to stay in the cockpit permanently.

NOTE: *Any part of the CDLS may be inoperative provided that the complete system is deactivated in the unlocked position and is not required by operational regulations for the intended flight.*

52-50-01-03 COCKPIT DOOR LOCKING SYSTEM – CALL PUSHBUTTON ON DOOR CALL PANEL

RI	NI	NR	PLACARD	O	M
C	1	0	YES	NO	NO

May be inoperative provided that the crew members interphone is operative.

52-50-01-04 COCKPIT DOOR LOCKING SYSTEM – EMER PUSHBUTTON ON DOOR CALL PANEL

RI	NI	NR	PLACARD	O	M
C	1	0	YES	YES	NO

May be inoperative provided that the crew members interphone is operative.

OPERATIONAL (O) PROCEDURES

► Move the manual lock bolt(s) to CLOSE position.

NOTE: *When the door is locked with the manual bolt(s), the emergency access to the cockpit is unavailable.*

Two crew members are mandatory to stay in the cockpit permanently.

NOTE: *Any part of the CDLS may be inoperative provided that the complete system is deactivated in the unlocked position and is not required by operational regulations for the intended flight.*

52-50-01-05 COCKPIT DOOR LOCKING SYSTEM – OPEN LIGHT ON DOOR CALL PANEL

RI	NI	NR	PLACARD	O	M
C	1	0	YES	NO	NO

May be inoperative.

52-50-01-06 COCKPIT DOOR LOCKING SYSTEM – DENY LIGHT ON DOOR CALL PANEL

RI	NI	NR	PLACARD	O	M
C	1	0	YES	NO	NO

May be inoperative provided that the crew members interphone is operative.

←

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56-11-01 WINDSHIELD (INCLUDING MAIN WINDSHIELD, FWD AND AFT WINDOWS)

56-11-01A (A)

RI	NI	NR	PLACARD	O	M
A	6	6	NO	YES	YES

Structural damages may be present provided:
(a) Flight is conducted unpressurized, and
(b) Vision through affected cockpit windshield is acceptable to the pilot, and
(c) Operations are limited to one calendar day.

NOTE: *A damaged cockpit window must be assessed in accordance with Structural Repair Manual Refer to SRM 56-10-00. If it is assessed as acceptable by maintenance personnel, the vision through this cockpit window must also be assessed as being acceptable by the pilot.*

OPERATIONAL (O) PROCEDURES
Unpressurized Flight
Limitation
Maximum operating altitude ..... 10 000 ft
Normal Procedures
► OVBD VALVE .....FULL OPEN
► CAB PRESS MODE SEL..... MAN
► MAN RATE knob .....MAX INCREASE
► BLEED VALVE .....ON

NOTE: *Control aircraft vertical speed to take into account passenger comfort.*

←

MAINTENANCE (M) PROCEDURES
Refer to SRM 56.10.00.

←

- OR -

56-11-01 **WINDSHIELD (INCLUDING MAIN WINDSHIELD, FWD AND AFT WINDOWS** *(Cont'd)*

56-11-01B **(B)**

RI	NI	NR	PLACARD	O	M
C	6	6	NO	NO	YES

External ply may be cracked, provided that vision through affected cockpit windshield is acceptable to the pilot.

NOTE: *A cockpit window with minor damages, as described in Refer to SRM 56-10-00, is considered operative, but the vision through this cockpit window must be acceptable to the pilot.*

MAINTENANCE (M) PROCEDURES
Refer to SRM 56.10.00.

56-21-01-01 PASSENGER COMPARTMENT WINDOW – PASSENGER CONFIGURATION

RI	NI	NR	PLACARD	O	M
A	-	-	NO	YES	YES

One or more cabin windows may be cracked provided that:
(a) Flight is conducted unpressurized, and
(b) Operations are limited to one calendar day.

OPERATIONAL (O) PROCEDURES
Unpressurized Flight
Limitation
Maximum operating altitude ..... 10 000 ft
Normal Procedures
► OVBD VALVE .....FULL OPEN
► CAB PRESS MODE SEL..... MAN
► MAN RATE knob .....MAX INCREASE
► BLEED VALVE .....ON

NOTE: *Control aircraft vertical speed to take into account passenger comfort.*

MAINTENANCE (M) PROCEDURES
Refer to SRM 56.21.00.

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61-21-02 PEC CHANNEL

RI	NI	NR	PLACARD	O	M
A	4	2	YES	YES	NO

Two may be inoperative provided that:
(a) One channel is operative on each engine, and
(b) Reversers are not used when operated on narrow runways (width <30 m (98 ft)), and
(c) Operations are limited to two calendar days.

OPERATIONAL (O) PROCEDURES
On the engine affected by a PEC SGL CH light illuminated:
Other channel test:
Engine not running – PL on GI - CL on FUEL S.O.
► Set WOW on FLT position on RH Maintenance panel
► Advance CL to AUTO
► Check PEC FAULT illuminates after 30 s
► Retard CL to FUEL S.O.
► Check PEC FAULT extinguishes after 30 s
► Set WOW on NORM position on RH Maintenance panel

Overspeed governor test:
Engine running - PL above FI - CL on AUTO – Make sure NP is above 77%
► Select PEC to OFF
► Slowly advance PL up to reach NP = 102.5 % +/- 1%.
► Increase TQ by 5 % and check NP unchanged.
► Retard PL to GI.
► Select PEC to ON and check PEC OFF extinguishes.
NOTE: *Low pitch stop is automatically checked when propeller is unfeathered.*

61-21-06 PROPELLER SPEED SELECTION IN PIU

RI	NI	NR	PLACARD	O	M
C	2	0	NO	YES	NO

May be inoperative provided that both CL are set to 100 % OVRD.

NOTE: *If affected side cannot be identified, the Propeller Speed Selection of both PIUs should be considered as inoperative.*

OPERATIONAL (O) PROCEDURES

During Preliminary Cockpit Preparation check:

- **Before each affected engine start**
  - ► PL ..... GI
  - ► CL ..... FUEL S.O.
  - ► GUST LOCK ..... OFF
  - ► CABIN TEMPERATURE..... <30 °C
  - ► AVIONICS VENT EXHAUST MODE..... NORM
  - ► STICK PUSHER / SHAKER pb..... OFF
  - ► WOW SWITCH..... FLT

  NOTE: *Depending on angle-of-attack, the aural stall warning may trigger during the test.*

  - ► CL ..... AUTO
- **After 30 s**
  - ► PEC FAULT LIGHT ..... CHECK TURN ON
  - ► CL ..... FUEL S.O.
  - ► PEC..... RESET
  - ► PEC FAULT LIGHT ..... CHECK TURN OFF
  - ► WOW SWITCH..... NORM

During propeller unfeathering

- ► SGL CH INDICATION ..... CHECK TURN ON
- ► SGL CH INDICATION ..... CHECK TURN OFF

61-22-02-01 ATPCS

RI	NI	NR	PLACARD	O	M
C	1	0	YES	YES	NO

One may be inoperative provided that:
(a) At least one AFU is checked operative, and
(b) Operations are conducted in compliance with applicable AFM Deviation Guide.

OPERATIONAL (O) PROCEDURES

- **After engines start check AFU operative**
  - ► TQ Pointer .....CHECK OPERATIVE

Refer to AFM - DEVIATION GUIDE SUPPLEMENTS Dispatch with ATPCS Off.

61-22-02-02 UPTRIM FUNCTION

RI	NI	NR	PLACARD	O	M
C	2	0	YES	YES	NO

May be inoperative provided that:
(a) Autofeather is checked operative before each takeoff, and
(b) Operations are conducted in compliance with applicable AFM Deviation Guide.

OPERATIONAL (O) PROCEDURES
Refer to AFM - DEVIATION GUIDE SUPPLEMENTS Dispatch with Uptrim Function Inoperative.

61-22-02-03 AUTOFEATHER FUNCTION

RI	NI	NR	PLACARD	O	M
C	2	0	YES	YES	NO

May be inoperative provided that:
(a) Both AFU are checked operative,
(b) EEC/uptrim is operative, and
(c) Operations are conducted in compliance with applicable AFM Deviation Guide.

OPERATIONAL (O) PROCEDURES

- **After engines start check AFU operative**
  - ► TQ Pointer .....CHECK OPERATIVE

Refer to AFM - DEVIATION GUIDE SUPPLEMENTS Dispatch with Autofeather System Inoperative.

61-22-03 ATPCS ARM LIGHT

RI	NI	NR	PLACARD	O	M
-	1	0	YES	NO	NO

May be inoperative provided that ATPCS is considered inoperative.
Refer to Item 61-22-02-01 ATPCS.

61-42-01 NP INDICATIONS

RI	NI	NR	PLACARD	O	M
C	2	1	YES	NO	NO

One may be inoperative provided that aircraft is not operated into known or forecast icing conditions.

61-50-01 PROPELLER BRAKE *(IF INSTALLED)*

RI	NI	NR	PLACARD	O	M
D	-	0	YES	NO	YES

May be inoperative provided that it is deactivated.

NOTE: *Hotel mode is not available.*

MAINTENANCE (M) PROCEDURES

Refer to MP ATR-A-61-51-60-02ZZZ-560Z-A.

61-50-02 PROPELLER BRAKE SWITCH GUARD *(IF INSTALLED)*

RI	NI	NR	PLACARD	O	M
-	-	0	YES	NO	NO

May be inoperative provided that propeller brake is considered inoperative.
Refer to Item 61-50-01 Propeller Brake (if installed).

61-51-01 PROP BRAKE UNLK LIGHT *(IF INSTALLED)*

RI	NI	NR	PLACARD	O	M
-	-	0	YES	NO	NO

May be inoperative provided that propeller brake is considered inoperative.
Refer to Item 61-50-01 Propeller Brake.

61-51-02 PROP BRK LIGHT *(IF INSTALLED)*

PROP BRK Local Light (if installed)

RI	NI	NR	PLACARD	O	M
C	1	0	YES	NO	NO

May be inoperative provided that propeller brake is not used.

61-51-03 PROP BRK READY LIGHT *(IF INSTALLED)*

RI	NI	NR	PLACARD	O	M
C	1	0	YES	YES	NO

May be inoperative provided that propeller brake is not used.

OPERATIONAL (O) PROCEDURES

- **If propeller brake releasing is needed**
  - ► BLUE HYD PRESS ..... CHECK 3000 PSI
  - ► PROP BRK..... RELEASE
  - ► UNLOCK LIGHT.....CHECK EXTINGUISHES AFTER 6 S
- **If PROP BRK warning is activated**
  - ► CL .....FUEL S.O.

61-60-01 PROPELLER SYSTEM – PROP FEATH PUMP AND PROP OVSP TEST SWITCHES (RH MAINTENANCE PANEL)

RI	NI	NR	PLACARD	O	M
C	2	0	NO	NO	NO

May be inoperative provided test switch remains in the neutral and guarded position.

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73-13-01 IDLE GATE FAIL LIGHT....73-1

73-15-01 FUEL HEATER

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GUARDED SWITCHES....73-11

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73-13-01 IDLE GATE FAIL LIGHT

RI	NI	NR	PLACARD	O	M
C	1	0	YES	YES	NO

May be inoperative.

OPERATIONAL (O) PROCEDURES

Normal Procedures

- **AFTER TAKEOFF**
  - ► IDLE GATE lever .....CHECK PUSHED IN
- **AT LANDING**
  - ► IDLE GATE lever .....CHECK PULLED OUT

73-15-01 FUEL HEATER

73-15-01A LOW Fuel Temperature Reported (A)

RI	NI	NR	PLACARD	O	M
C	2	0	YES	NO	YES

May be inoperative provided that:
(a) Fuel anti-icing additives meets the engine manufacturer's specification, and
(b) Engine oil level is checked before the first flight of each day.

MAINTENANCE (M) PROCEDURES

- **IF TOO LOW TEMP**
  Fuel anti-icing additive for all fuel:
  To be added in compliance with Engine Maintenance Manual (Refer to Engine Maintenance Manual 72-00-00 – Anti-icing Additives Table).

Check of oil level:

► Open left hinged cowl (435AL or 445AL).
► Check that oil level is to "FULL".
► Close cowl(s).

- OR -

73-15-01B LOW Fuel Temperature Reported (B)

RI	NI	NR	PLACARD	O	M
C	2	0	YES	NO	NO

May be inoperative provided that flight can be conducted in atmospheric conditions above 0° SAT.

- OR -

73-15-01C High Fuel Temperature Reported (C)

RI	NI	NR	PLACARD	O	M
C	2	0	YES	NO	NO

May be inoperative provided that no fault in other fuel systems (ATA 28) is reported or is under MEL dispatch.

73-15-01 FUEL HEATER Cont'd)

- OR -

73-15-01D Fuel Temperature Indicator Inoperative (D)

RI	NI	NR	PLACARD	O	M
C	2	0	YES	NO	NO

May be inoperative provided that:
(a) Flight can be conducted in atmospheric conditions above 0° SAT, and
(b) No other fault in fuel systems (ATA 28) is reported or is under MEL dispatch.

73-23-01 PWR MGT ROTARY SELECTOR

RI	NI	NR	PLACARD	O	M
A	2	1	YES	YES	NO

One may be inoperative provided that operations are limited to two calendar days.

OPERATIONAL (O) PROCEDURES

■ **If One rotary selector stuck in TO position**

► CL 1 + 2, for the whole flight.....100 % OVRD

● **For CLIMB and CRUISE**

► Valid rotary selector.....CLIMB / CRUISE

► TQ affected side ..... According to the non affected one

■ **One rotary selector stuck in MCT positon**

► ATPCS is lost: Refer to ATPCS (and Associated TEST) (O)

► CL 1 + 2, for the whole flight.....100 % OVRD

● **For TAKEOFF**

► Inhibit CONFIG WARNING by depressing MASTER WARNING pb when applying power.

NOTE: ► *EWD alert will not disappear.*

● **CLIMB and CRUISE**

► Valid rotary selector.....CLIMB / CRUISE

► TQ affected side ..... According to the non affected one

■ **One rotary selector stuck in CLB or CRZ position**

► ATPCS is lost: Refer to ATPCS (and Associated TEST) (O)

● **For TAKEOFF**

► CL 1 + 2 .....100 % OVRD

► Inhibit CONFIG WARNING by depressing MASTER WARNING pb when applying power.

NOTE: ► *EWD alert will not disappear.*

● **For CLIMB and CRUISE**

► Valid rotary selector.....CLIMB / CRUISE

► CL 1 + 2 .....AUTO

► TQ affected side ..... According to the non affected one

● **For FINAL APPROACH**

► CL 1 + 2 .....100 % OVRD

73-23-02-01 ELECTRONIC ENGINE CONTROL - EEC

RI	NI	NR	PLACARD	O	M
A	2	1	YES	YES	NO

One EEC, counter and associated TQ target bug may be inoperative provided that:
(a) The DC HYD pump is operative, and
(b) The associated pack and bleed valve are operative, and
(c) Operations are conducted in compliance with applicable AFM Deviation Guide, and
(d) The operations are limited to two calendar days.

NOTE: *Disregard HBV FLAG on TQ 1(2) indication.*

OPERATIONAL (O) PROCEDURES
Refer to AFM - DEVIATION GUIDE SUPPLEMENTS Dispatch with One EEC OFF.

73-23-05 ENGINE RATING SELECTION

73-23-05A (A)

RI	NI	NR	PLACARD	O	M
C	1	0	YES	YES	YES

May be inoperative provided that:
(a) Operations are conducted with BOOST ON,
(b) ERIU is deactivated, and
(c) BOOST ON light is on.

OPERATIONAL (O) PROCEDURES

- **(A) Engine Rating Selection**
  Limitations
  BOOST: USE
- **(B) BOOST Function**
  Limitations
  BOOST Selection: Prohibited

MAINTENANCE (M) PROCEDURES

ERIU De-activation

► Open, secure, and tag the following C/B ENG IND/CTL/RATING/ERIU SUPPLY.

- OR -

73-23-05B BOOST FUNCTION (B)

RI	NI	NR	PLACARD	O	M
D	1	0	YES	YES	NO

May be inoperative provided that:
(a) Operations are conducted with BOOST OFF,
(b) BOOST ON light is off, and
(c) BOOST FAULT light is off.

OPERATIONAL (O) PROCEDURES

- **(A) Engine Rating Selection**
  Limitations
  BOOST: USE
- **(B) BOOST Function**
  Limitations
  BOOST Selection: Prohibited

73-31-01 FUEL FLOW INDICATION

RI	NI	NR	PLACARD	O	M
B	2	1	YES	NO	NO

One may be inoperative.

73-31-02 FUEL USED

RI	NI	NR	PLACARD	O	M
C	2	1	YES	NO	NO

One may be inoperative provided that associated tank FUEL QTY indicator is operative.

73-34-01 FUEL CLOG ALERT

RI	NI	NR	PLACARD	O	M
A	2	1	YES	NO	NO

One may be inoperative provided that:

(a) Associated engine parameters are monitored during the flight, and

(b) Operations are limited to one calendar day and aircraft does not depart an airport where repair or replacement may be carried out.

73-35-01 FUEL TEMPERATURE INDICATION

RI	NI	NR	PLACARD	O	M
-	2	0	YES	NO	NO

May be inoperative provided that associated fuel heater is considered inoperative:
Refer to Item 73-15-01 Fuel Heater - Fuel Temperature Indicator Inoperative (D).

73-60-01 ENG - RH MAINTENANCE PANEL GUARDED SWITCHES

RI	NI	NR	PLACARD	O	M
C	-	0	NO	NO	NO

May be inoperative provided test switch remains in the neutral and guarded position.

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CONTENTS

74-21-01 IGNITION SYSTEM (A OR B).....74-1

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74-21-01 IGNITION SYSTEM (A OR B)

RI	NI	NR	PLACARD	O	M
A	4	2	YES	NO	NO

Any in excess of two may be inoperative provided that:
(a) Operations are limited to three calendar days, and
(b) One is operating on each engine and used.

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CONTENTS

75-20-01 HBV DISCHARGE FLAPPER 75-1

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75-20-01 HBV DISCHARGE FLAPPER

RI	NI	NR	PLACARD	O	M
D	4	0	NO	NO	NO

All may be inoperative.

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←

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77-60-01 DATA BUS (RH MAINTENANCE PANEL) 77-10

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77-00-02 AFU

RI	NI	NR	PLACARD	O	M
B	2	1	YES	YES	YES

One may be inoperative provided that:
(a) AFU is deactivated,
(b) ATPCS is considered inoperative, and
(c) Operations are conducted in compliance with applicable AFM Deviation Guide.
←

OPERATIONAL (O) PROCEDURES
Refer to AFM - DEVIATION GUIDE SUPPLEMENTS Dispatch with ATPCS Off.

MAINTENANCE (M) PROCEDURES
This task is performed to deactivate the AFU and then to avoid an untimely autofeather in flight:
► Open "C/B ENG CTL/ENG 1 (2) / AFU".
► Open "C/B ENG CTL/ENG 1 (2) / FEATH CTL & AFU"
► Open nacelle cowling left side.
► Disconnect AFU connector.
► Secure and cap electrical connectors.
► Close nacelle cowling.
► Close previously opened C/Bs.

77-11-01-02 NH INDICATOR

RI	NI	NR	PLACARD	O	M
C	2	1	YES	YES	NO

One Indicator (pointer and counter) may be inoperative provided that both engines TQ indicator and NP indicator are operative.

OPERATIONAL (O) PROCEDURES

► When starting affected engine, select CL FTR 5 s after START pb has been depressed.

77-11-02 NL INDICATOR

RI	NI	NR	PLACARD	O	M
C	2	0	YES	NO	NO

One or both may be inoperative provided that associated NH indicator is operative and monitored.

77-13-01-01 TQ INDICATIONS – DIGITAL COUNTER

RI	NI	NR	PLACARD	O	M
C	2	0	YES	YES	NO

One or both Digital counters may be inoperative provided that:
(a) Both EECs are operative,
(b) HBV is checked operative, and
(c) Both TQ pointers are operative.

←

OPERATIONAL (O) PROCEDURES

- **During flight preparation check HBV operative**
  - ► HBV FLAG ON TQ DIGITAL INDICATION..... CHECK NOT DISPLAYED
  - ► ---.- ON TORQUE DIGITIAL INDICATION ..... CHECK NOT DISPLAYED

77-13-01-02 TQ INDICATIONS – POINTER & DIGITAL COUNTER

RI	NI	NR	PLACARD	O	M
C	4	2	YES	YES	NO

Both Pointer and Digital counter on the same ENG may be inoperative provided that:
(a) ATPCS checked operative,
(b) Both EECs are operative,
(c) All NH and NP Indications are operative, and
(d) Operations are conducted in compliance with applicable AFM Deviation Guide.
←

OPERATIONAL (O) PROCEDURES

Refer to AFM - DEVIATION GUIDE SUPPLEMENTS Dispatch with TQ Indications Inoperative (Both Pointer Digital Counter on the Same ENG).

77-13-01-03 TQ INDICATIONS – POINTER

RI	NI	NR	PLACARD	O	M
-	2	1	YES	YES	NO

One pointer may be inoperative provided that:
(a) Associated AFU is considered inoperative, Refer to Item 77-00-02 AFU.
(b) Both Digital counters are operative,
(c) Both EECs are operative, and
(d) HBV is checked operative.

OPERATIONAL (O) PROCEDURES

- **During flight preparation check HBV operative**
  - ► HBV FLAG ON TQ DIGITAL INDICATION ..... CHECK NOT DISPLAYED
  - ► ---.- ON TORQUE DIGITAL INDICATION ..... CHECK NOT DISPLAYED

77-14-01 TQ TARGET INDICATION

RI	NI	NR	PLACARD	O	M
C	2	0	NO	YES	NO

One or both may be inoperative.

OPERATIONAL (O) PROCEDURES

- **If one TQ target indication is inoperative**
  ► TQ TARGET (affected side): USE OPERATIVE TARGET INDICATION
- **If both TQs target indication are inoperative**
  TQ TARGET 1+2: USE TQ TABLE (Refer to QRH PER.2.Torque Value)

77-20-01 ITT INDICATIONS

RI	NI	NR	PLACARD	O	M
C	4	2	YES	NO	NO

Either pointer or counter may be inoperative on each indicator.

←

77-21-01 EXCESSIVE ITT ALERT

RI	NI	NR	PLACARD	O	M
A	2	0	YES	NO	NO

May be inoperative provided that:
(a) Associated ITT indicator is operative and monitored throughout the flight, and
(b) Operations are limited to three calendar days.

77-60-01 DATA BUS (RH MAINTENANCE PANEL)

RI	NI	NR	PLACARD	O	M
C	1	0	NO	NO	NO

May be inoperative provided test switch remains in the neutral and guarded position.

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79-32-01 OIL TEMPERATURE INDICATION 79-1

79-33-01 OIL PRESSURE INDICATION 79-2

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79-32-01 OIL TEMPERATURE INDICATION

RI	NI	NR	PLACARD	O	M
B	2	1	YES	NO	YES

One may be inoperative provided that:
(a) The associated fuel temperature indication is operative, and
(b) The Oil pressure indication is operative, and
(c) The affected engine oil level is checked before the first flight of each day.

MAINTENANCE (M) PROCEDURES
Refer to MP ATR-A-12-13-79-00ZZZ-310Z-A.

79-33-01 OIL PRESSURE INDICATION

RI	NI	NR	PLACARD	O	M
B	2	1	YES	NO	YES

One may be inoperative provided that the associated ENG OIL PRESS alert is checked operative.

MAINTENANCE (M) PROCEDURES

Refer to MP ATR-A-79-34-XX-00ZZZ-320Z-A.